Software Requirements Specification

for

E-LEARNING PLATFORM

Version 3.0 approved

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BONAFIDE CERTIFICATE

Certified that this project report "**E-LEARNING PLATFORM**" is the bonafide work of **KARNASSAGAR S (202109026), SIVA PRASANTH M(202109049), RAMKUMAR K (202109253)** who carried out the project work under my supervision.

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The project report submitted for the viva voce held on ………….

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Revision History

Name	Date	Reason For Changes	Version
Review - I	06/03/24	-	1.0.0
Review - II	17/04/24	-	2.0.0
Review - III	14/06/25	Inaccurate documentation	3.0.0

1. Introduction

1.1 Purpose

The purpose of the E-Learning Platform project is to provide an online environment for educational institutions to facilitate teaching and learning activities. In today's digital era, the demand for flexible and accessible learning solutions has grown significantly. This platform aims to bridge the gap between traditional classroom settings and modern online learning by offering a comprehensive suite of tools and features. It is designed to cater to the specific needs of administrators, teachers, and students by offering role-based functionalities.

1.2 Document Conventions

Bold: Headings, subheadings, and emphasis.
Italics: Terms defined in the glossary and references.
Monospace: Code snippets, commands, or system outputs.

1.3 Intended Audience and Reading Suggestions

This document is intended for the development team, project stakeholders, quality assurance team, and end-users including administrators, teachers, and students. The reading suggestions are provided in the document for different audience groups. Developers can refer to the provided source code for implementation details.

1.4 Product Scope

The E-Learning Platform will have three login roles: Admin, Teacher, and Student. Each role will have specific functionalities to perform various tasks related to course management, attendance, assessments, and communication. The platform will utilize MongoDB for data storage and retrieval.

1.5 References

E-Learning Platform Vision and Scope Document, Version 1.0, Author: [Insert Author Name], Date: [Insert Date], Location: [Insert Location]
E-Learning Platform User Interface Style Guide, Version 1.0, Author: [Insert Author Name], Date: [Insert Date], Location: [Insert Location]
E-Learning Platform System Requirements Specification, Version 1.0, Author: [Insert Author Name], Date: [Insert Date], Location: [Insert Location]
E-Learning Platform Use Case Document, Version 1.0, Author: [Insert Author Name], Date: [Insert Date], Location: [Insert Location]

2. Overall Description

2.1 Product Perspective

The E-Learning Platform is a standalone system designed to facilitate online education. It will interact with users through a web-based interface. As an integral part of the educational ecosystem, the platform aims to complement traditional teaching methods by providing a flexible and accessible online learning environment. It will support seamless integration with existing Learning Management Systems (LMS) and educational tools, ensuring a cohesive and enhanced learning experience for all users.

2.2 Product Functions

Admin:
View profile: View and edit user's own user profile details and avatar.
Manage Classes: Add, remove and view virtual classrooms for different subjects.
Manage Subjects: Add, remove and view virtual subjects for different classrooms.
Manage Teachers and Students: Add, remove and view Teacher and Student profiles whilst assigning them to classes and subjects.
Take Attendance: Record and track student attendance for each subject.
Provide Marks: Give/edit a student's mark for a subject they are registered in.
Publish Notices: Share important announcements, updates, and notifications with other admins, teachers and students.
View Complaints: View the complaints raised by Students

Teacher:
View profile: View and edit user's own user profile details and avatar.
Take Attendance: Record and track student attendance for each subject.
Provide Marks: Give/edit a student's mark for a subject they are registered in.
View Notices: View announcements published by administrators.

Student:
View profile: View and edit user's own user profile details and avatar.
View Attendance: Access and review their attendance records.
View Marks: Access and review their attendance performance.
Raise Complaints: Lodge complaints or concerns related to course content, assessments, or any other issues.
View Notices: View announcements published by administrators.

2.3 User Classes and Characteristics

Admin:
Role: Administrator
Characteristics:
Manages the overall system and course administration.
Responsible for user management, course creation, and system configuration.

Teacher:
Role: Instructor or Educator
Characteristics:
Conducts classes, assessments, and interactive sessions.
Evaluates students' academic performance and attendance.

Student:
Role: Learner or Participant
Characteristics:
Views their academic performance and attendance in a subject, and participates in virtual classrooms.

2.4 Operating Environment

The platform will operate in a web-based environment compatible with modern web browsers such as Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge. It will require a stable internet connection to ensure uninterrupted access and optimal performance. The platform will also be accessible on various devices, including desktop computers, laptops, tablets, and smartphones, providing users with the flexibility to learn anytime, anywhere.

2.5 Design and Implementation Constraints

Security Measures:
Implementation of robust security protocols to protect user data and prevent unauthorized access.
Encryption of sensitive information and secure data transmission to safeguard user privacy and confidentiality.

Scalability:
Design and architecture of the platform to accommodate a large number of users, courses, and concurrent sessions.
Optimization of database queries and server resources to ensure efficient performance and scalability.

Development of a responsive and adaptive design to ensure compatibility with various devices, screen sizes, and resolutions.
Cross-browser compatibility testing and optimization to provide a consistent and seamless user experience across different web browsers.We have used Basic model-Organic for our project.

a = 2.4
b = 1.05
c = 2.5
d = 0.3

E = a * (KLOC) ^ b person-months.

E = 2.4 * (150) ^ 1.05

E = 462.5 person-months.

T = c * (E) ^ d months.

T = 2.5 * (462.5) ^ 0.38

T = 25.74 months.

No of Persons required = E/T Persons.

No of Persons required = 462.5/25.74

No of Persons required = 18 Persons.

2.6 User Documentation

User documentation will include user manuals, online help, and tutorials. The delivery format will be web-based and accessible within the platform.

2.7 Assumptions and Dependencies

Assumed factors include stable internet connectivity for users.
Dependencies include third party libraries for certain functionalities.

3. External Interface Requirements

3.1 User Interfaces

Intuitive Web-Based Interfaces:

Admin Interface:

Dashboard displaying an overview of system status, user management, course creation, and other administrative functions.

Navigation menu for easy access to various functionalities like class management, course creation, user management, and system settings.

Interactive forms and input fields for data entry, modification, and deletion.

Platforms to house important announcements and updates, along with

complaints.

Teacher Interface:

Dashboard provides an overview of classes, student attendance, course materials, and assessments.

Navigation menu for accessing attendance management and grade submission.

Student Interface:

Dashboard displaying course enrollments, attendance records, and course grades.

Navigation menu for accessing course details and attendance records.

User-friendly interfaces for constructing and submitting complaints.

Real-time notifications for course updates, grades, and communication from teachers or administrators.

Consistent Layout and Navigation:

Standardized design elements, such as headers, footers, and navigation bars, to maintain consistency across different pages and user roles.

Responsive and adaptive design to ensure compatibility and optimal viewing experience on various devices and screen sizes.

3.2 Hardware Interfaces

No Specific Hardware Interfaces Required:

The platform is designed to operate on standard computing devices, including desktop computers, laptops, tablets, and smartphones, without the need for specialized hardware interfaces.

3.3 Software Interfaces

Integration with Authentication Systems:

Integration with Single Sign-On (SSO) systems or third-party authentication services for secure and seamless user login.

Implementation of OAuth or JWT for secure authentication and authorization processes.

Database Interfaces:

Integration with MongoDB for efficient and scalable data storage and retrieval.

Implementation of CRUD (Create, Read, Update, Delete) operations for managing user data, course details, attendance records, and other relevant information.

3.4 Communications Interfaces

Real-Time Communication Tools:

Integration with messaging APIs or platforms, such as WebSocket, to enable seamless and interactive communication within the platform.

4. System Features

4.1 Admin Features

View Profile:
View profile details: Admin can view their profile details and their profile picture/avatar.
Edit avatar: Admin can edit their profile picture/avatar when needed.

Manage Classes and Courses:
Manage Classes: Admin can create new classes by specifying the class name, subject, time, and other relevant details. The Admin can also delete any created classes.

Manage Subjects: Admin can create subjects by providing course details such as course name, description, department, and associated class. The Admin can also delete any created subjects.

Manage Teachers and Students:
Add New Teacher/Student: Admin can add new teachers or students by entering their personal and contact details.

Delete Teacher/Student: Admin can remove existing teacher or student profiles as needed.

Take Attendance:
Class-wise Attendance: Admin can mark and maintain attendance records for each class and view attendance reports.

Generate Attendance Reports: Admin can generate detailed attendance reports for students.

Provide Marks:
Enter Marks: Teachers can enter and update assessment marks for students.

Generate Grade Reports: Teachers can generate grade reports of students.

Edit Marks: Admin can edit previously registered marks of students.

Publish Notices:
Create Notices: Admin can create and publish notices regarding important updates, announcements, or events.

Delete Notices: Admin can remove any published notices if they are considered irrelevant.

Handle Complaints:
View complaints: Admin can view complaints submitted by students.
Solve complaints: Admin can establish that a complaint has been resolved

4.2 Teacher Features

View Profile:
View profile details: Teachers can view their profile details and their profile picture/avatar.
Edit avatar: Teachers can edit their profile picture/avatar when needed.

Take Attendance:
Mark Attendance: Teachers can mark and update attendance records for their respective classes.
View Attendance Reports: Teachers can view and monitor attendance reports for students in their classes.

Provide Marks:
Enter Marks: Teachers can enter and update assessment marks for students.
Generate Grade Reports: Teachers can generate grade reports of students.
Edit Marks: Teachers can edit previously registered marks of students.

4.3 Student Features

View Profile:
View profile details: Students can view their profile details and their profile picture/avatar.
Edit avatar: Students can edit their profile picture/avatar when needed.

View Marks and Attendance:
Check Grades: Students can view their assessment marks and grades for each course.
Access Attendance Records: Students can check and monitor their attendance records for each class.

Raise Complaints:
Submit Complaints: Students can raise complaints or concerns regarding classes, assessments, or other platform-related issues.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Platform Responsiveness:

The platform should load quickly, with a maximum load time of 2-3 seconds for all pages and functionalities.

The system should respond to user actions promptly, with a response time of less than 1 second for basic operations.

Concurrent User Support:

The platform should support a minimum of 1000 concurrent users to ensure smooth operation during peak times.

Load balancing mechanisms should be in place to distribute incoming traffic and maintain optimal performance.

5.2 Safety Requirements

Data Encryption:

All user data transmitted over the platform should be encrypted using secure encryption algorithms such as AES (Advanced Encryption Standard).

Data stored in the database should be encrypted to protect against unauthorized access and data breaches.

Access Control Measures:

Role-based access control (RBAC) should be implemented to restrict access to sensitive information based on user roles (Admin, Teacher, Student).

5.3 Security Requirements

User Authentication Mechanisms:

Secure user authentication methods such as password hashing, salting, and multi-factor authentication (MFA) should be implemented.

Password policies, including minimum length, complexity, and expiration, should be enforced for user accounts.

Regular Security Audits and Updates:

Regular security audits should be conducted to identify and address potential vulnerabilities.The platform should be updated with the latest security patches and updates to protect against new and emerging threats.

5.4 Software Quality Attributes

Usability:

Intuitive and user-friendly interfaces should be designed for easy navigation and accessibility across different devices and screen sizes.

Clear and concise instructions, tooltips, and help guides should be provided to assist users in using the platform effectively.

Reliability:

The system should be stable and available 24/7, with a minimum uptime of 99.9%.

Automated backup and recovery mechanisms should be in place to ensure data integrity and availability in the event of system failures or data loss.

Maintainability:

The platform should be modular and scalable, allowing for easy updates, modifications, and maintenance of system components.

Documentation and code commenting should be maintained to facilitate understanding and modification of the system by developers and administrators.

5.5 Business Rules

Admin Role:

Admins can manage all courses, classes, and users within their assigned college.

Admins can assign roles for teachers and students.

Admins can manage attendance and marks of students.

Admins can manage notices and student-submitted complaints.

Teacher Role:

Teachers can access courses, classes, and students they are assigned to.

Teachers can manage attendance and marks of students they are assigned to.

Teachers can view notices uploaded by administrators.

Student Role:

Students can access and view classes and subjects they are enrolled in.

Students can view their grades, attendance of subjects they are enrolled in.

Students can raise their complaints to administrators.

Students can view notices uploaded by administrators.

6. Blackbox Testing

Three major modules:

- Admin
- Teacher
- Student

We are going to test each module by unit-testing and then we will check integration-testing of the module by combining those functionalities with other modules.

6.1 Testing of Admin Module:

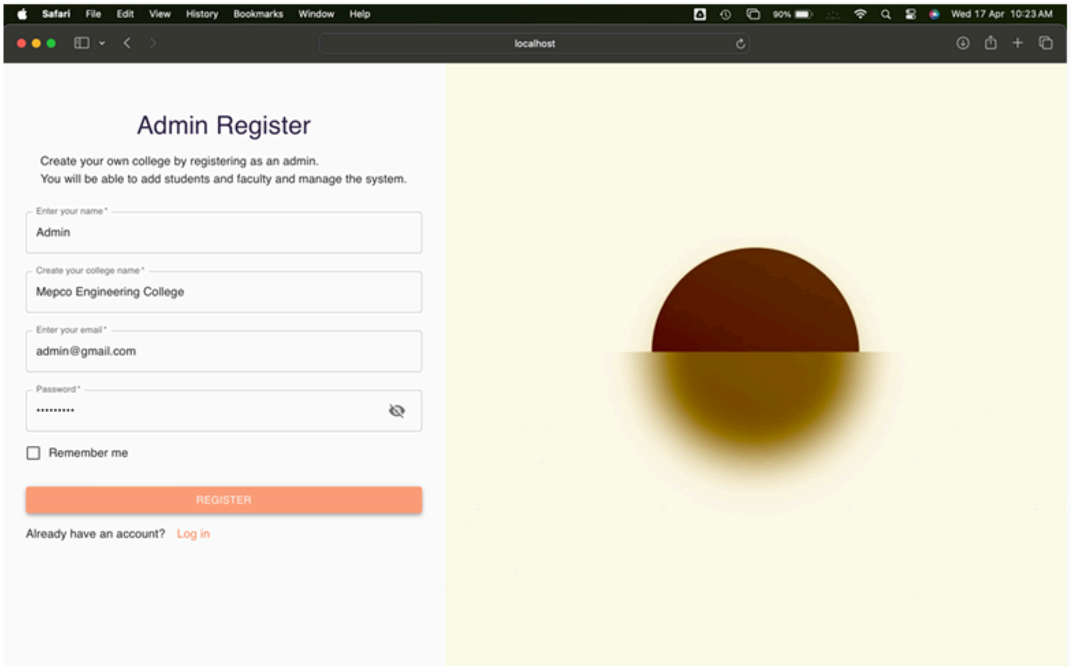
List of Modules:

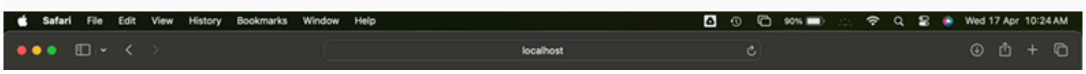
- Signup Page
- Login Page
- Admin SideBar
- Dashboard
- Class Creation
- Course Creation
- Teacher Login Id creation
- Student Login Id creation
- Notice
- Complain
- Logout

6.1.1 Unit Testing:

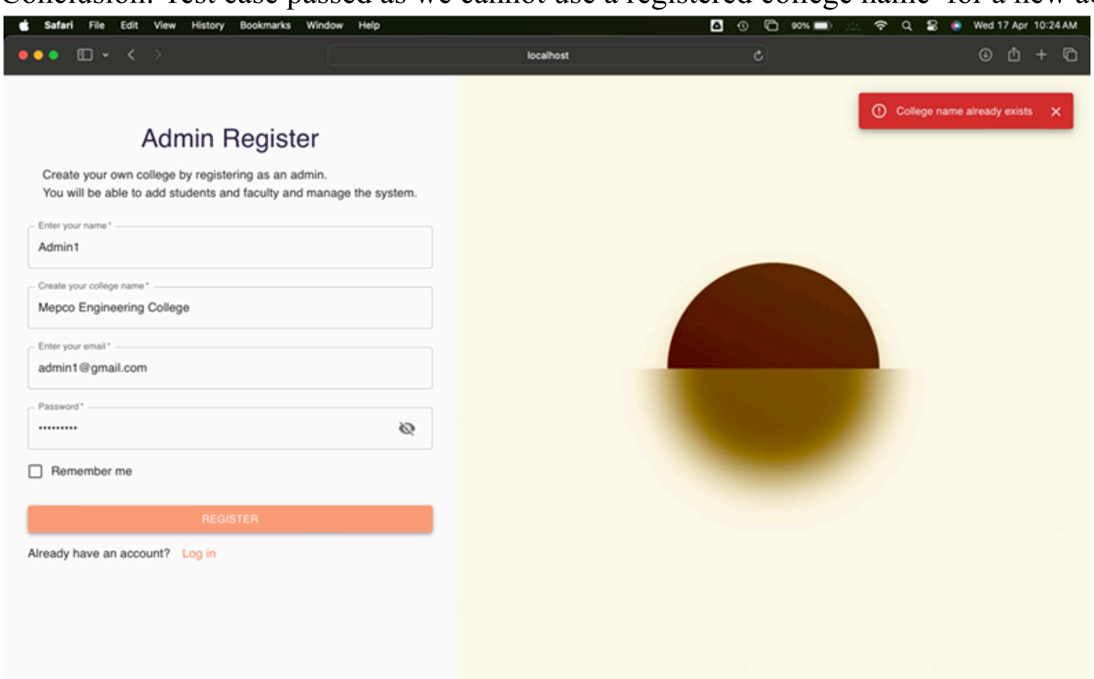
6.1.1.1 Signup Page:

Test case: Check for Signup page loading.
Conclusion: Test case passed as Signup page loaded



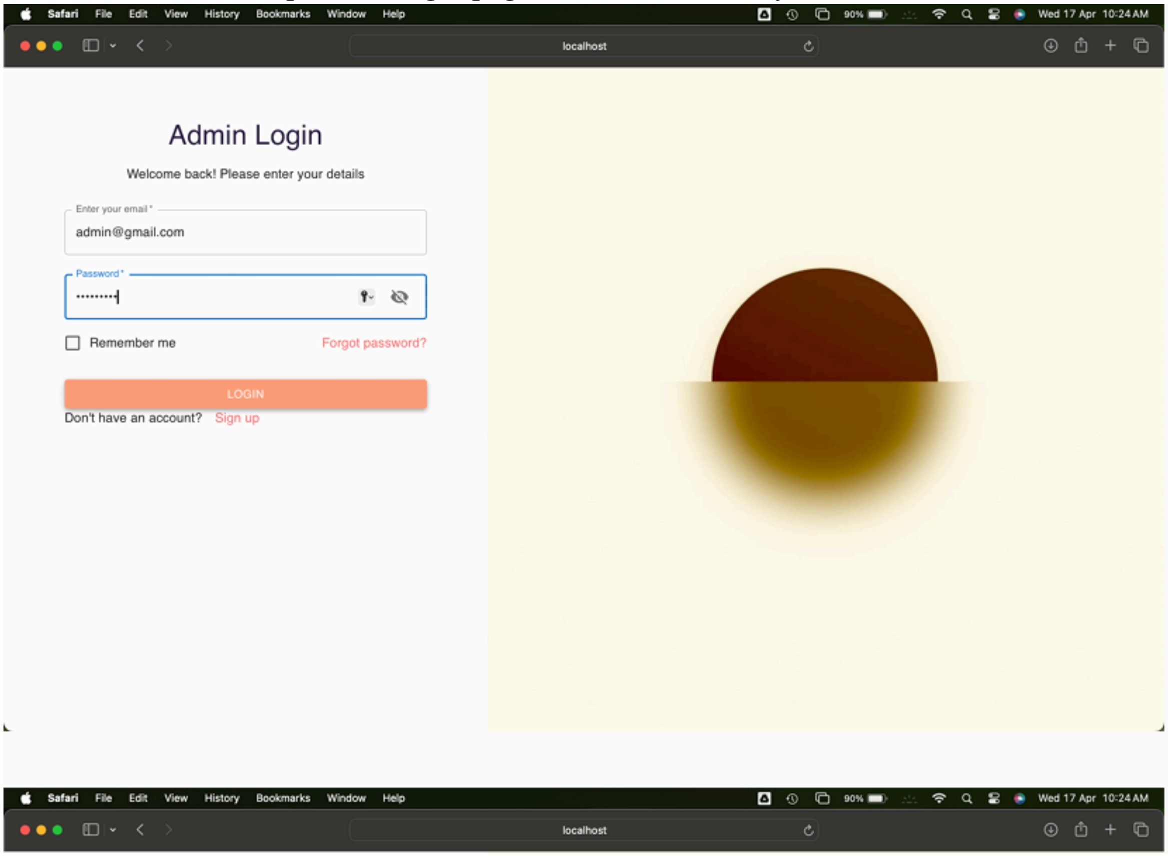


Test case: Check for Signing up with a registered college name.
Conclusion: Test case passed as we cannot use a registered college name for a new admin id.

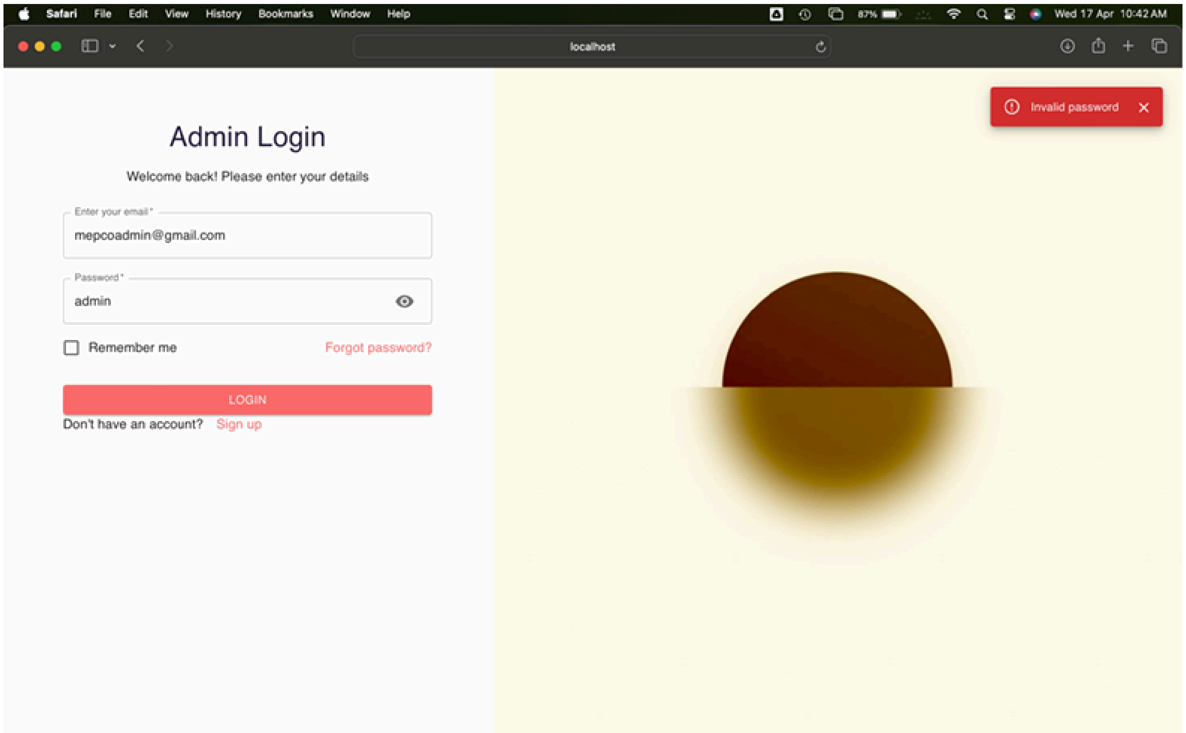


6.1.1.2 Login Page:

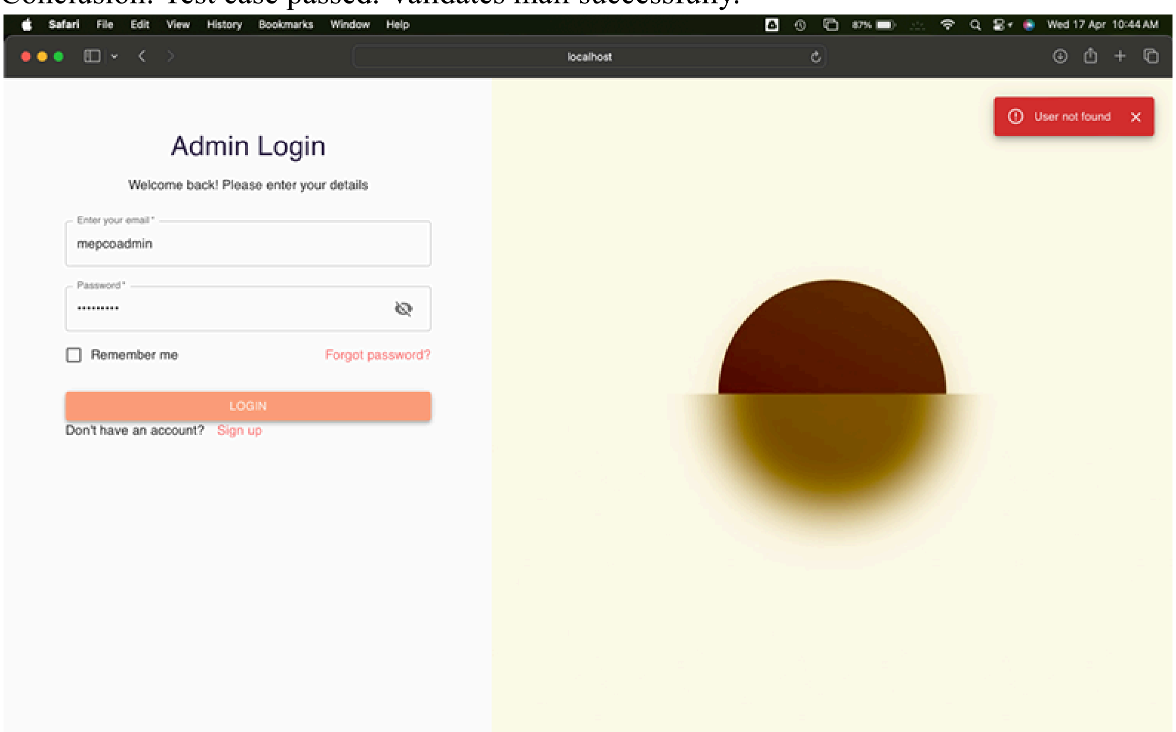
Test case: Login page loading
Conclusion: Test case passed. Login page loaded successfully.



Test case: Check for password validation
Conclusion: Test case passed. Validates password successfully.



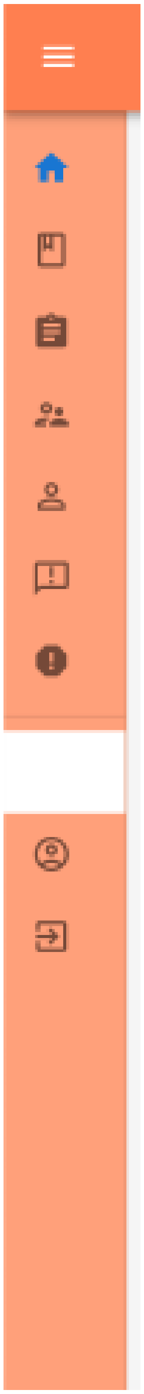
Test case: Check for mail validation
Conclusion: Test case passed. Validates mail successfully.



6.1.1.3 Admin Sidebar:

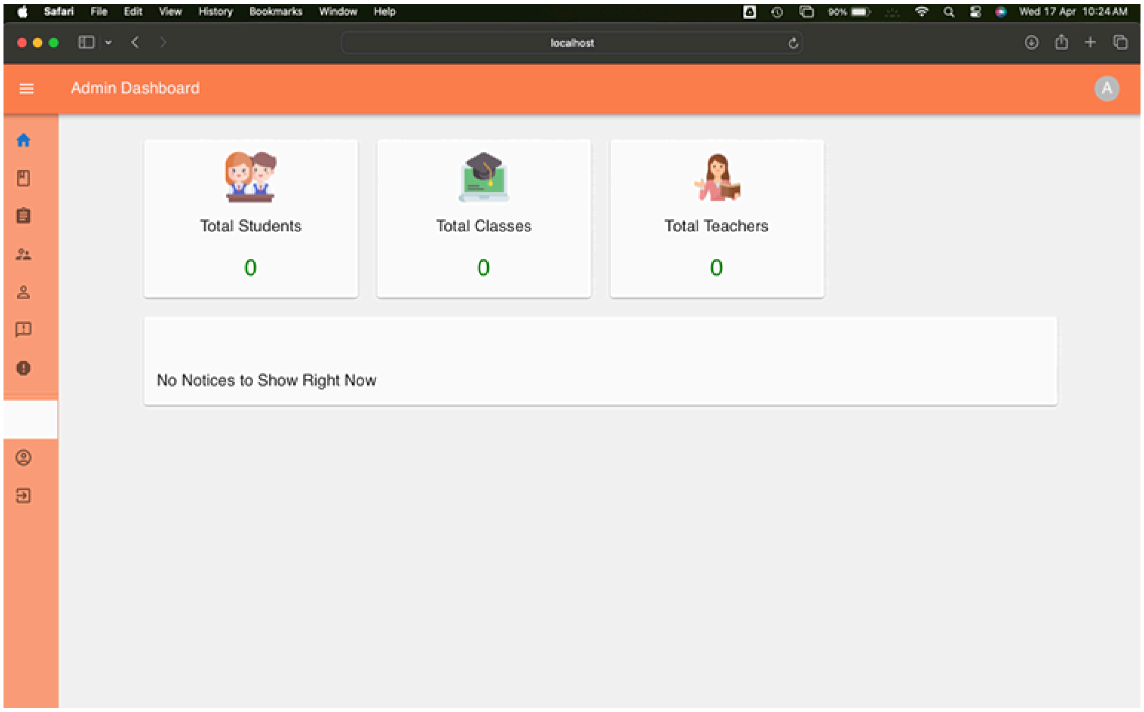
Test case: Check for loading of Admin NavBar

Conclusion: Test case passed. Admin NavBar loaded successfully.



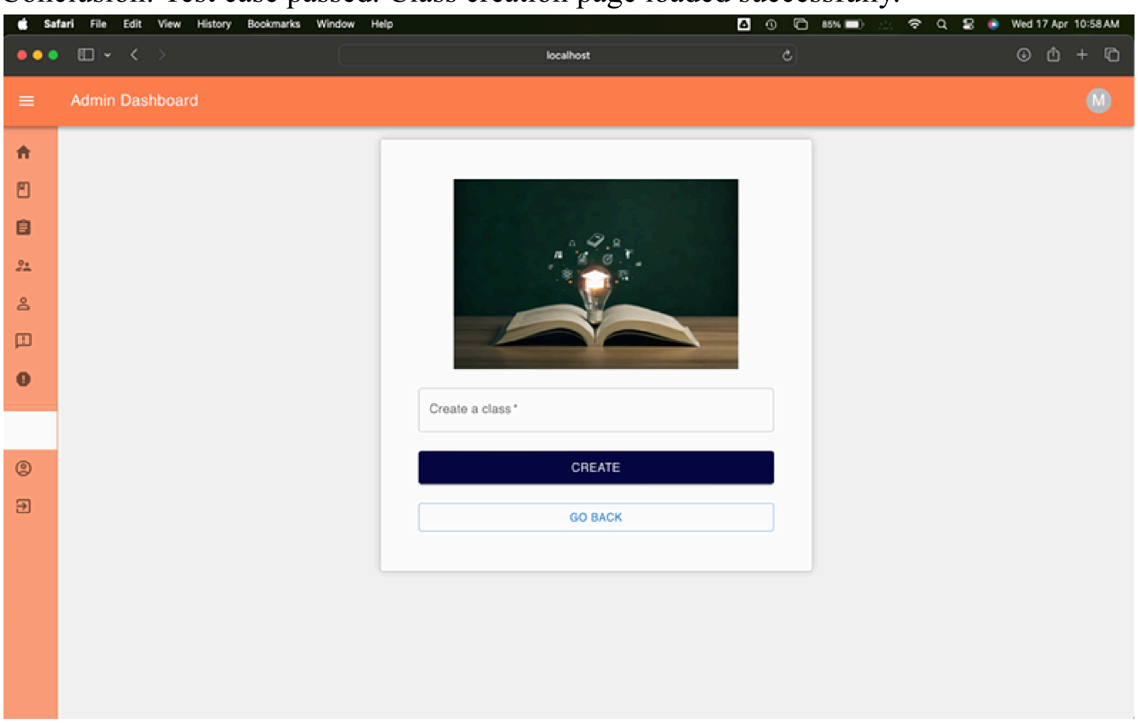
6.1.1.4 Dashboard:

Test case: Check for Admin Dashboard loading.
Conclusion: Test case passed. Admin Dashboard loaded successfully.

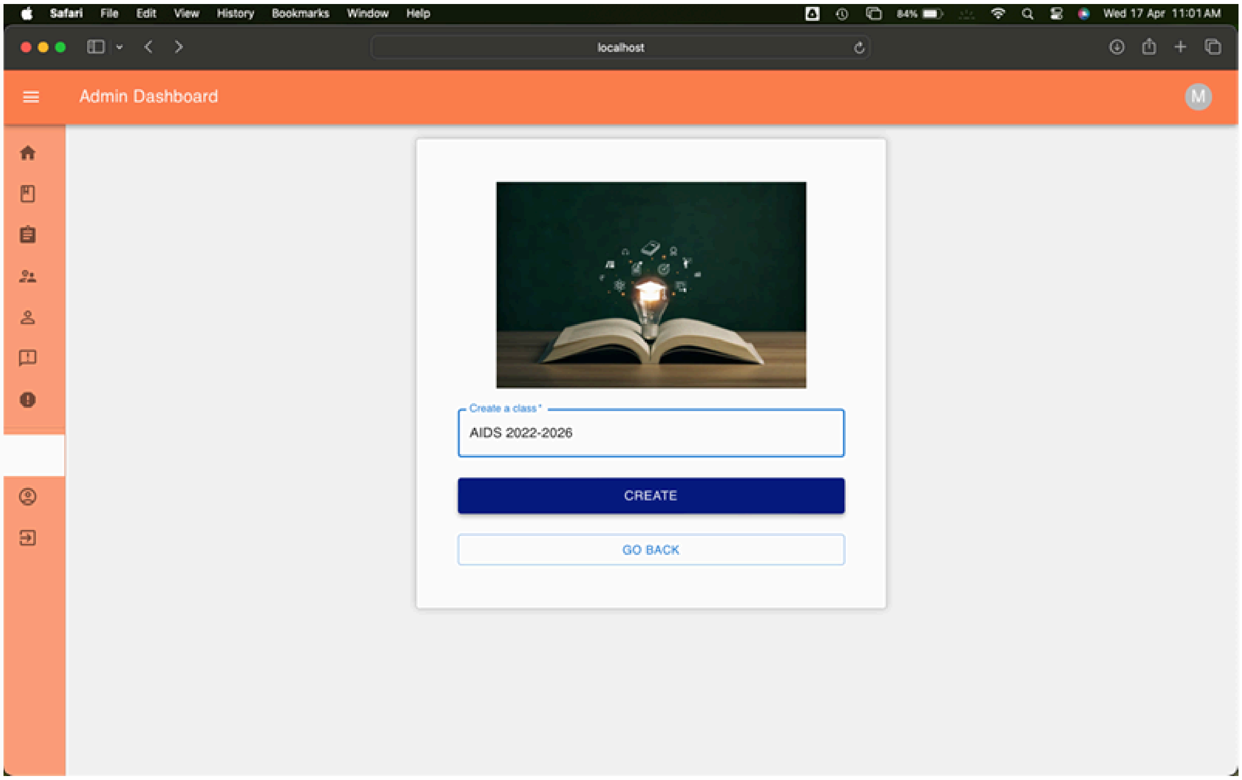


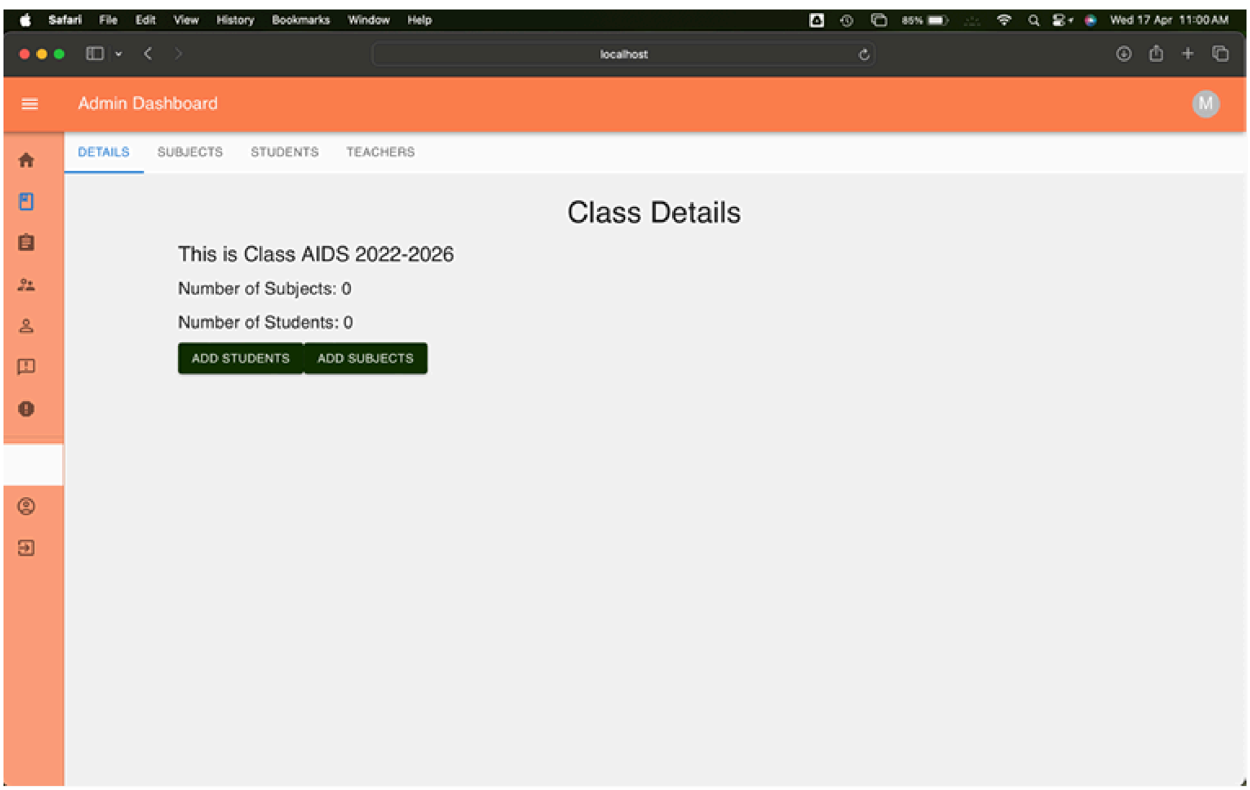
6.1.1.5 Class Creation:

Test case: Check for class creation page loading.
Conclusion: Test case passed. Class creation page loaded successfully.



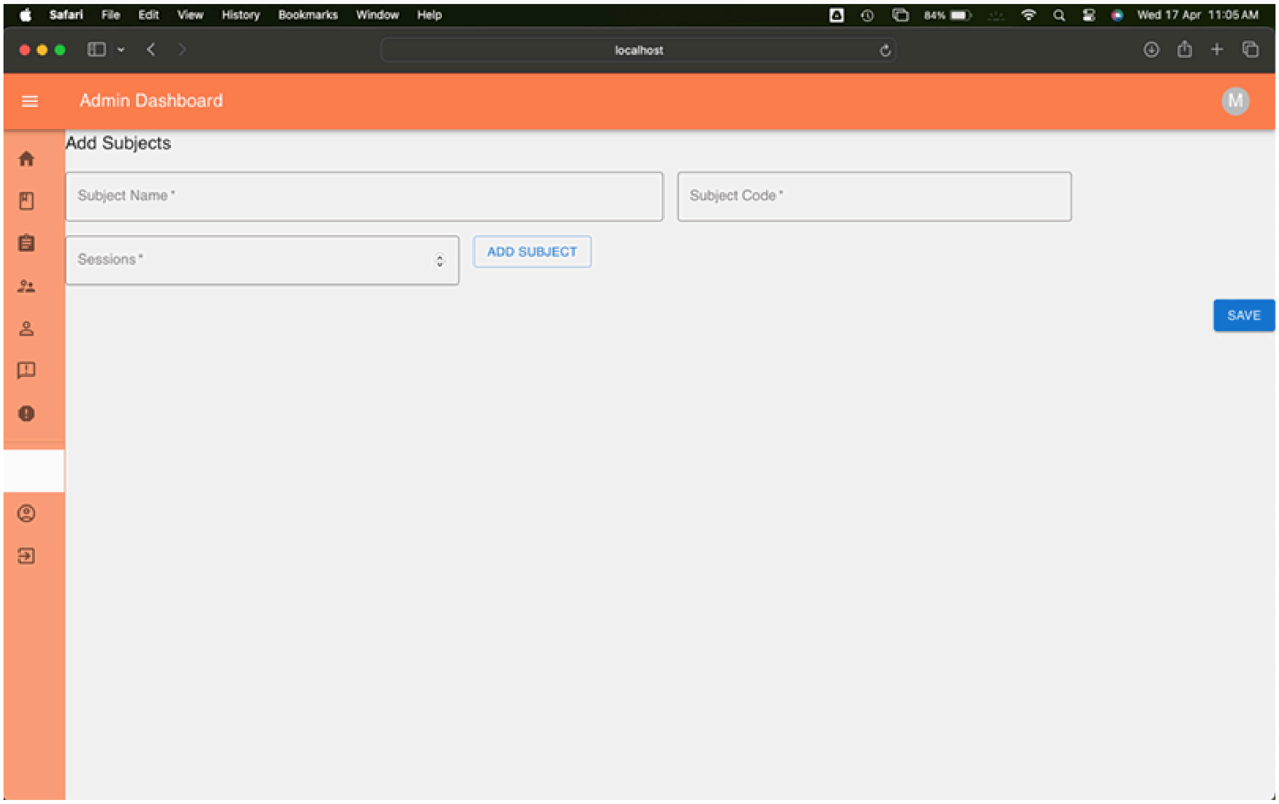
Test case: Check for class creation successful.
Conclusion: Test case passed. Class creation was successful.



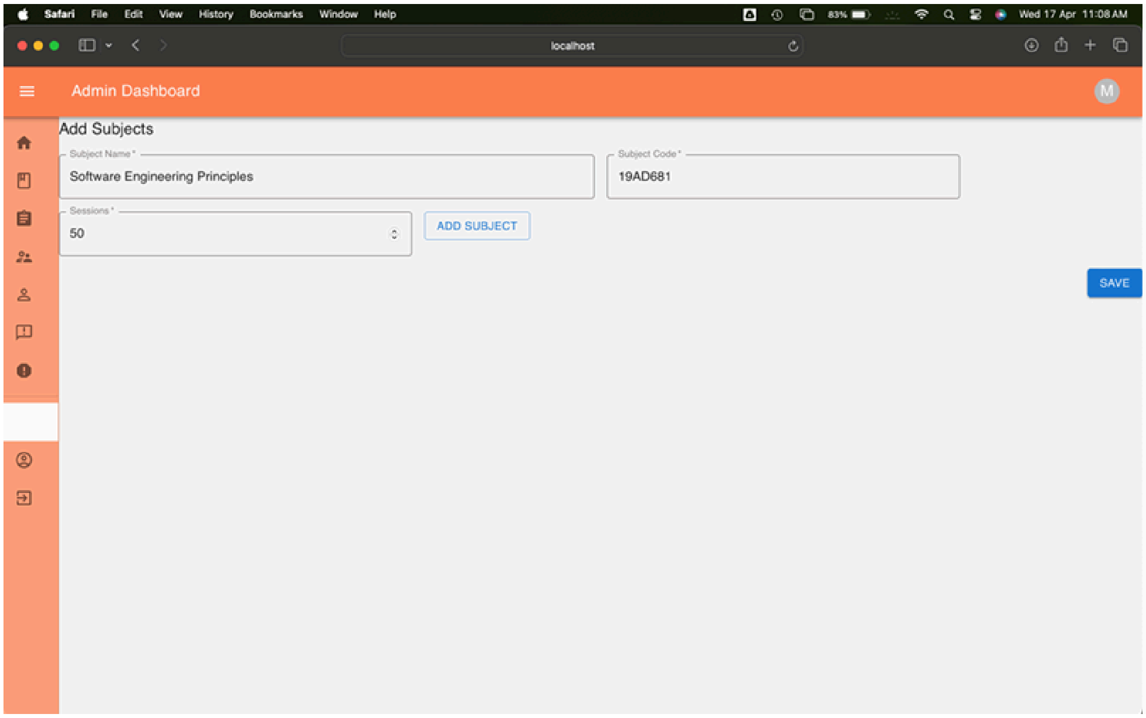


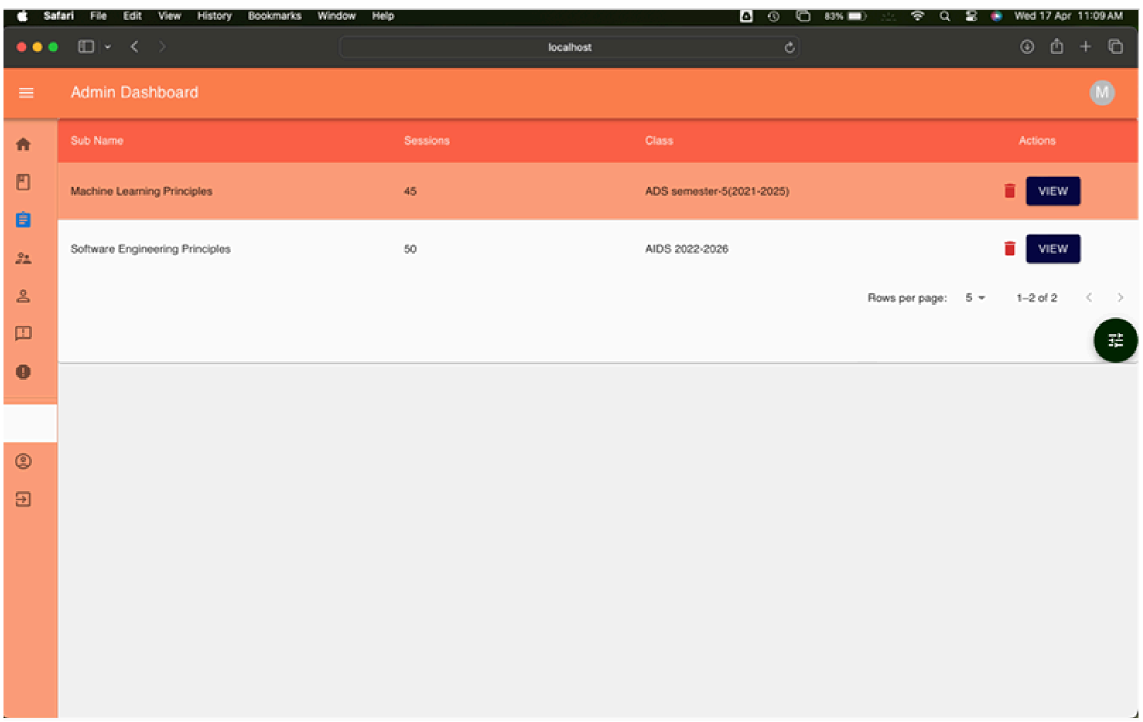
6.1.1.6 Course Creation:

Test case: Check for course creation page loading.
Conclusion: Test case passed. Course creation page loading successful.

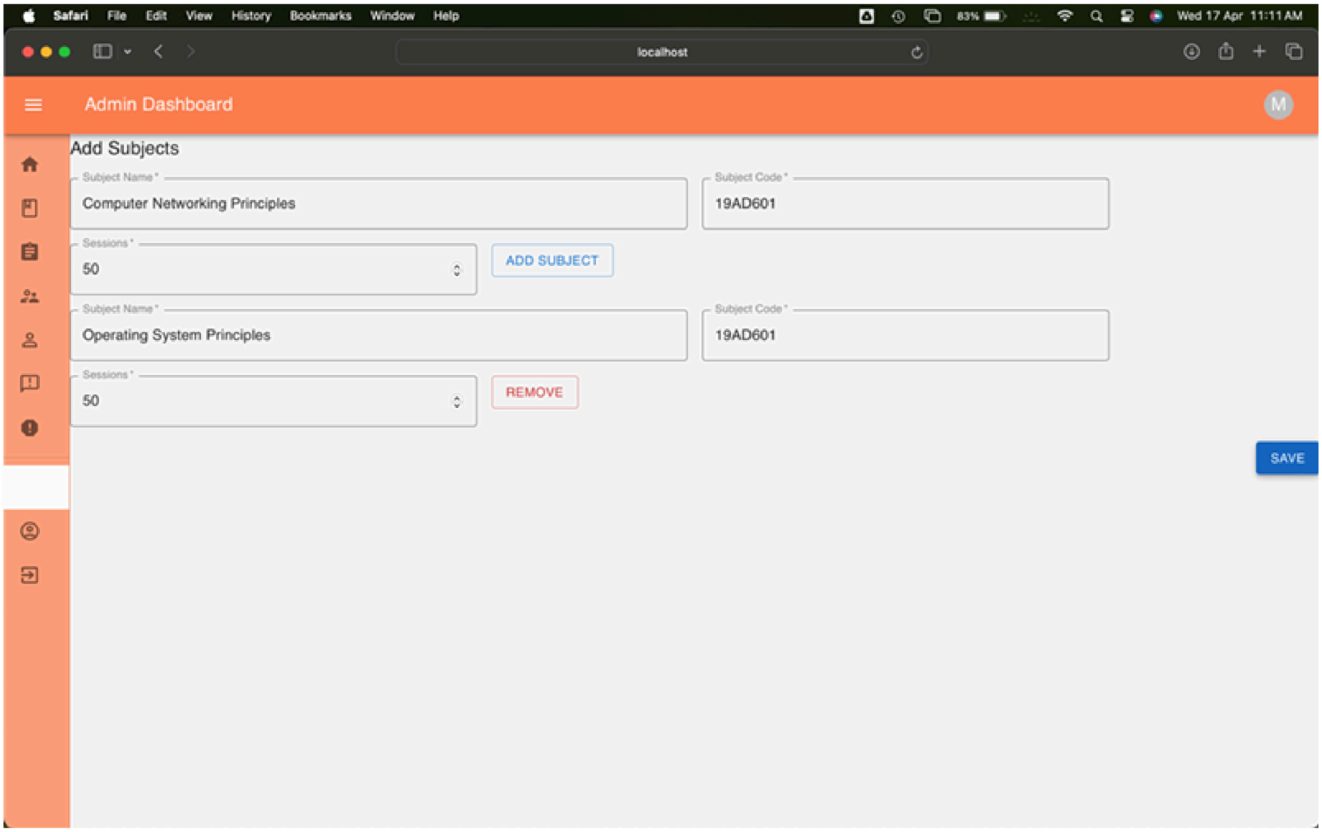


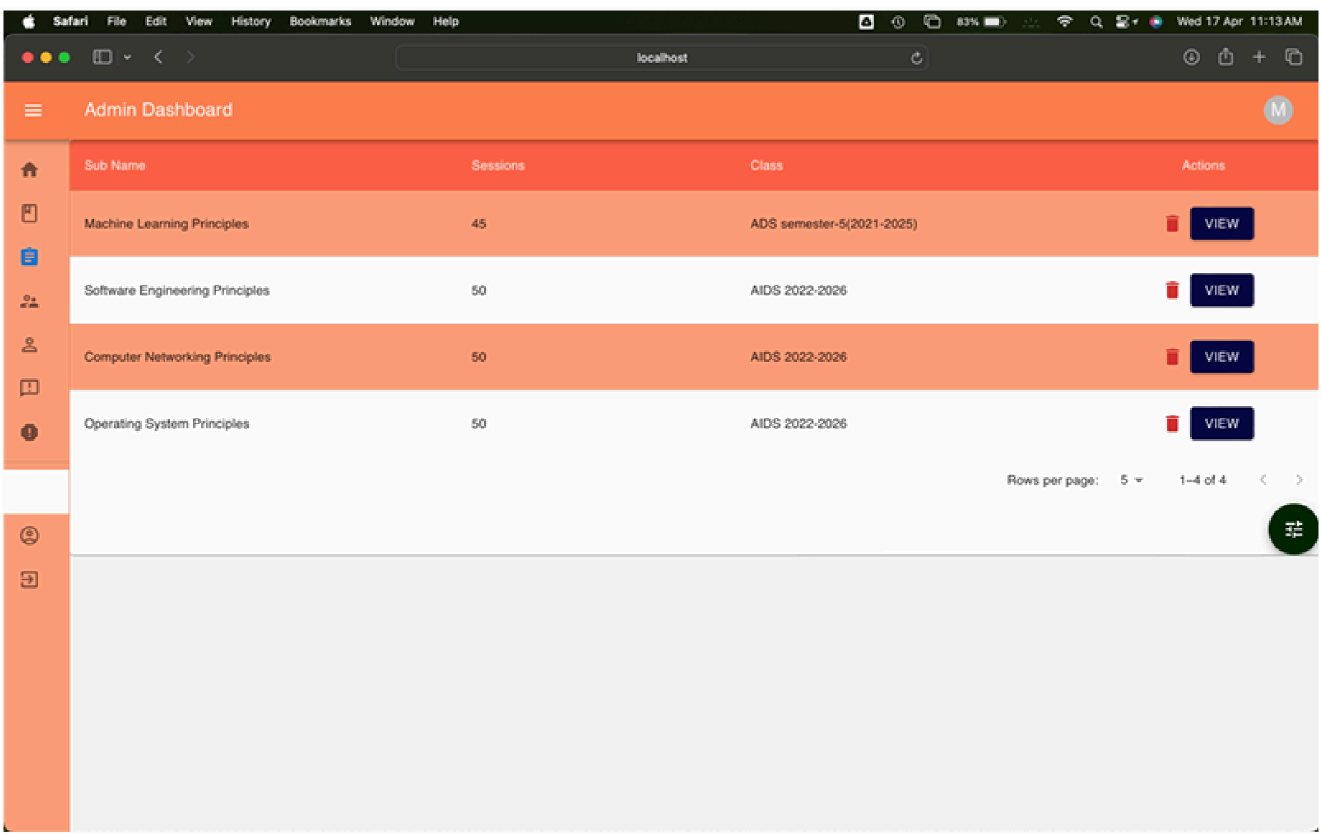
Test case: Check for course creation successfully.
Conclusion: Test case passed. Course creation was successful



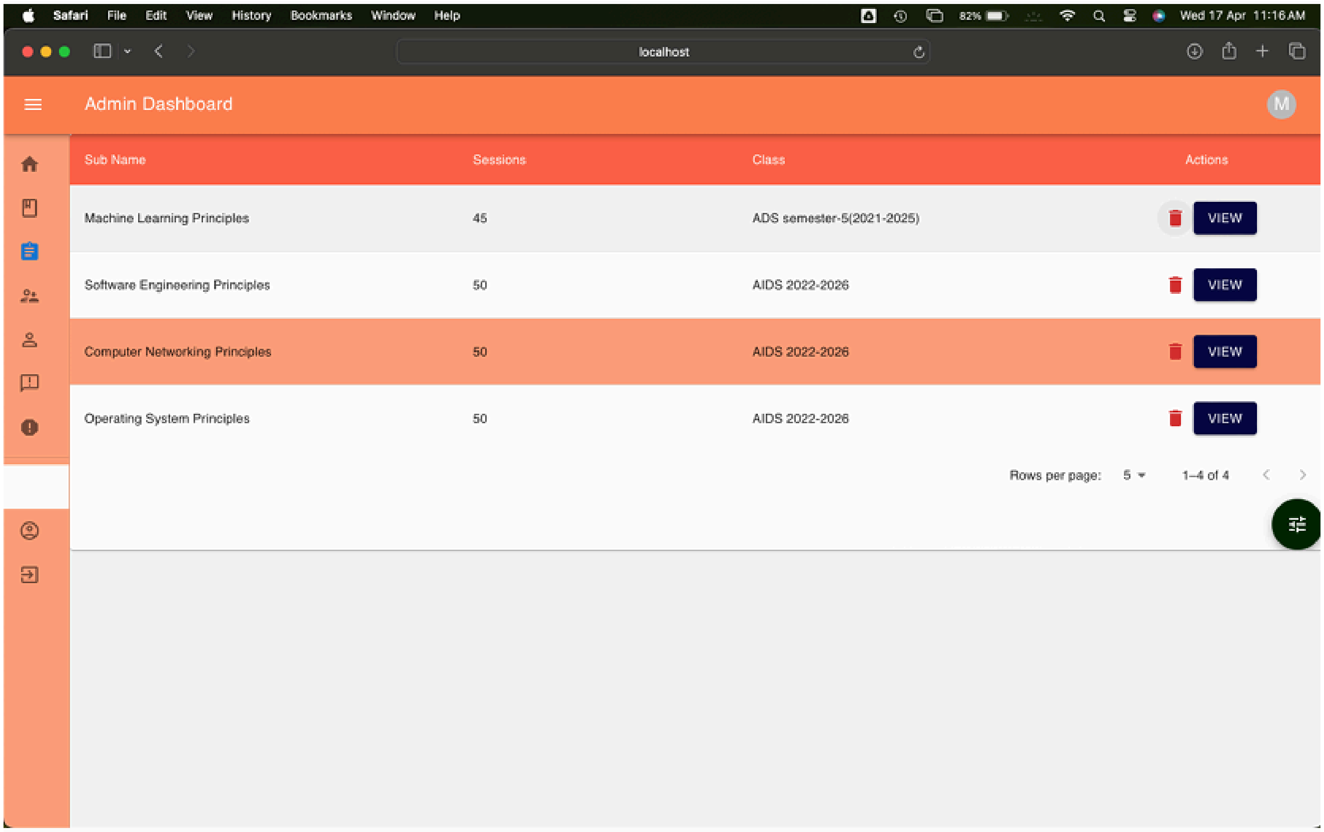


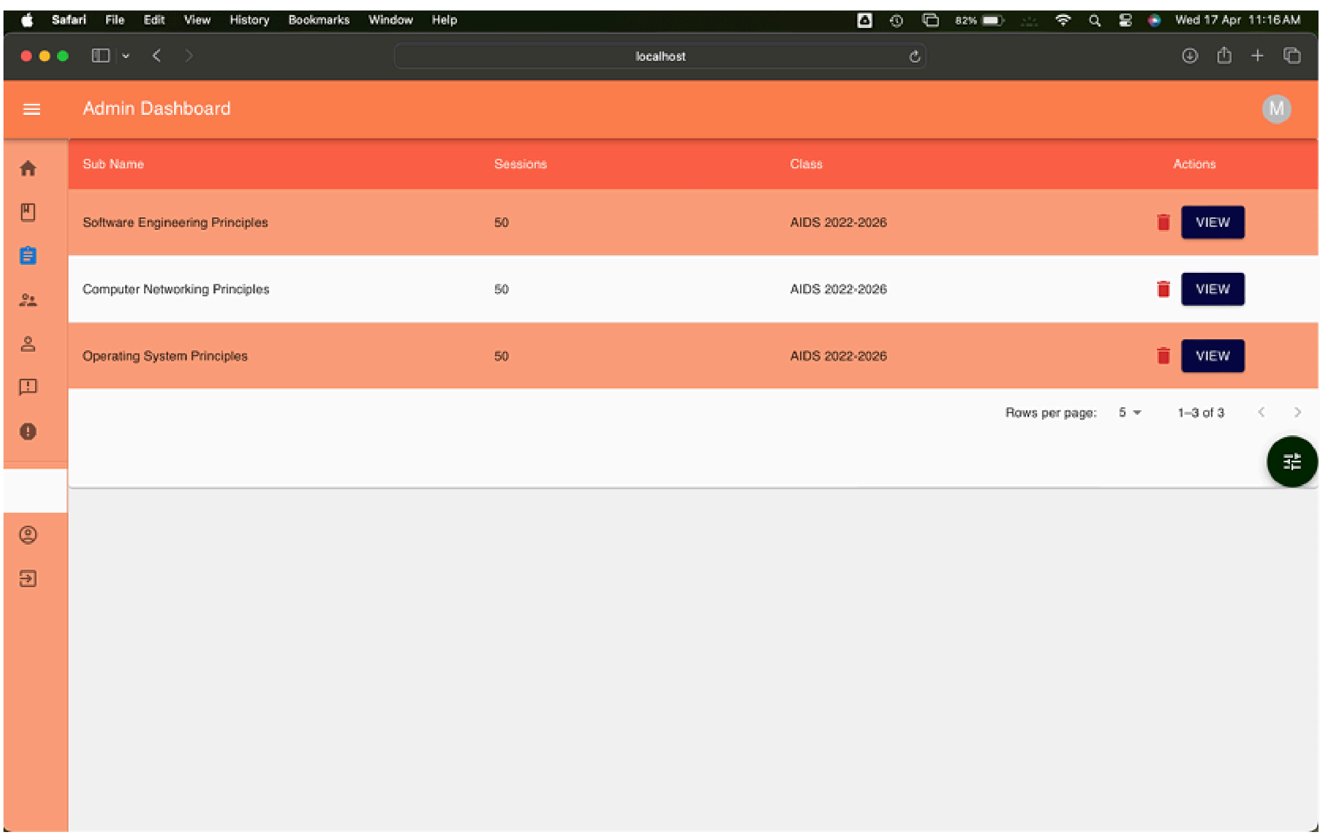
Test case: Check for multiple course creation
Conclusion: Test case passed. Multiple course creation successful.





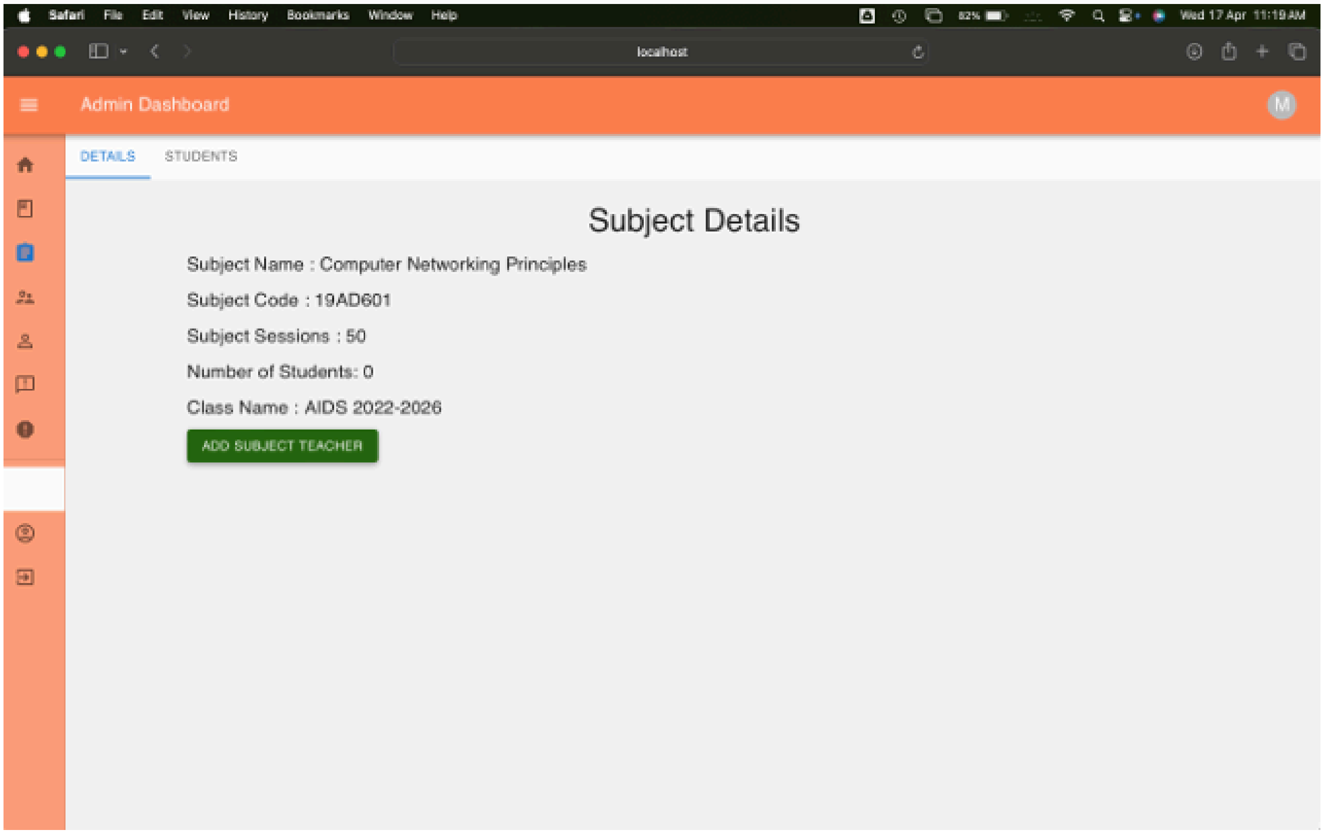
Test case: Check for deletion of a course.
Conclusion: Test case passed. Course deletion successful.

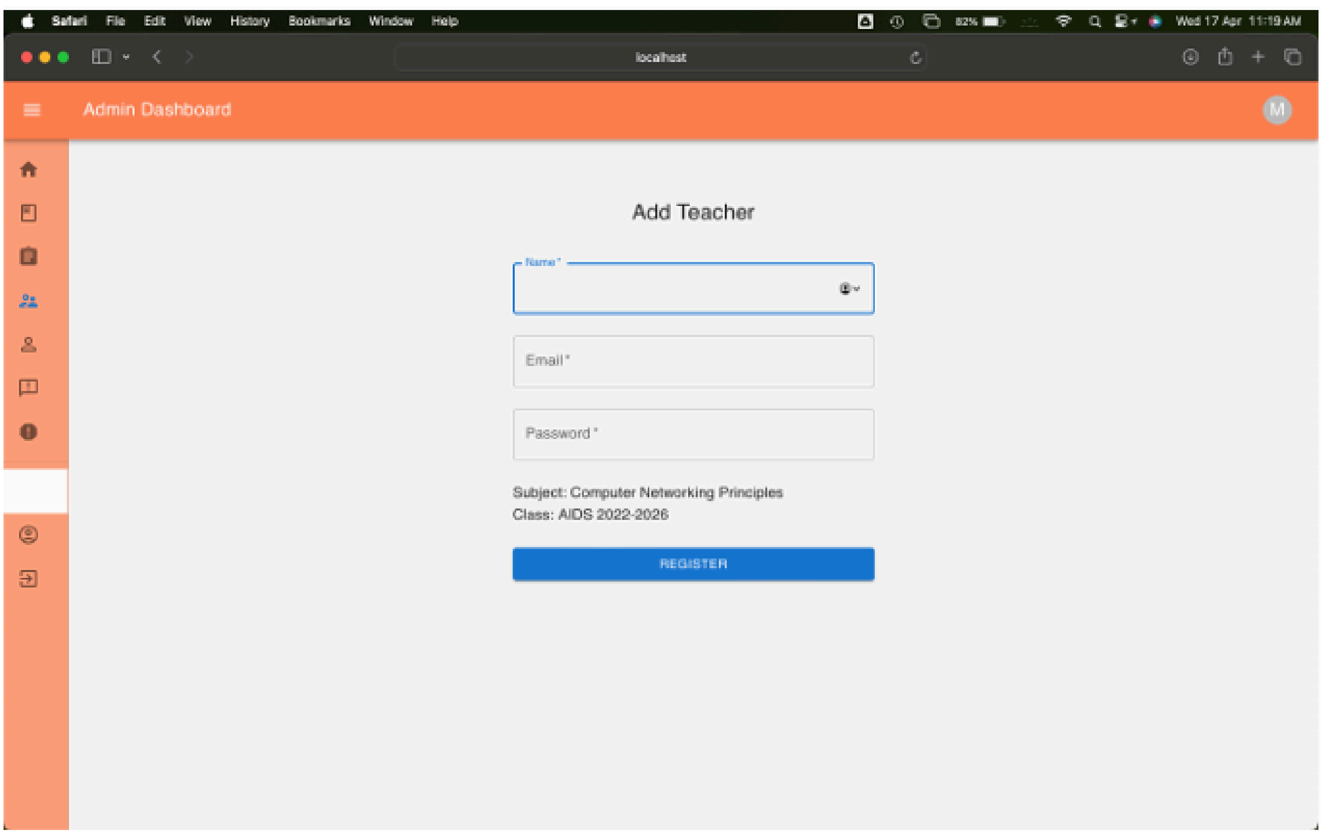




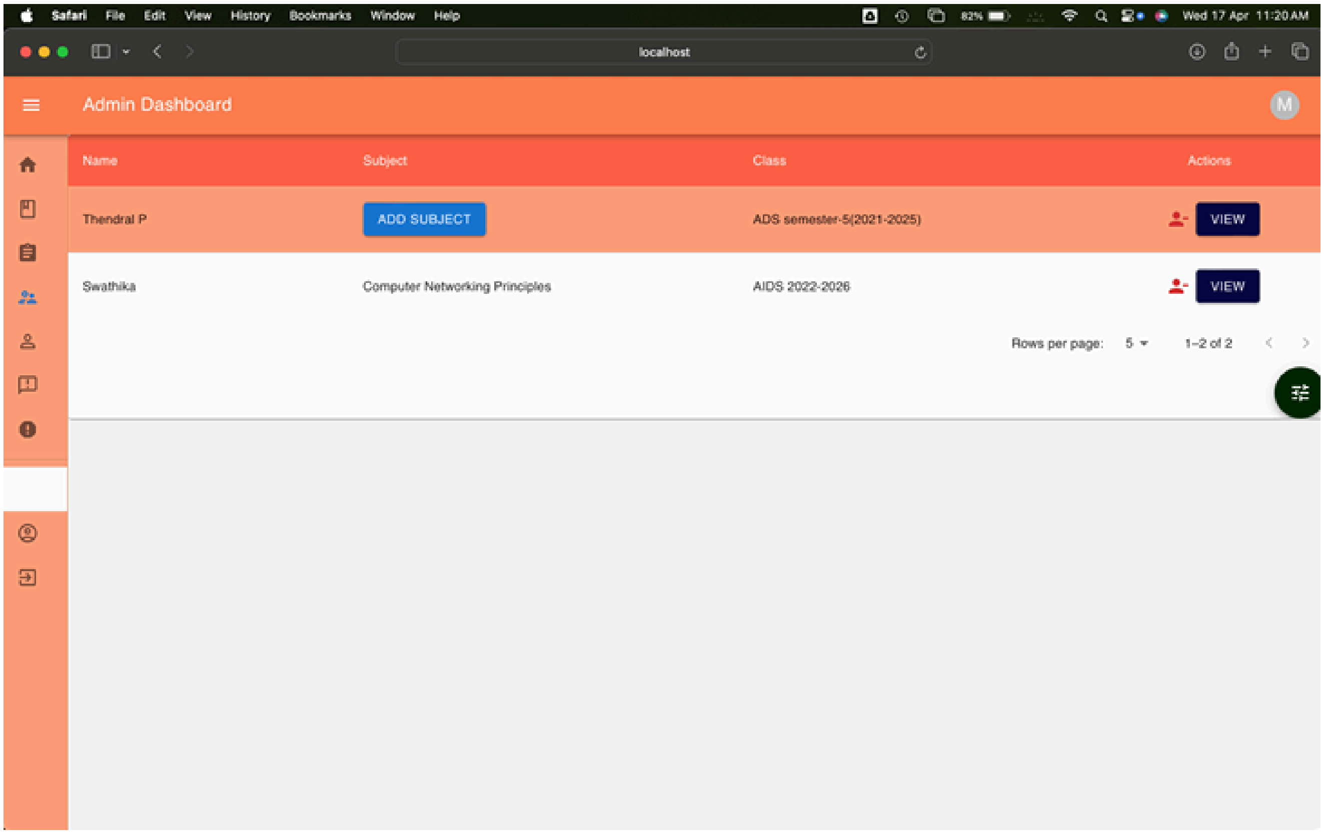
6.1.1.7 Teacher Login ID Creation:

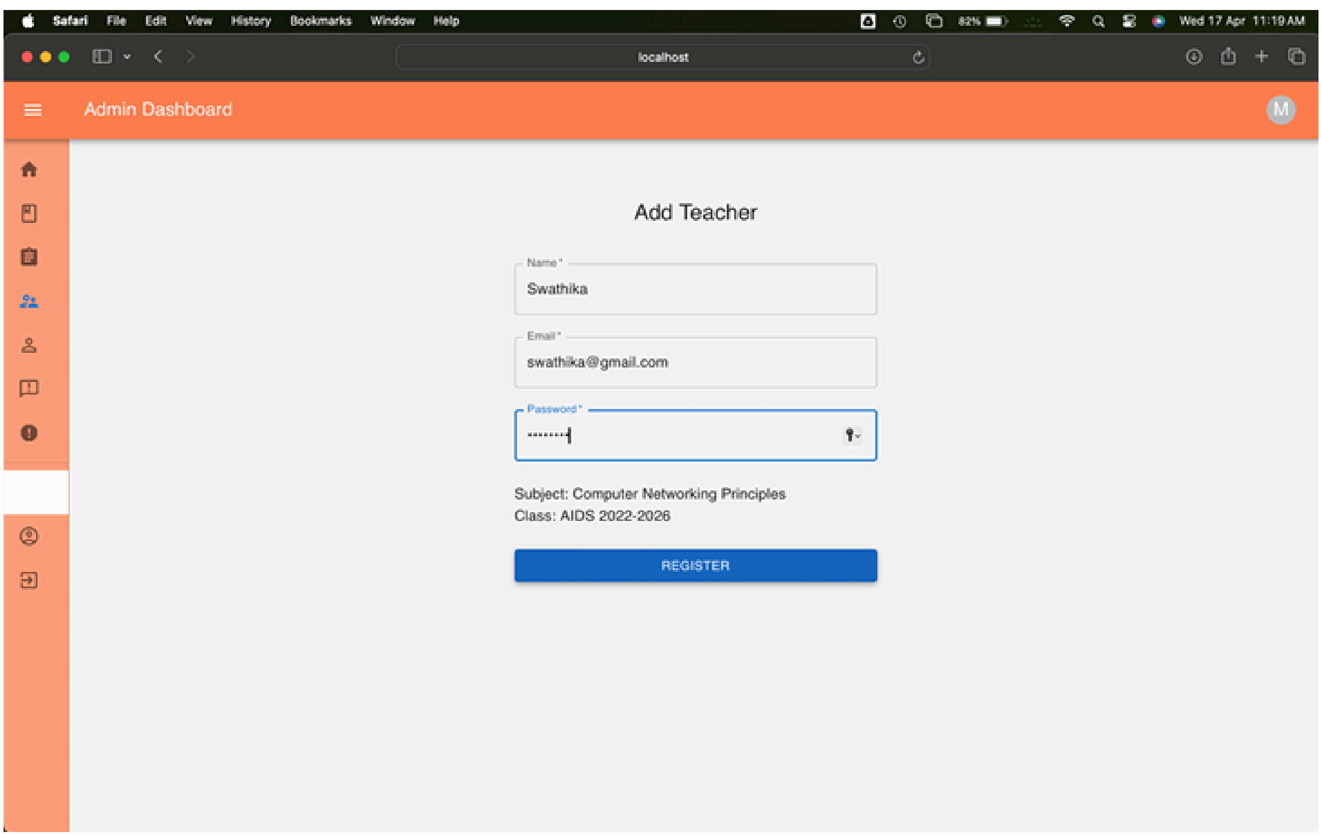
Test case: Check for Teacher login id creation successful.
Conclusion: Test case passed. Teacher Id creation successful.

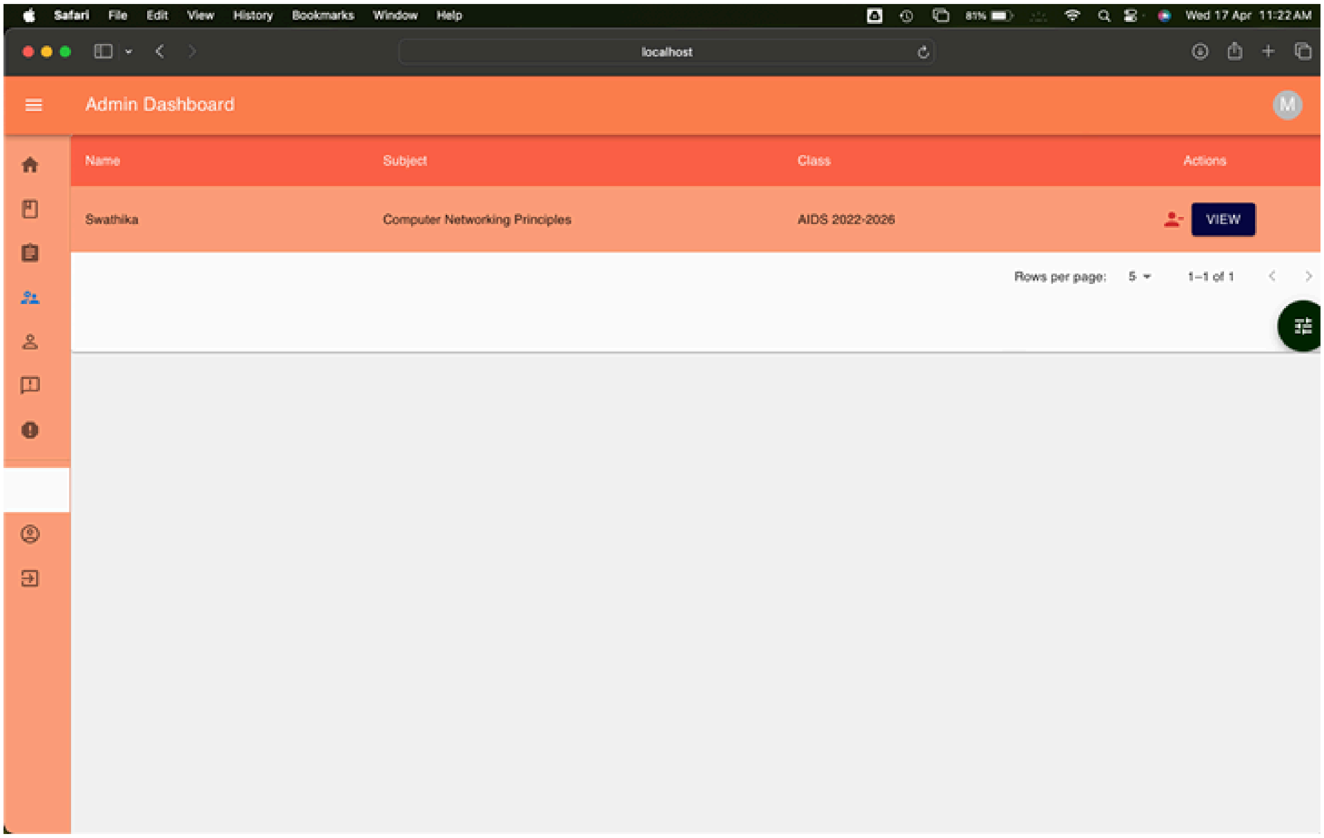




Test case: Check for Teacher Login Id deletion.
Conclusion: Test case passed. Teacher Login id deleted successfully.






Safari File Edit View History Bookmarks Window Help
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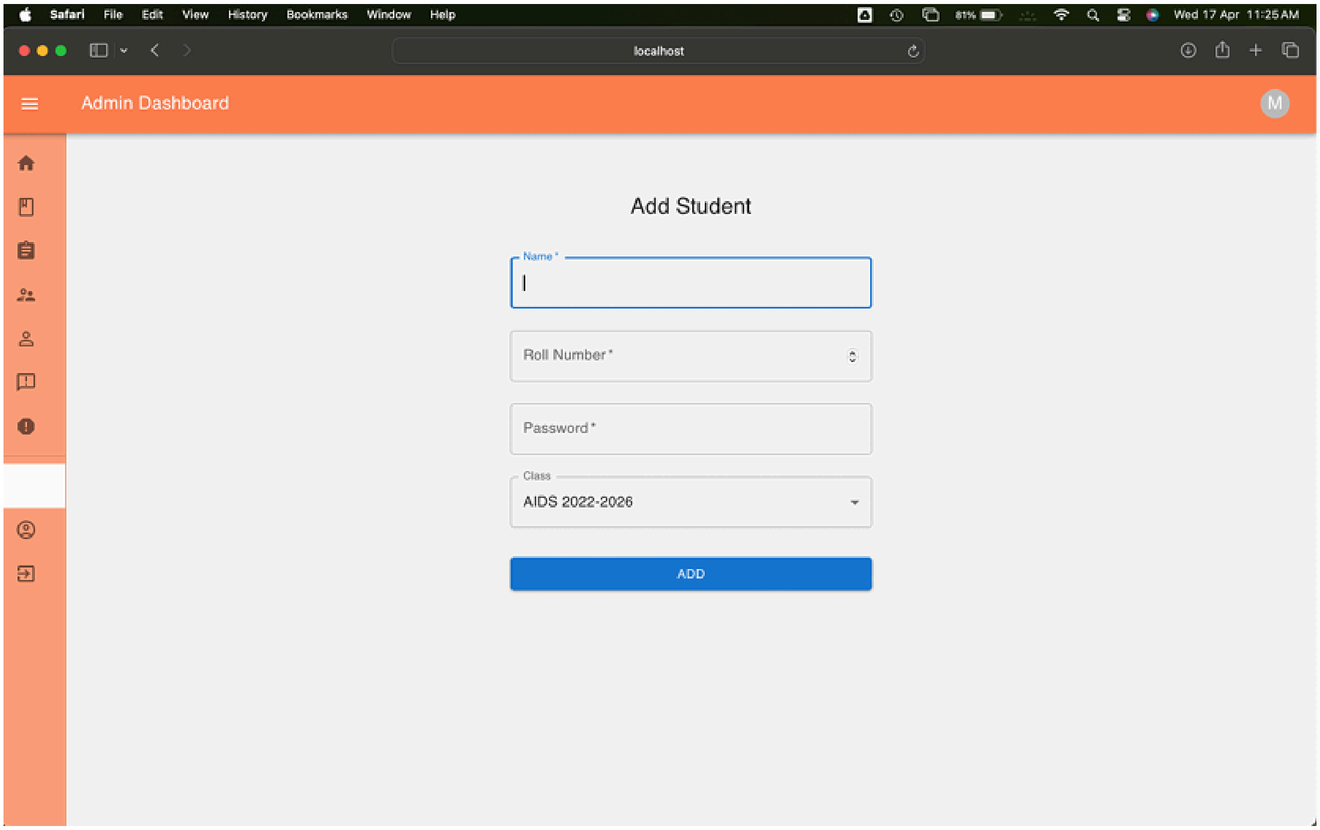
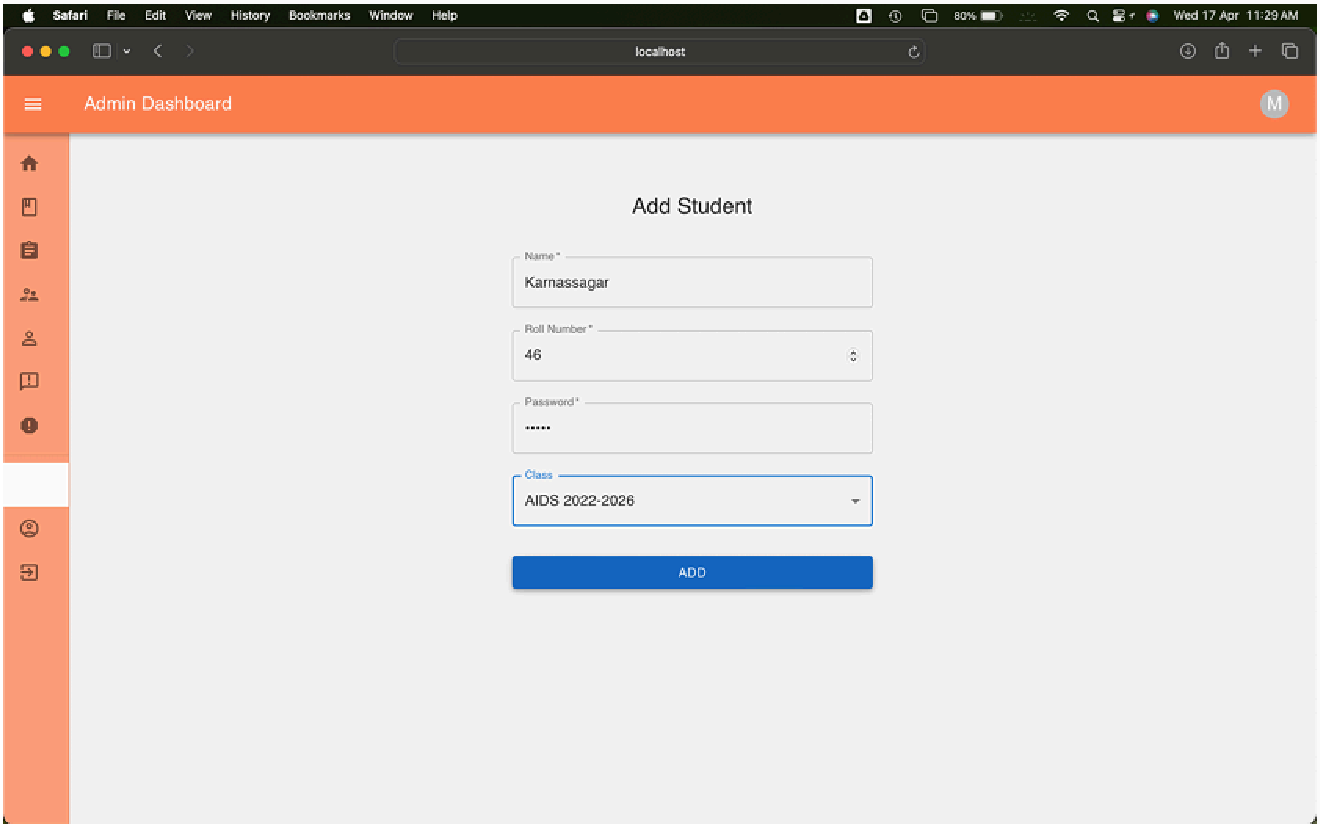
localhost
Admin Dashboard
Name
Subject
Class
Actions
Swathika
Computer Networking Principles
AIDS 2022-2026
VIEW
Rows per page: 5
1–1 of 1

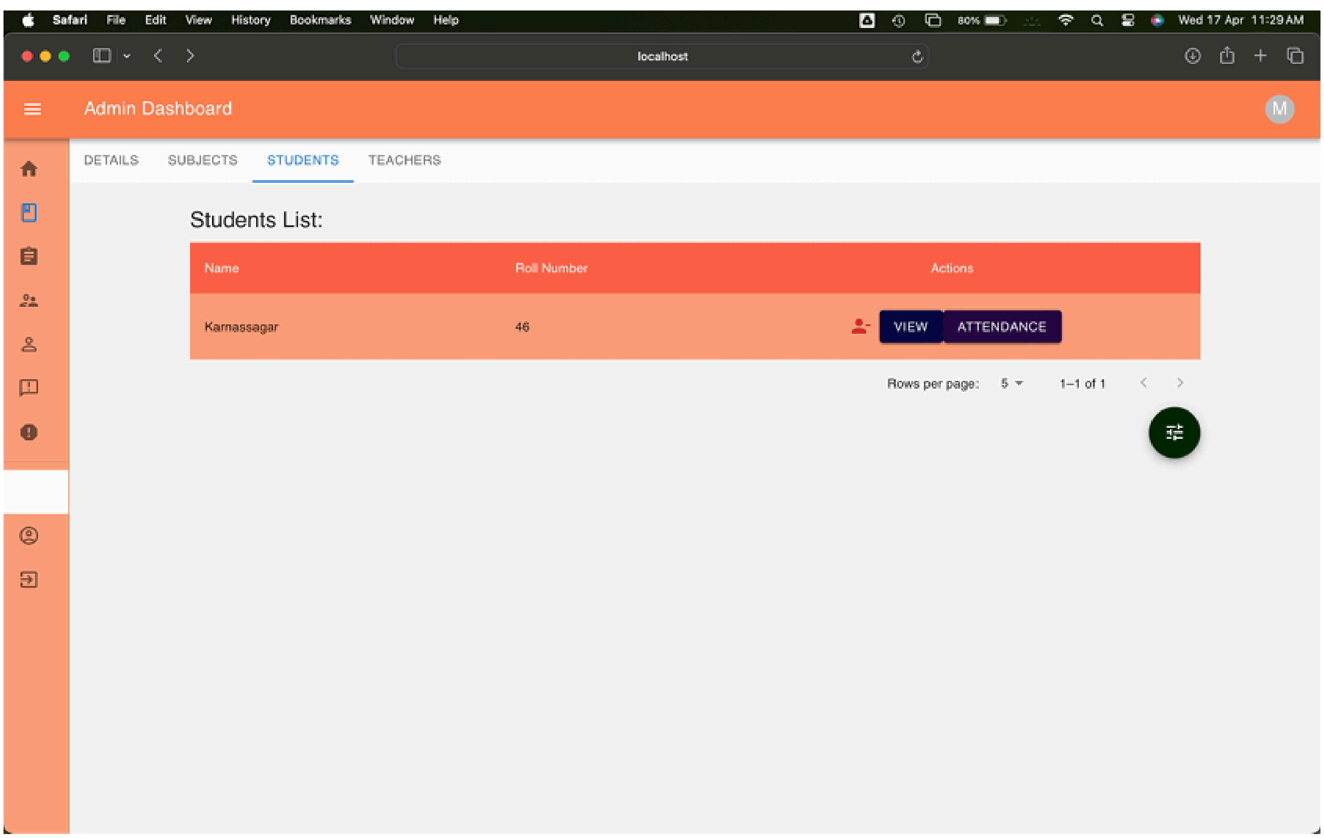
6.1.1.8 Student Login Id Creation:

Test case: Check for student login id creation page loading.
Conclusion: Test case passed. Student login id creation page loading successful.

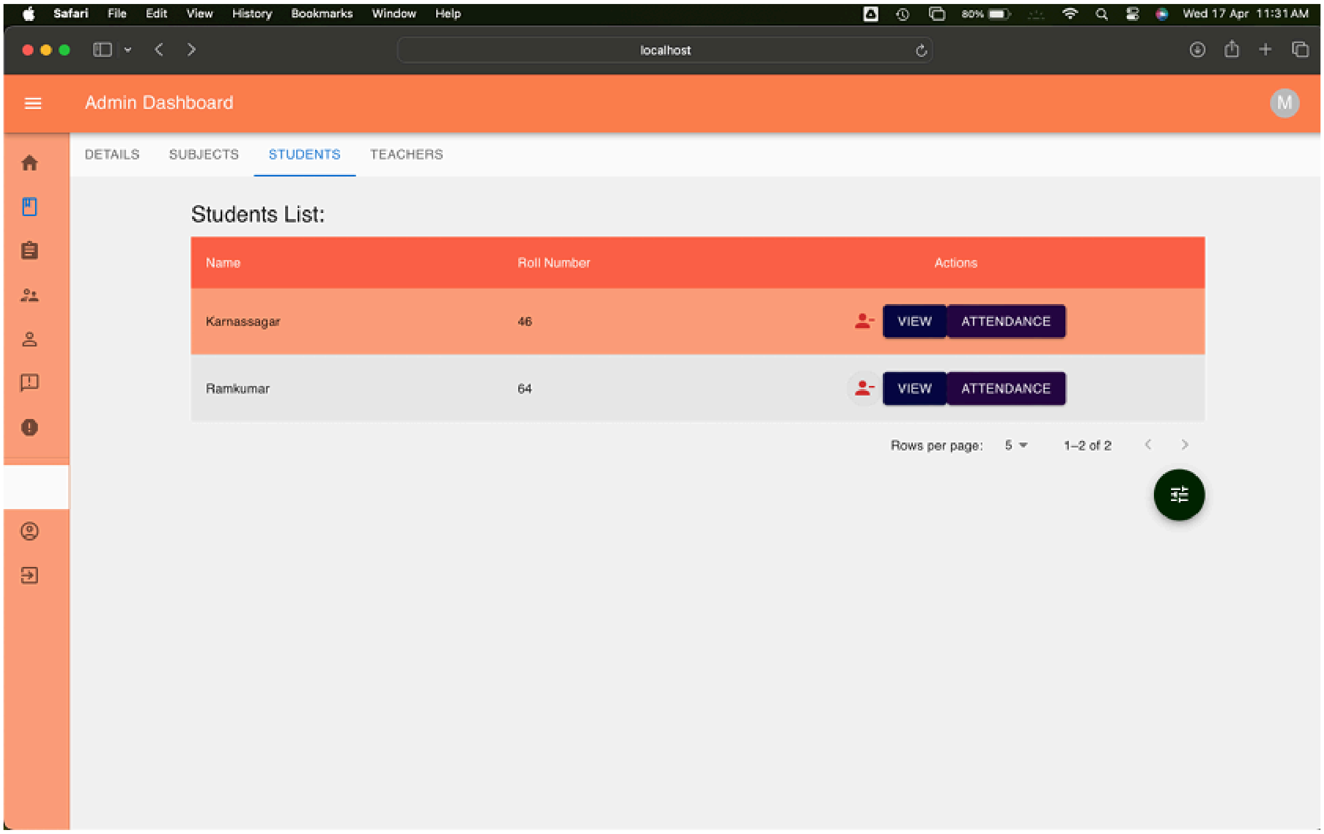


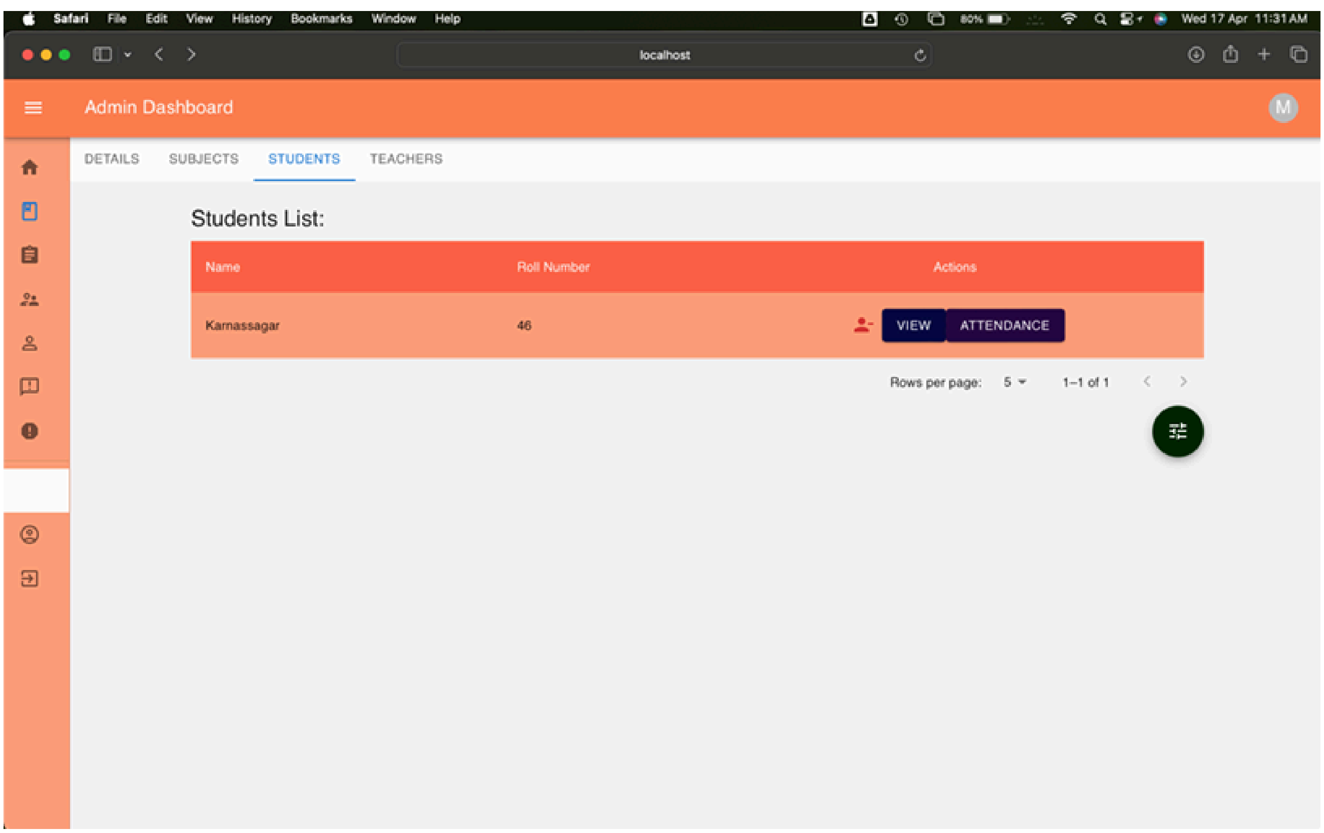
Test case: Check for Student login id creation.
Conclusion: Test case passed. Student login id created successfully.



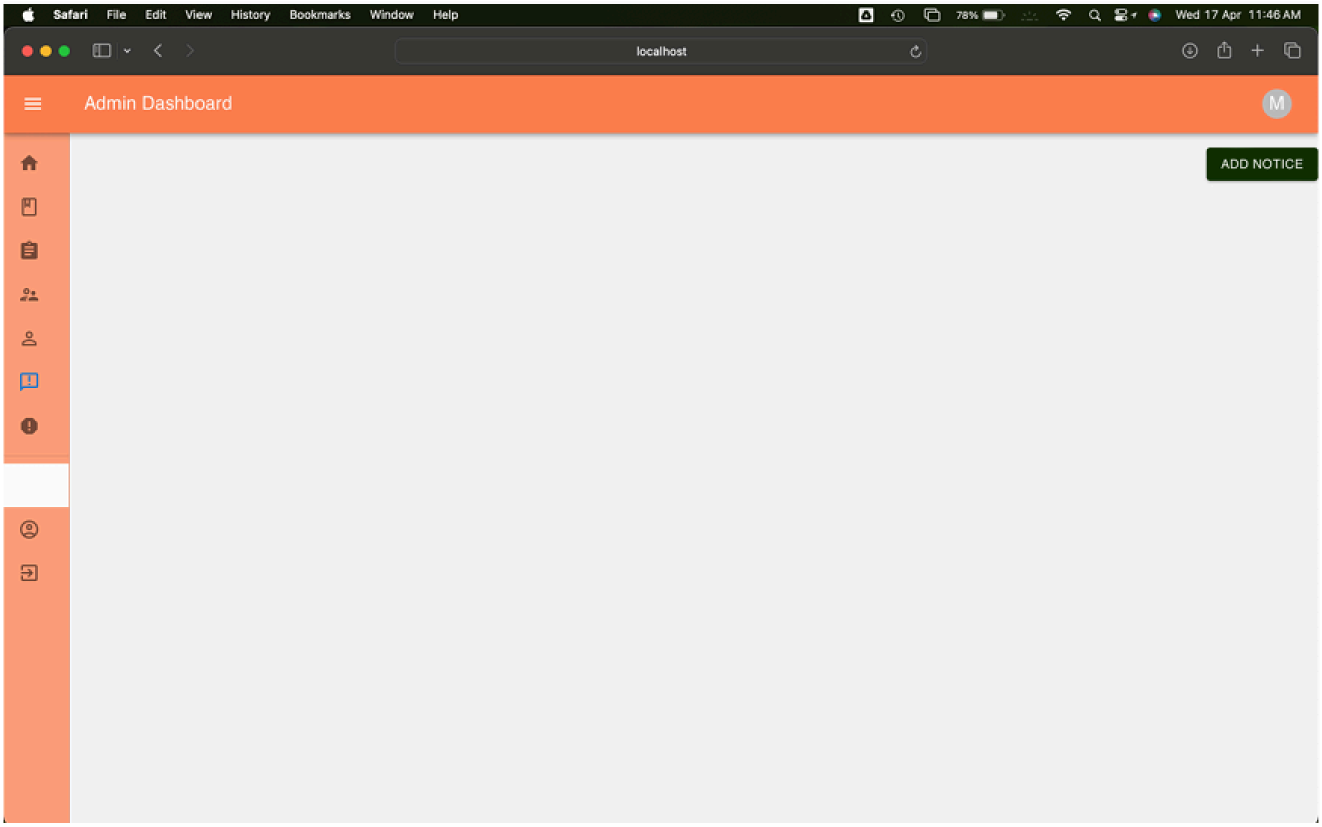


Test case: Check for student login id deletion.
Conclusion: Test case passed. Student login id deletion successful.

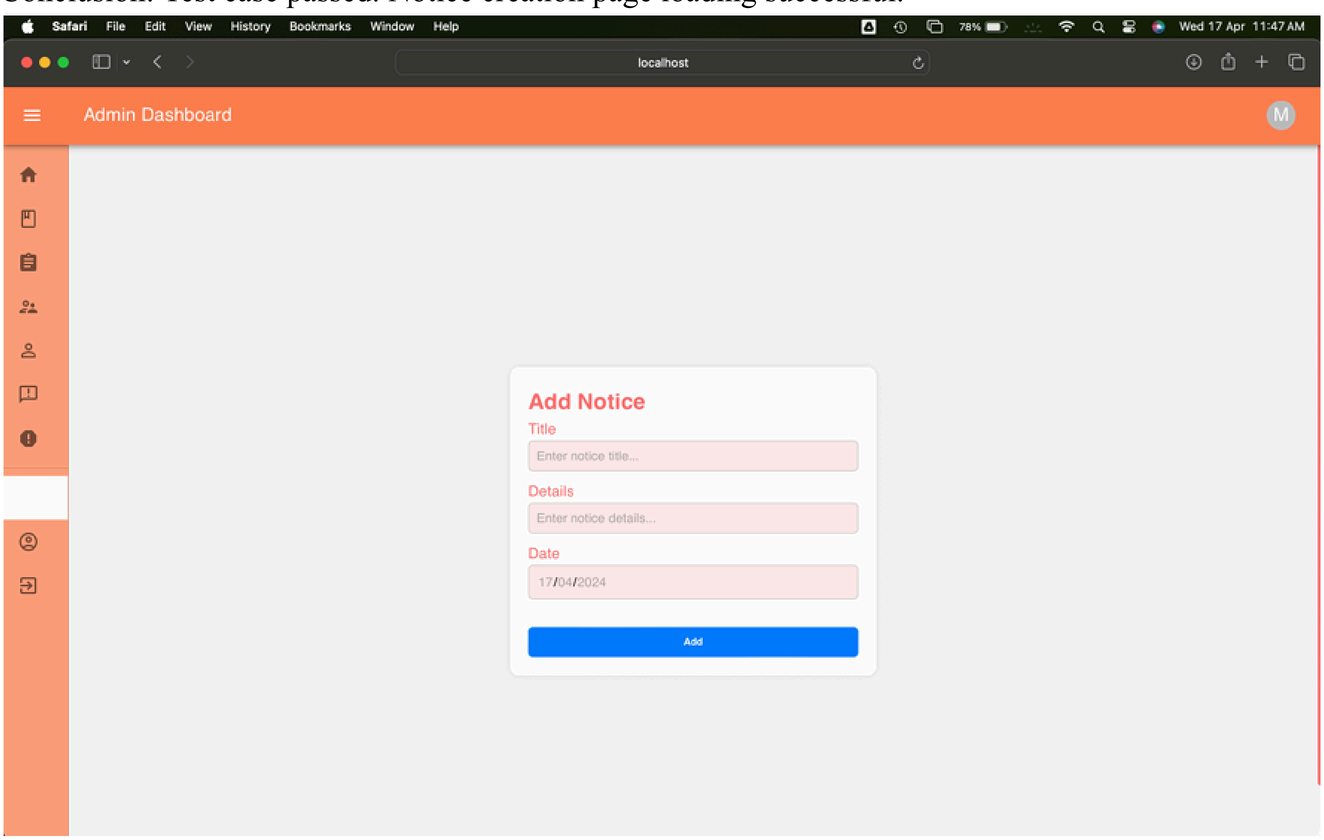




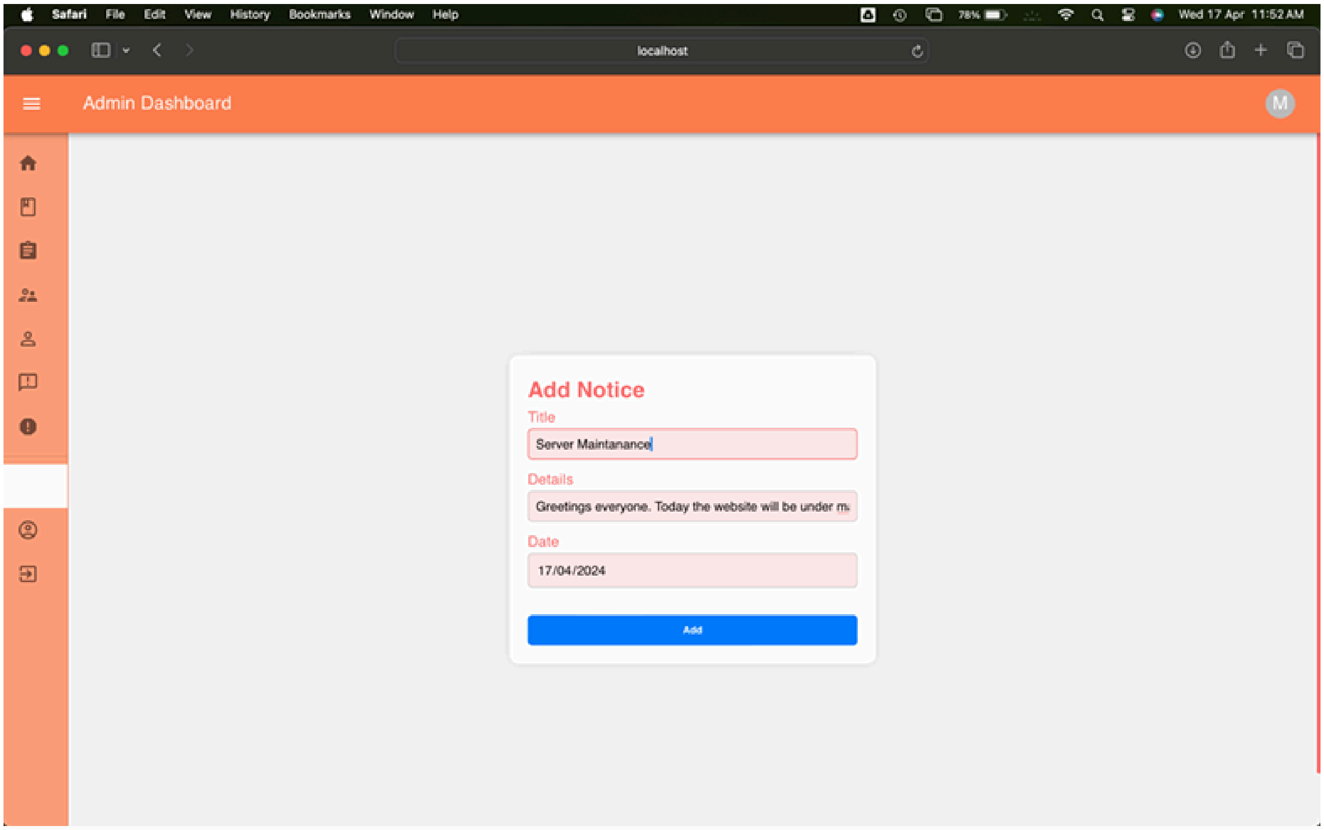
Test case: Check for Notice page loading.
Conclusion: Test case passed. Notice page loading is successful.

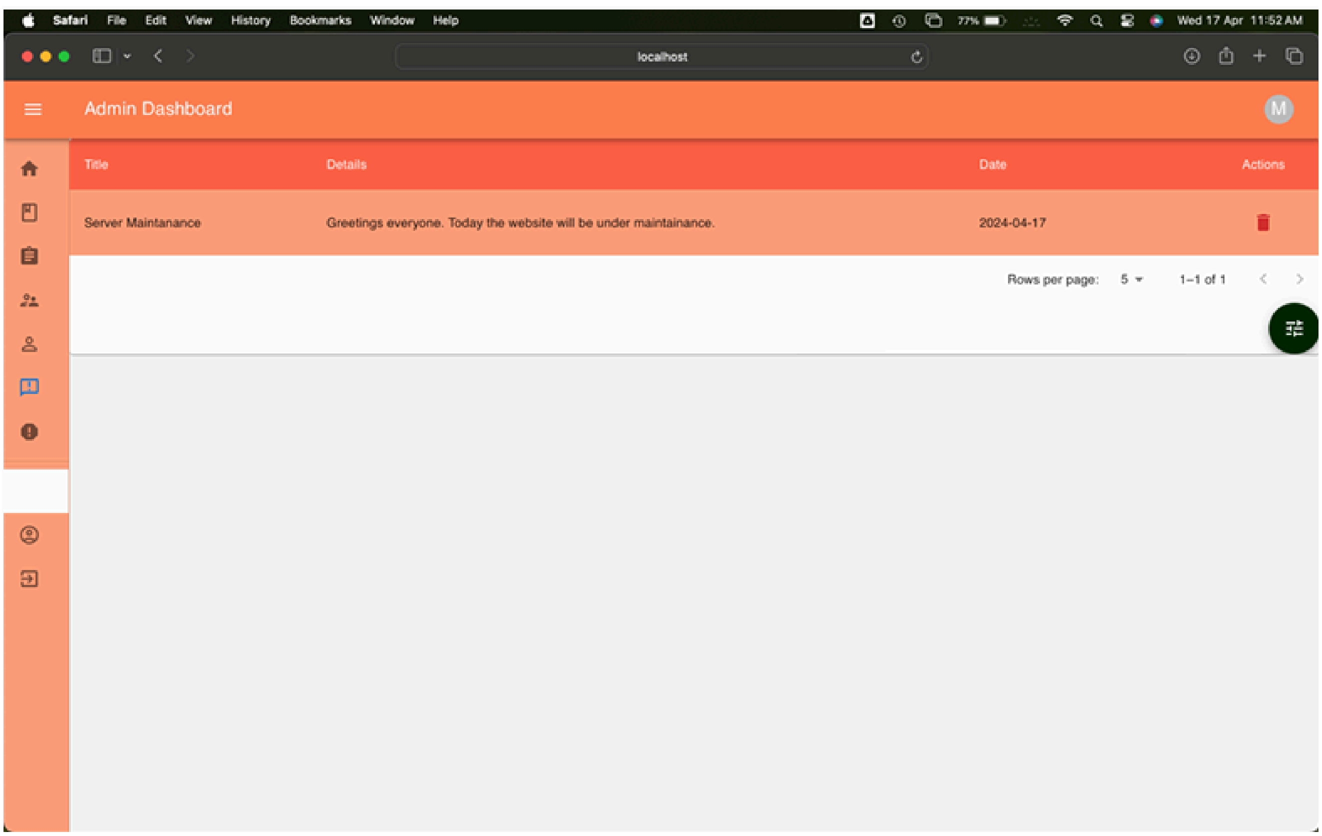


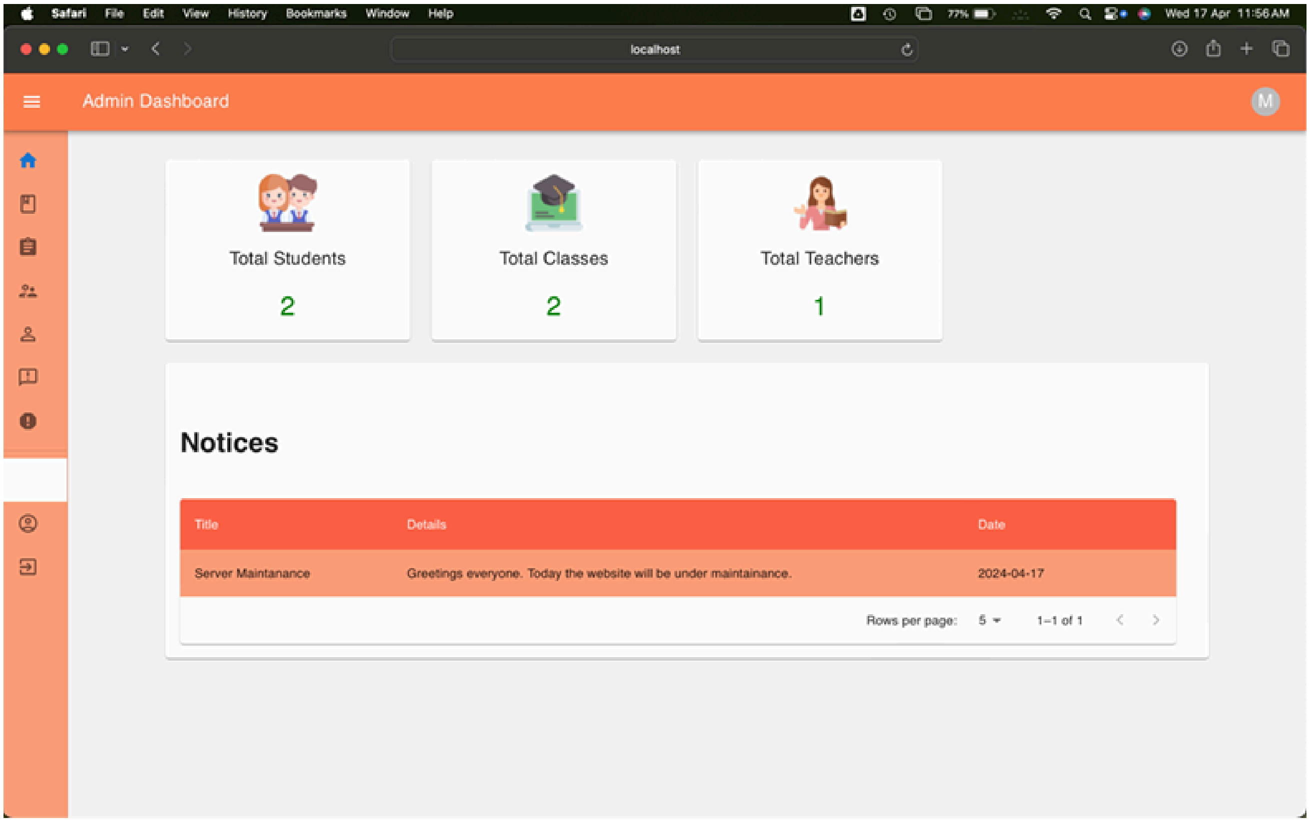
Test case: Check for Notice creation page loading.
Conclusion: Test case passed. Notice creation page loading successful.



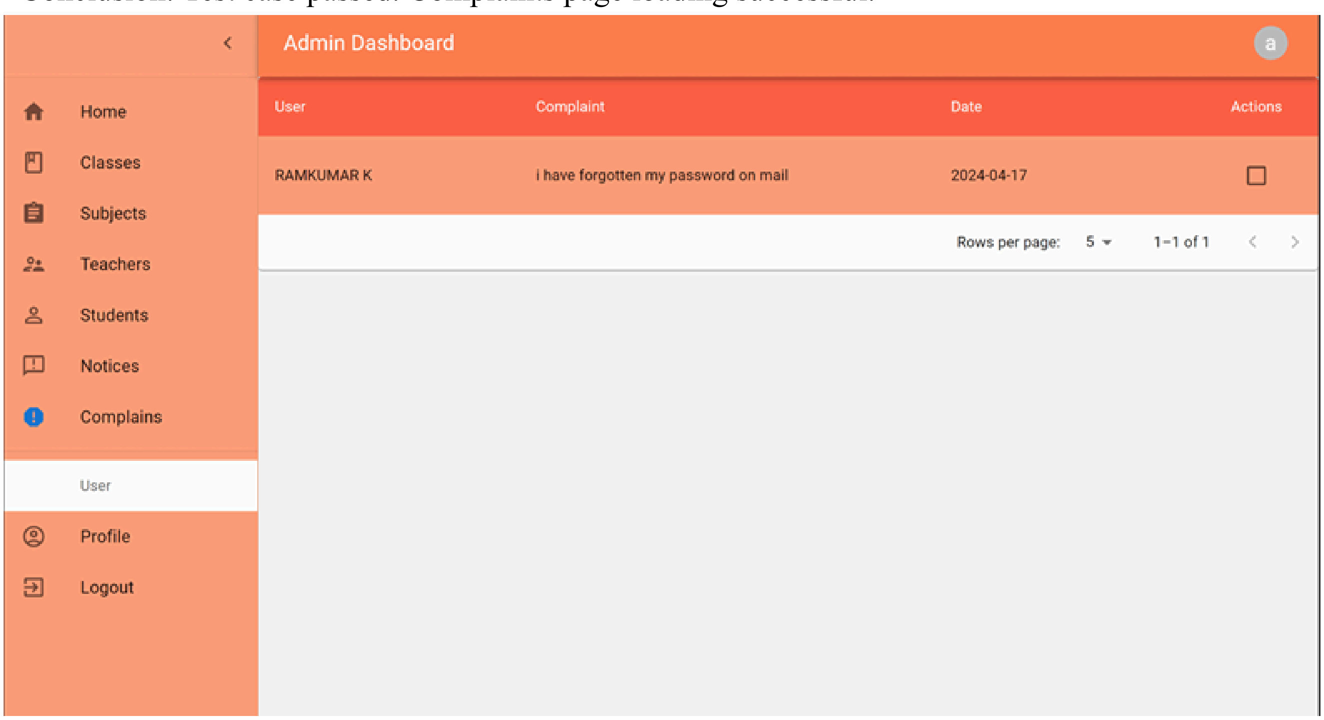
Test case: Check for creation of notice.
Conclusion: Test case passed. Notice creation is successful.





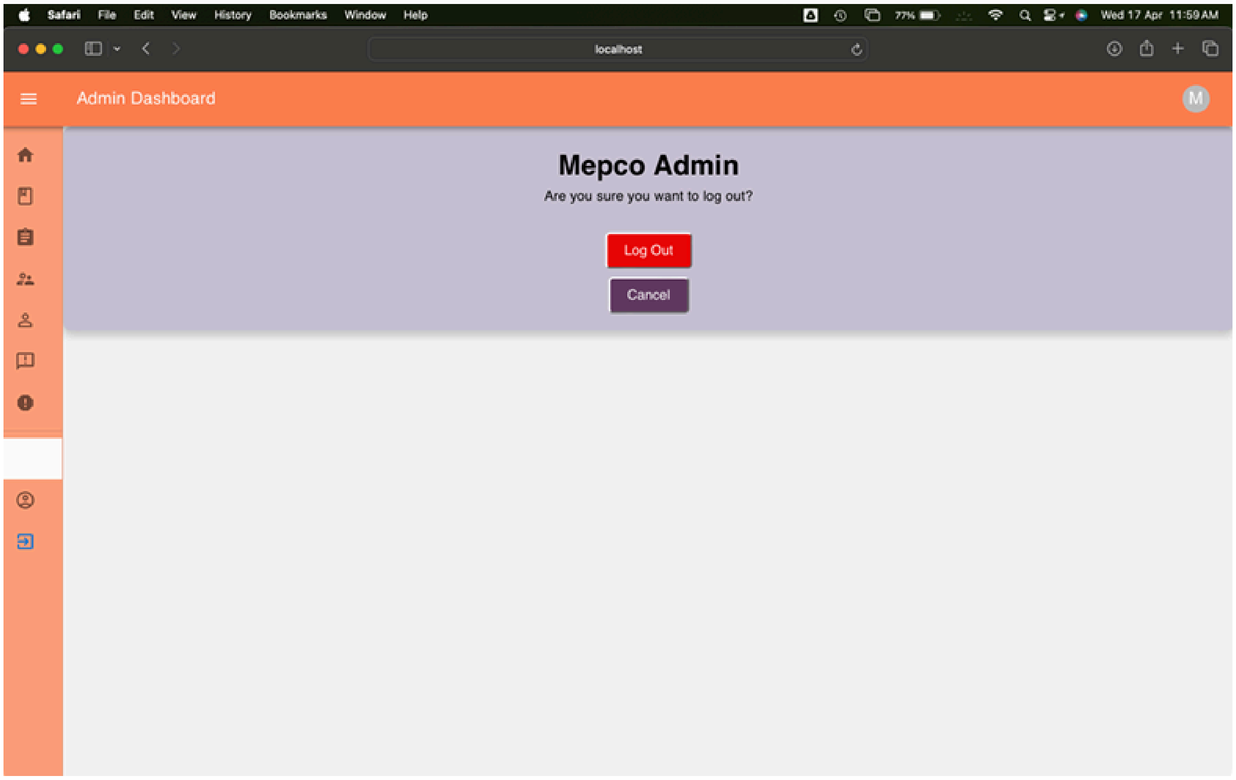


Test case: Check for Complain page loading.
Conclusion: Test case passed. Complaints page loading successful.

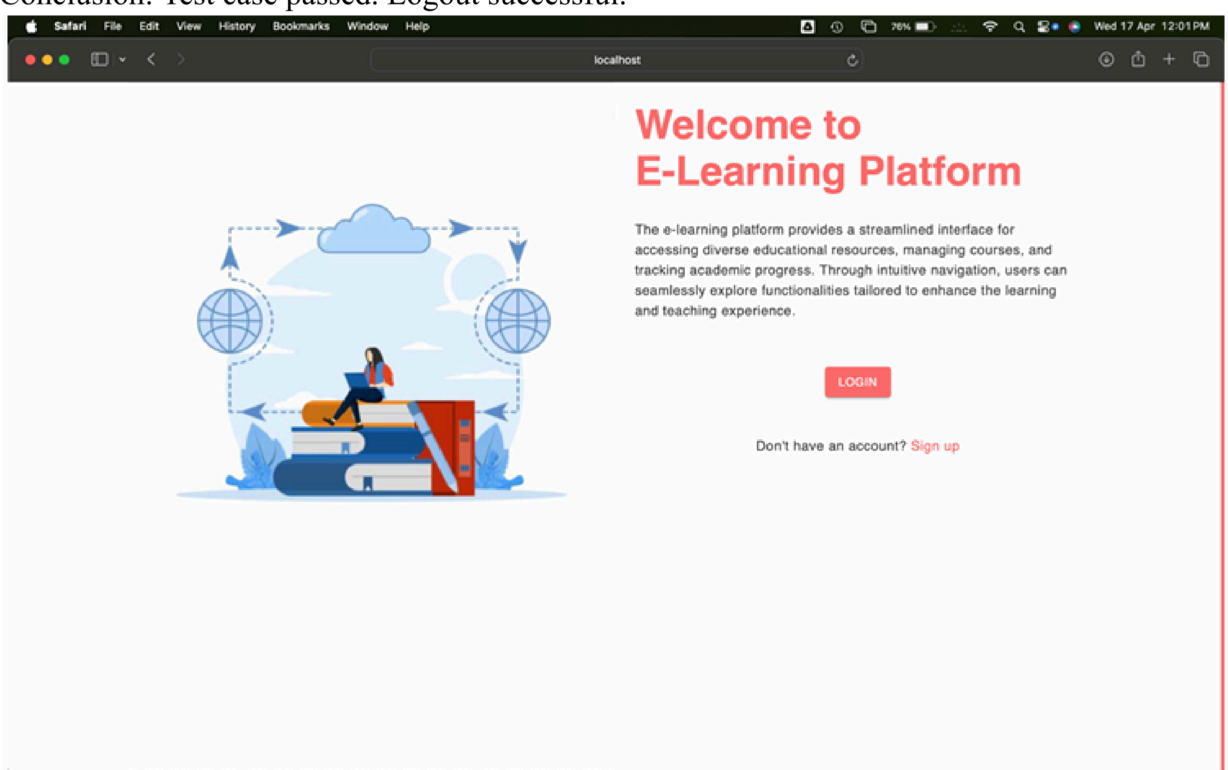


6.1.1.9 Logout:

Test case: Check for Logout page loading.
Conclusion: Test case passed. Logout page loading successful.



Test case: Check for successful logout.
Conclusion: Test case passed. Logout successful.



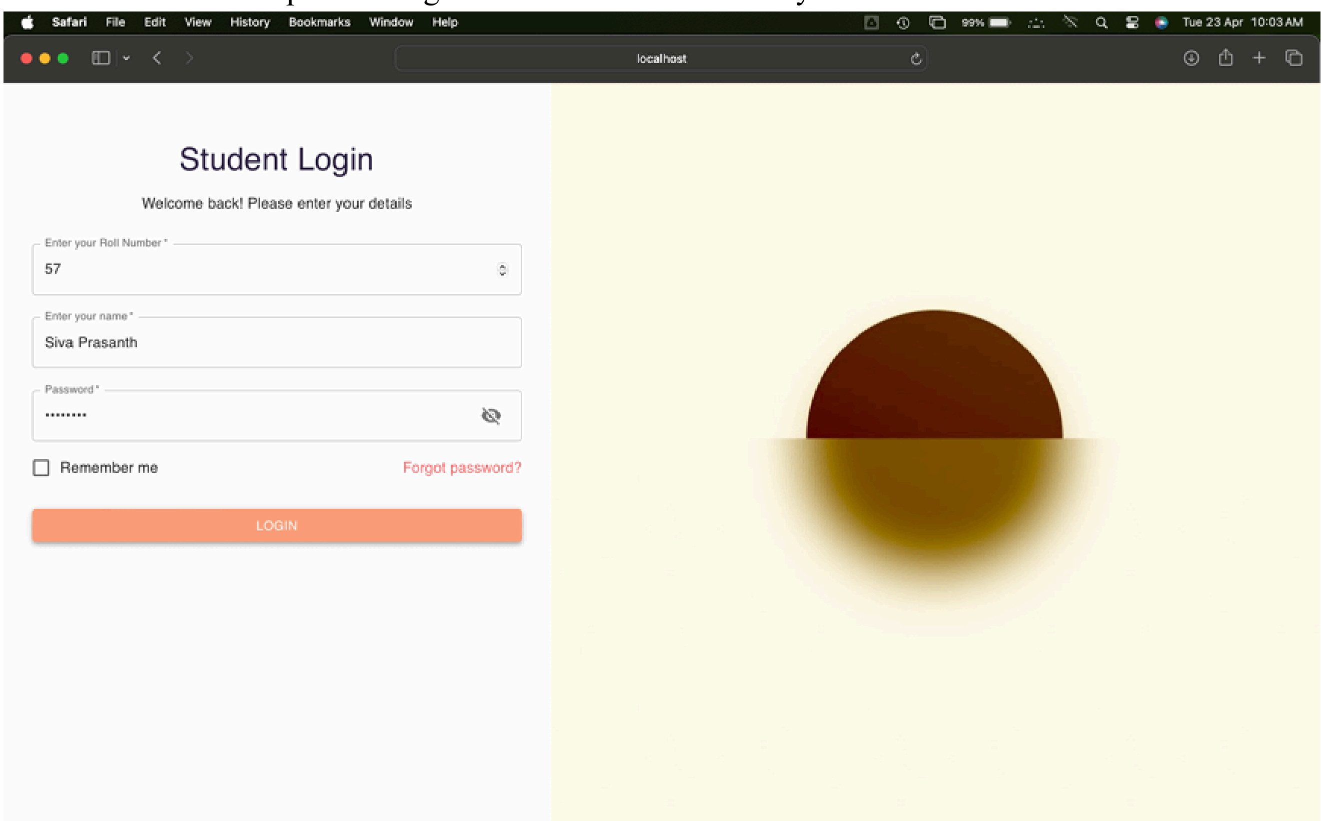
6.2 Testing of Student Module:

List of Modules:
• Login
• Dashboard
• Subject
• Attendance
• Complain
• Profile
• Logout

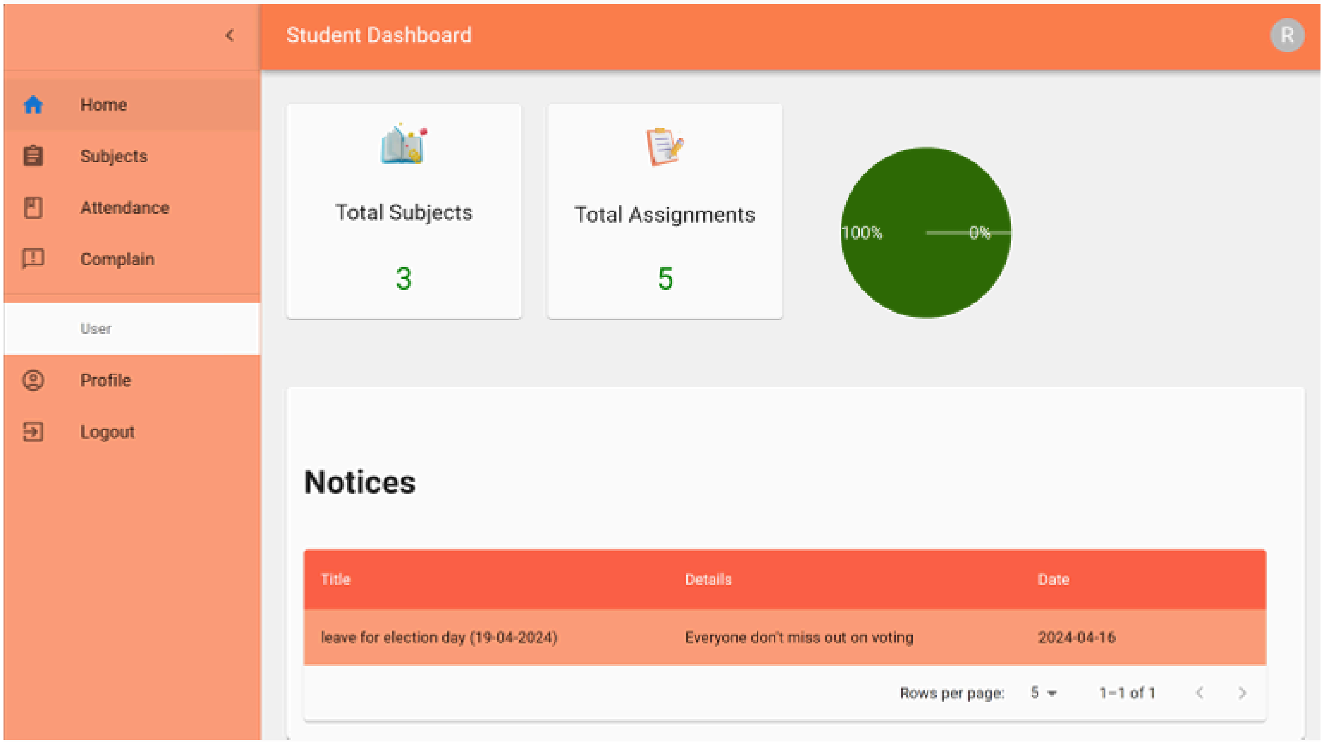
6.2.1 Unit Testing

6.2.1.1 Login

Test case: Check for Login screen loading.
Conclusion: Test case passed. Login screen loaded successfully.

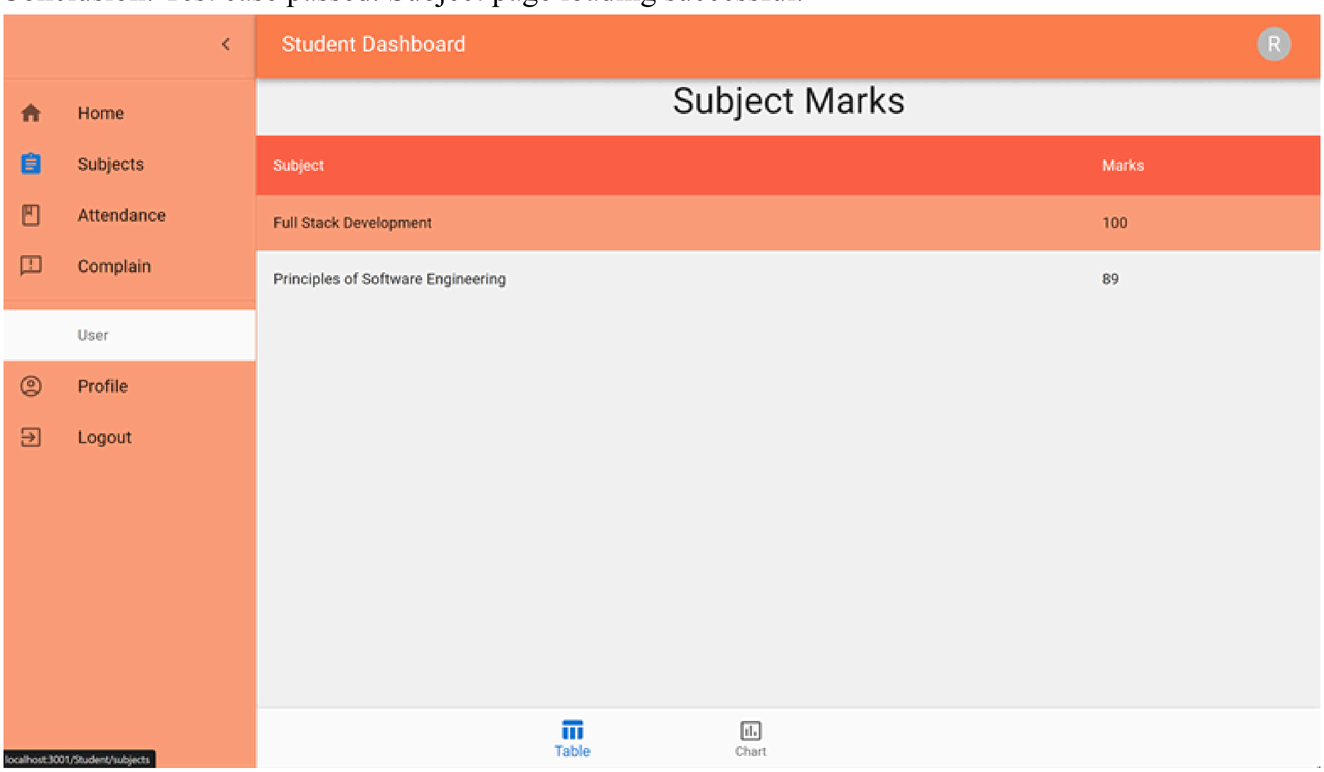


Test case: Check for login successful.
Conclusion: Test case passed. Login successful.

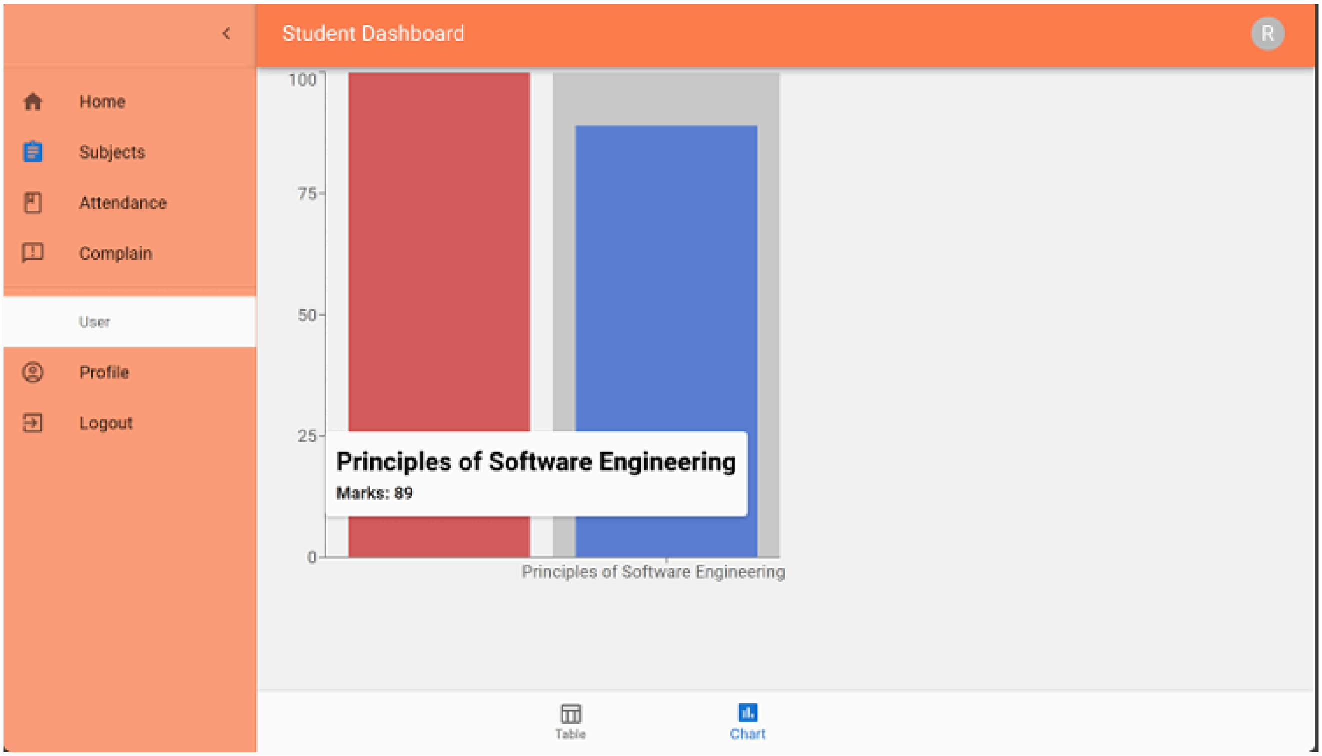


6.2.1.2 Subject

Test case: Check for subject page loading.
Conclusion: Test case passed. Subject page loading successful.

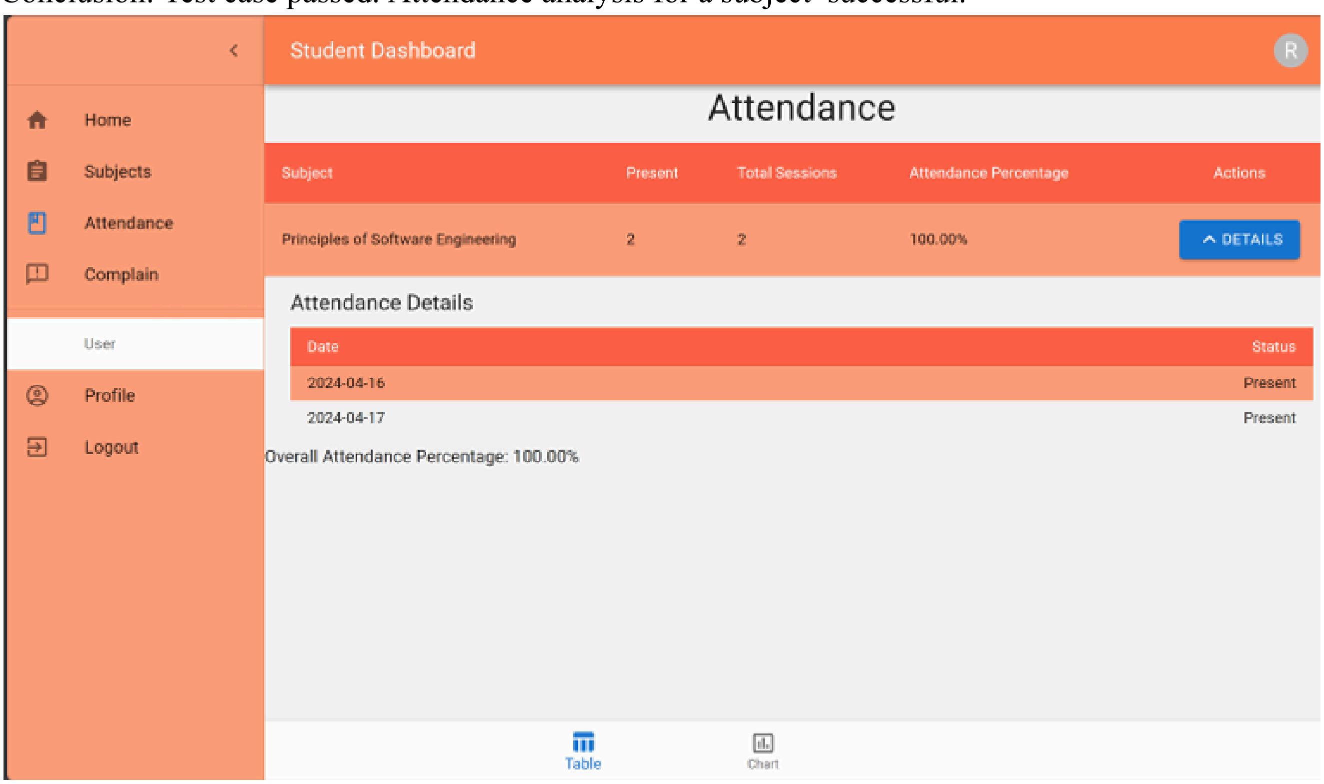


Test case: Check for Mark analysis.
Conclusion: Test case passed. Mark analysis successful.



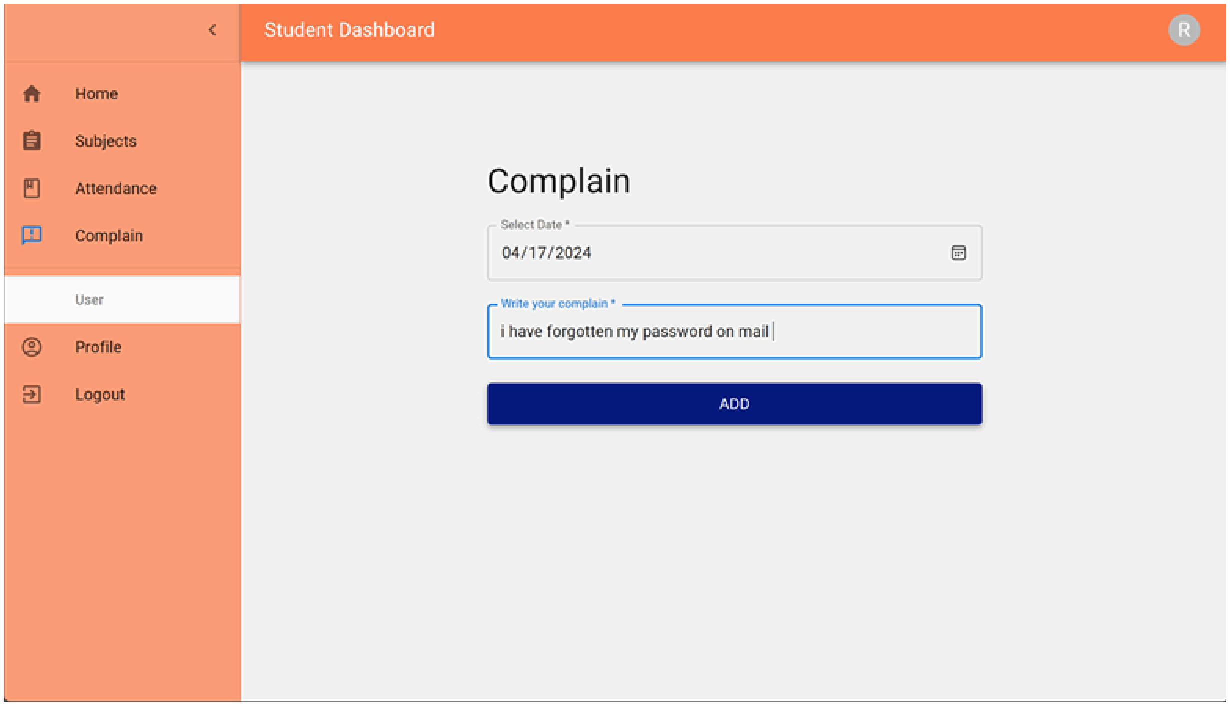
6.2.1.3 Attendance

Test case: Check for attendance analysis for a subject.
Conclusion: Test case passed. Attendance analysis for a subject successful.



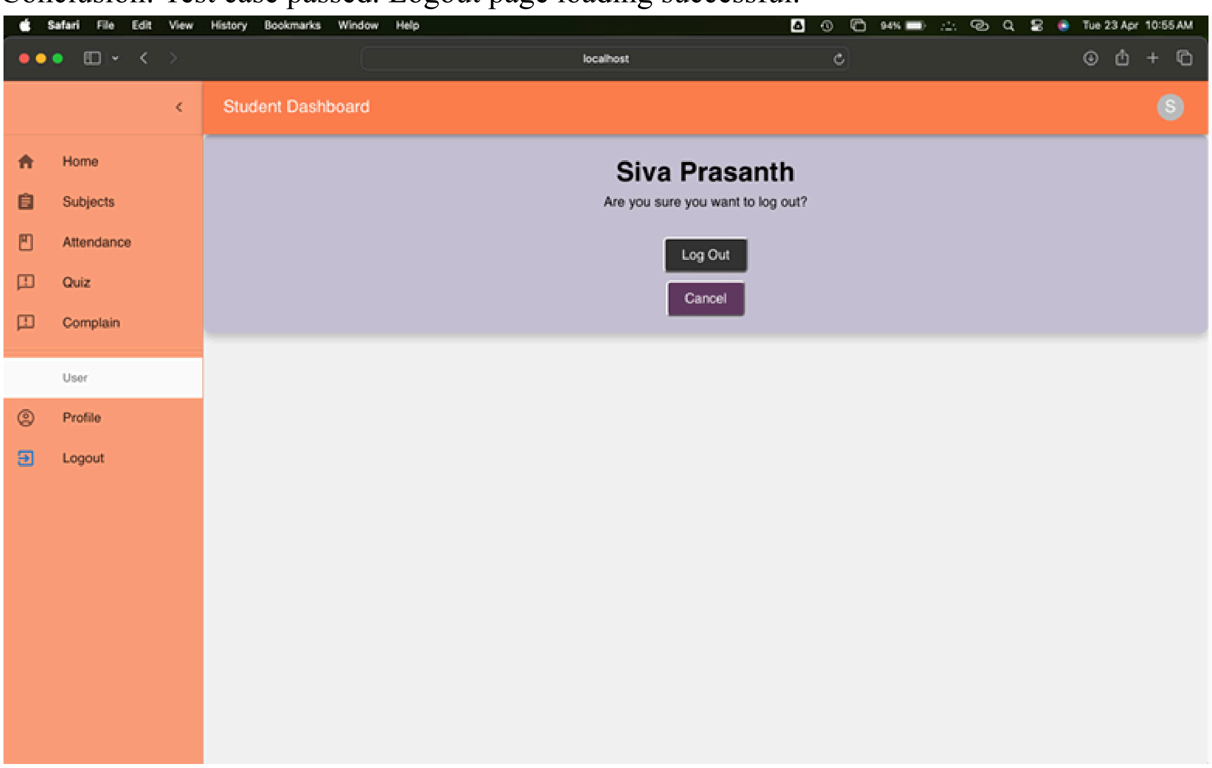
6.2.1.4 Complaint

Test case: Check for complaint creation.
Conclusion: Test case passed. Complaint creation successful.



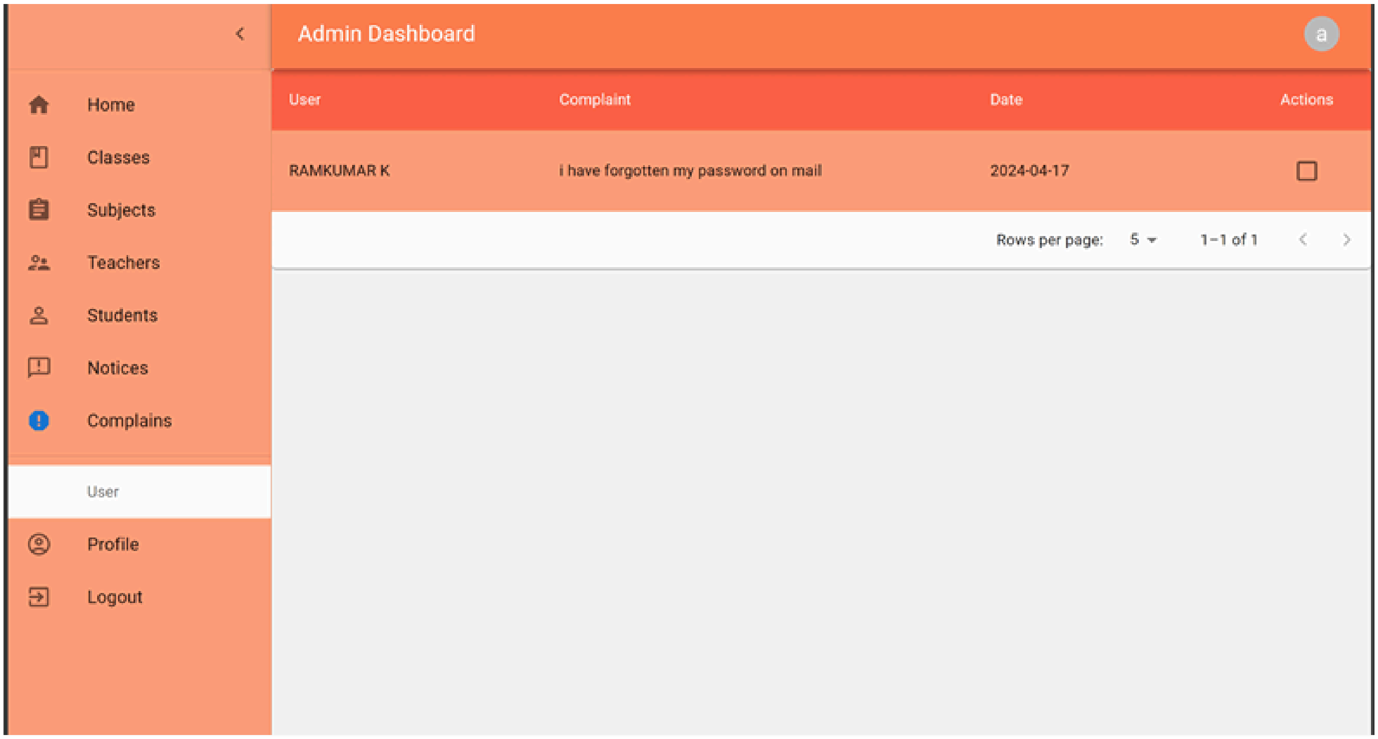
6.2.1.5 Logout

Test case: Check for logout page loading.
Conclusion: Test case passed. Logout page loading successful.

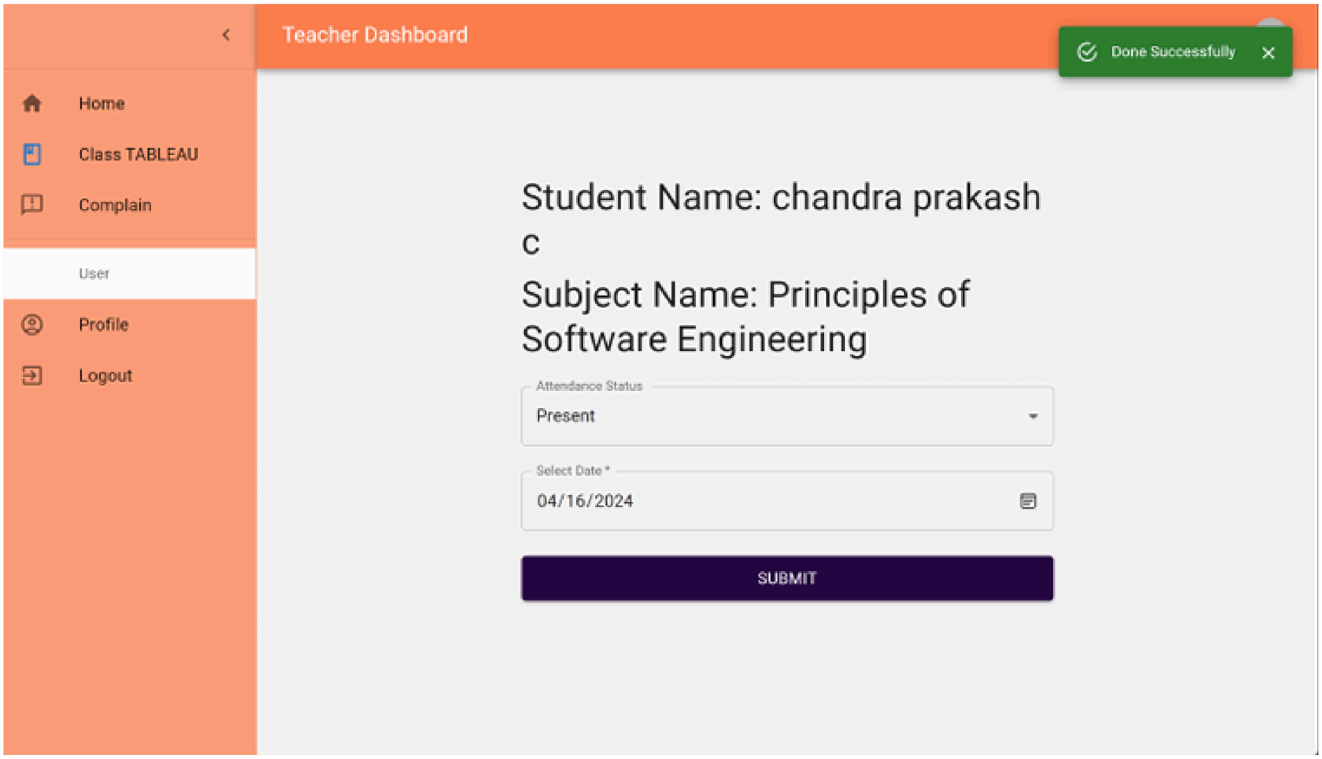


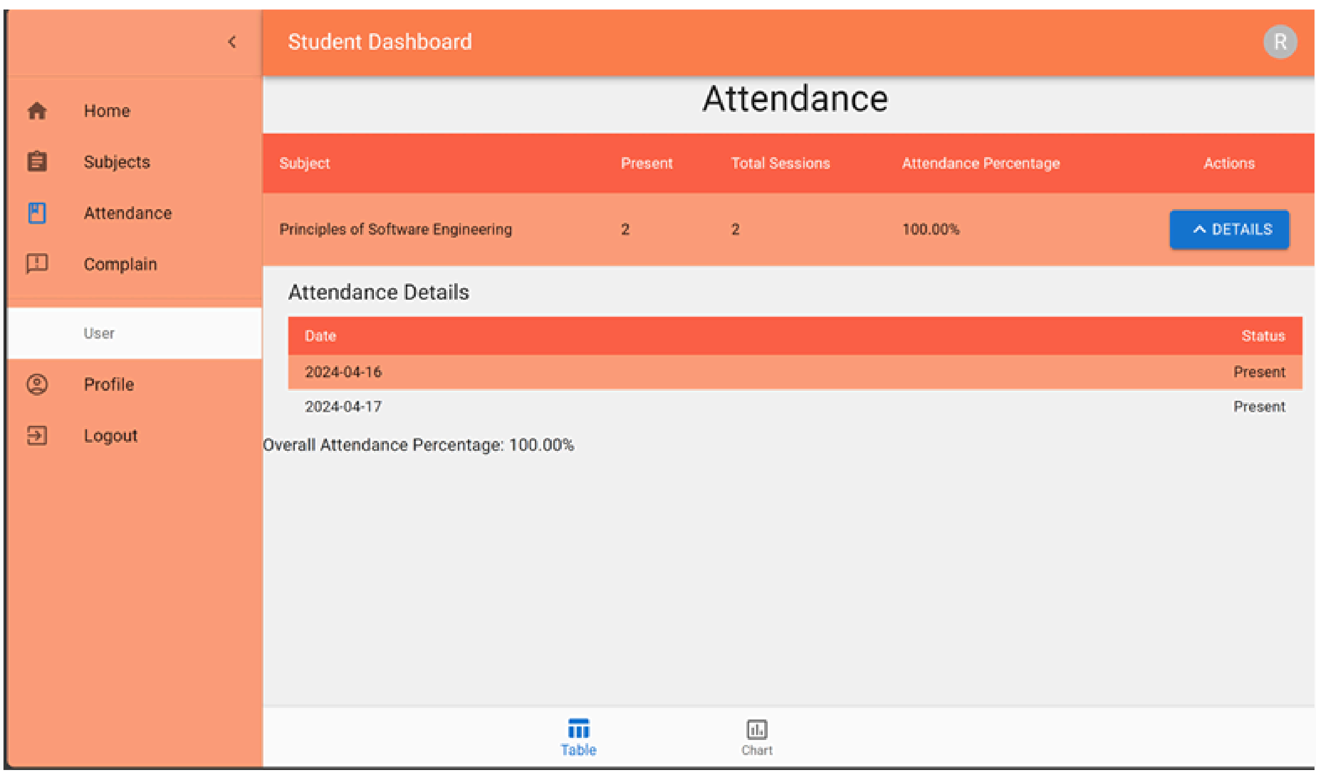
6.2.2 Integration Testing

Test case: Check for complaint raise in admin.
Conclusion: Test case passed. Complaint raise in admin successful.

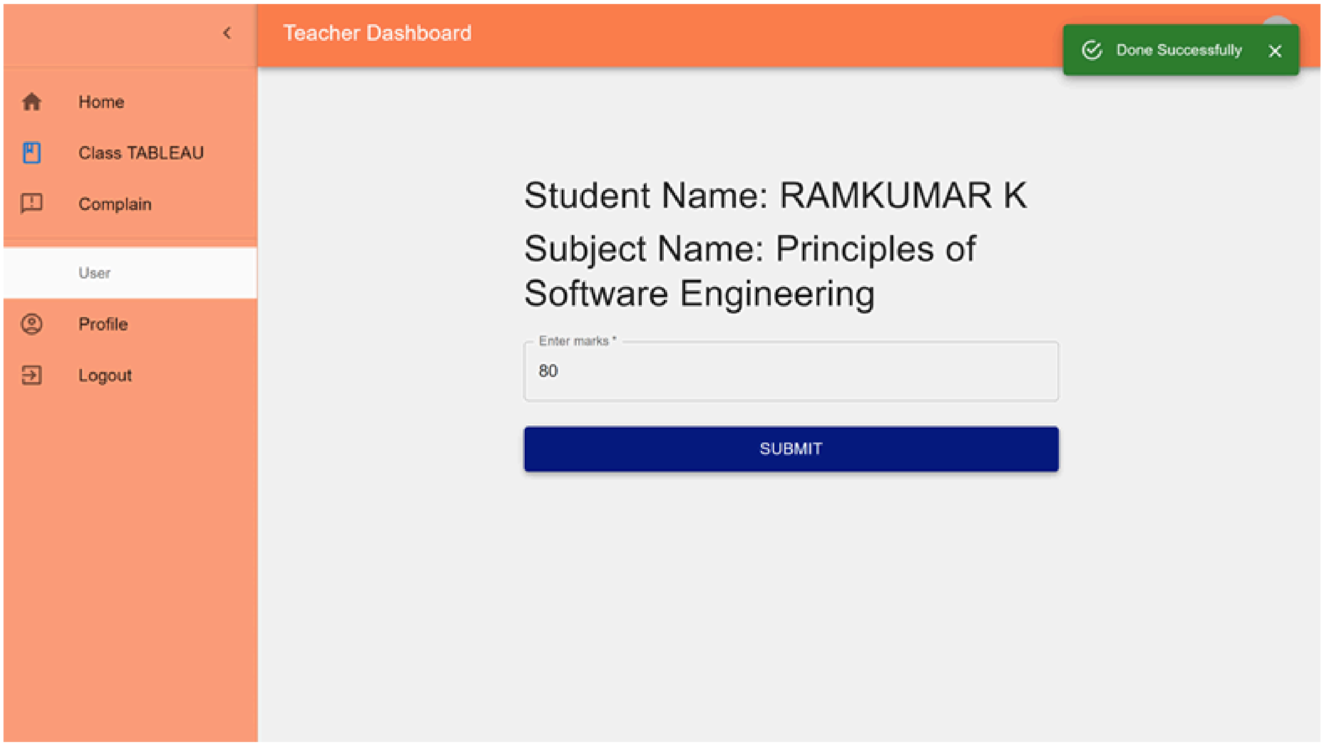


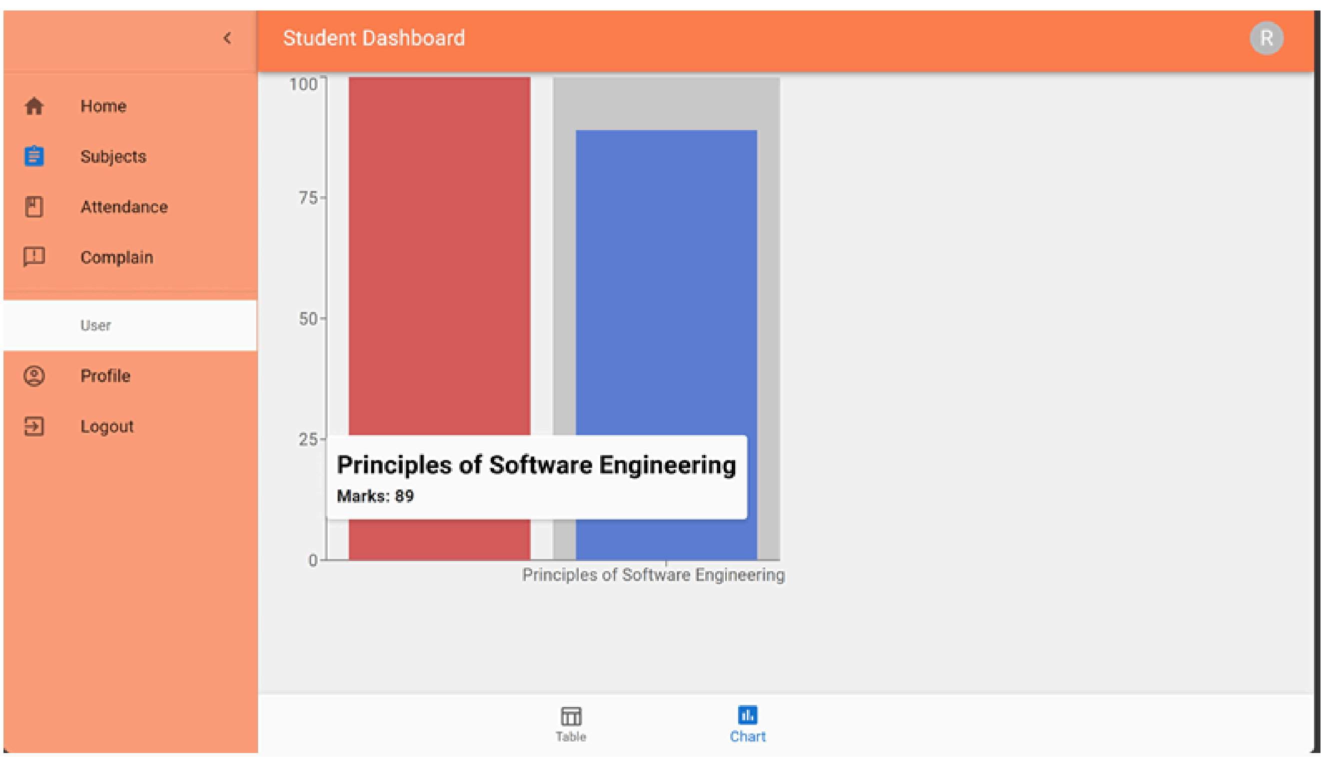
Test case: Check for attendance updates in students.
Conclusion: Test case passed. Attendance update for a student successful.





Test case: Check for mark updation in a student.
Conclusion: Test case passed. Mark update for a student successful.





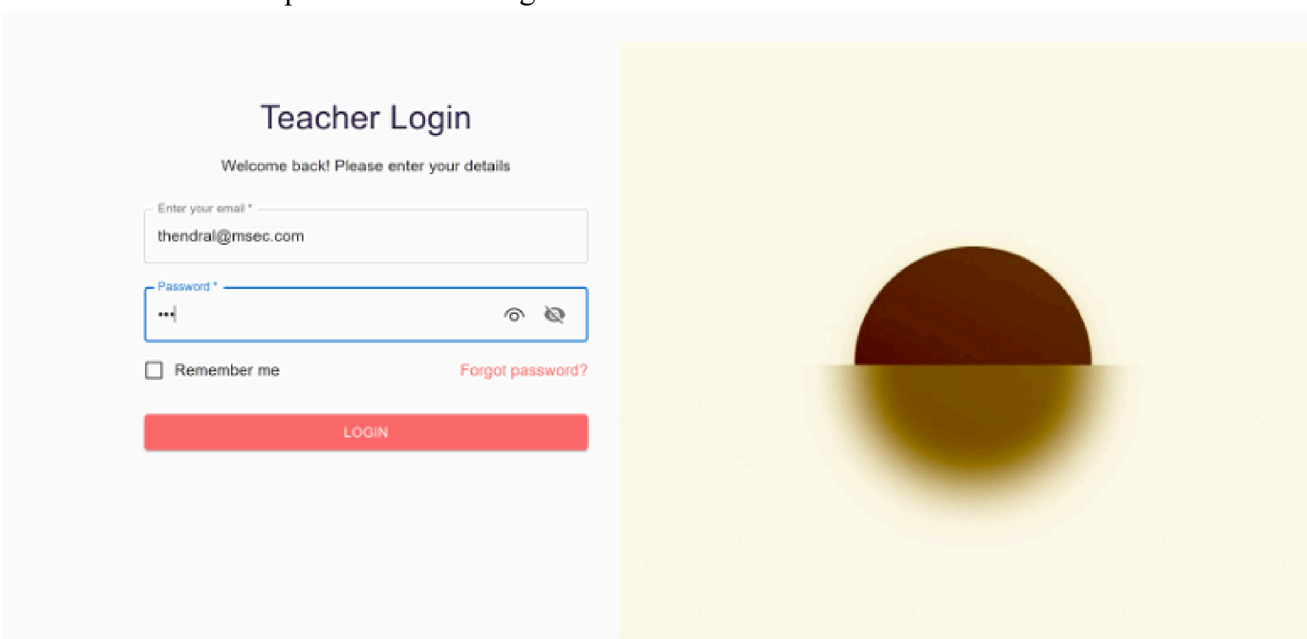
6.3 Testing of Teacher Module

List of Modules:
• Login
• Attendance
• Marks
• Subject
• Quiz
• Logout

6.3.1 Unit Testing

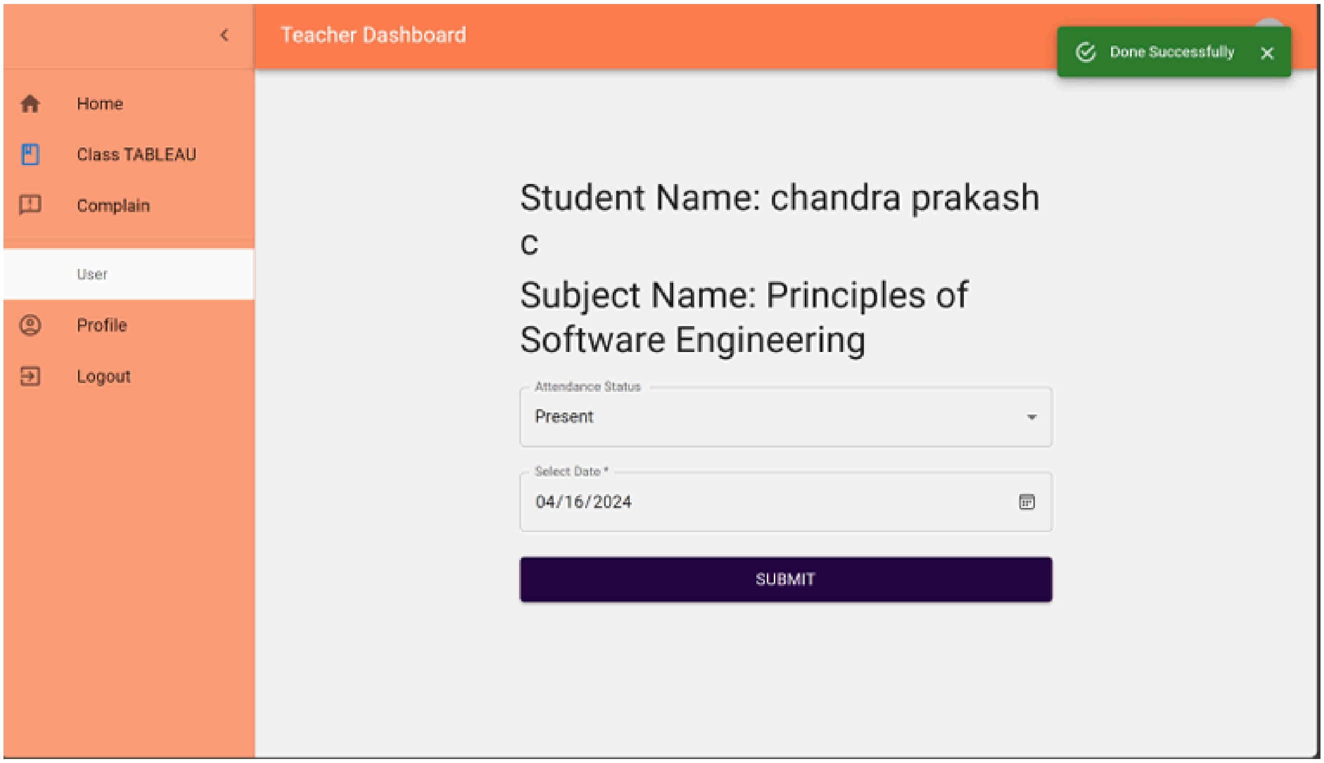
6.3.1.1 Login

Test case: Check for teacher login.
Conclusion: Test case passed. Teacher login successful.

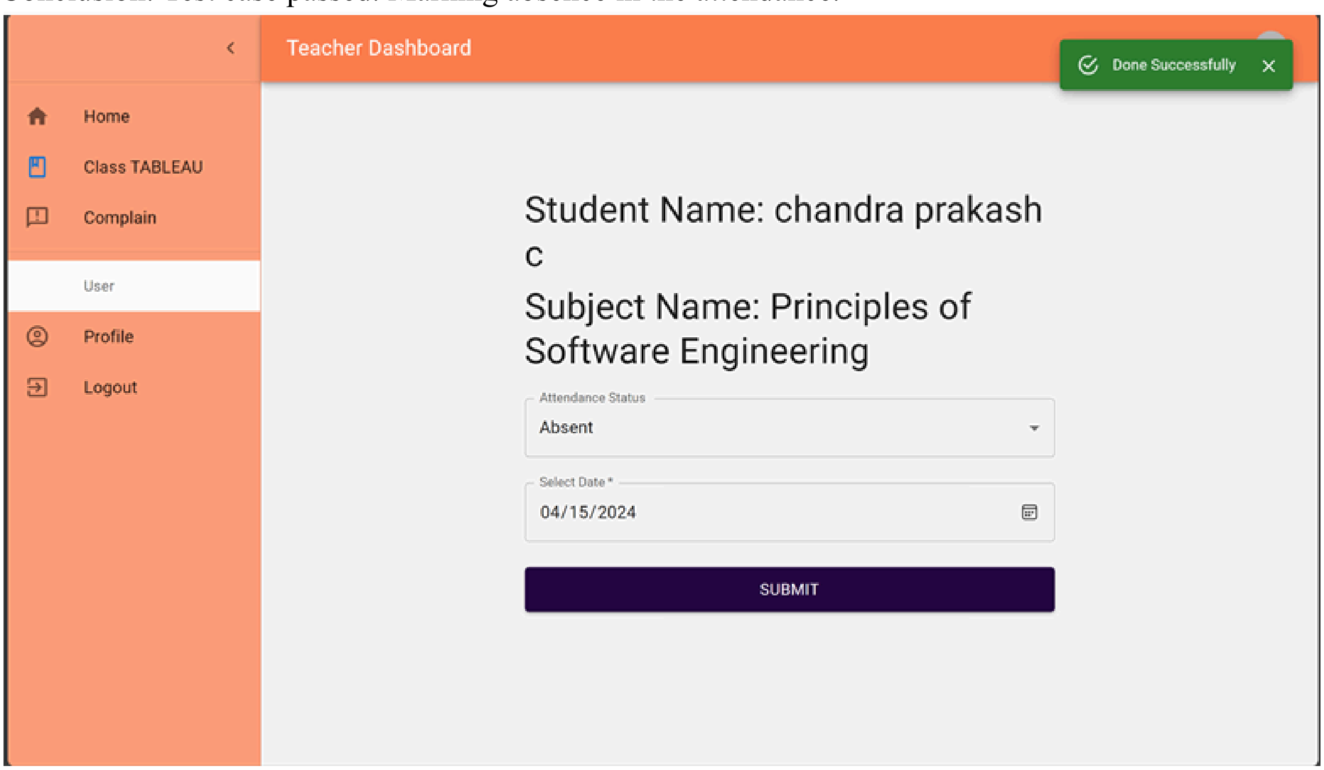


6.3.1.2 Attendance

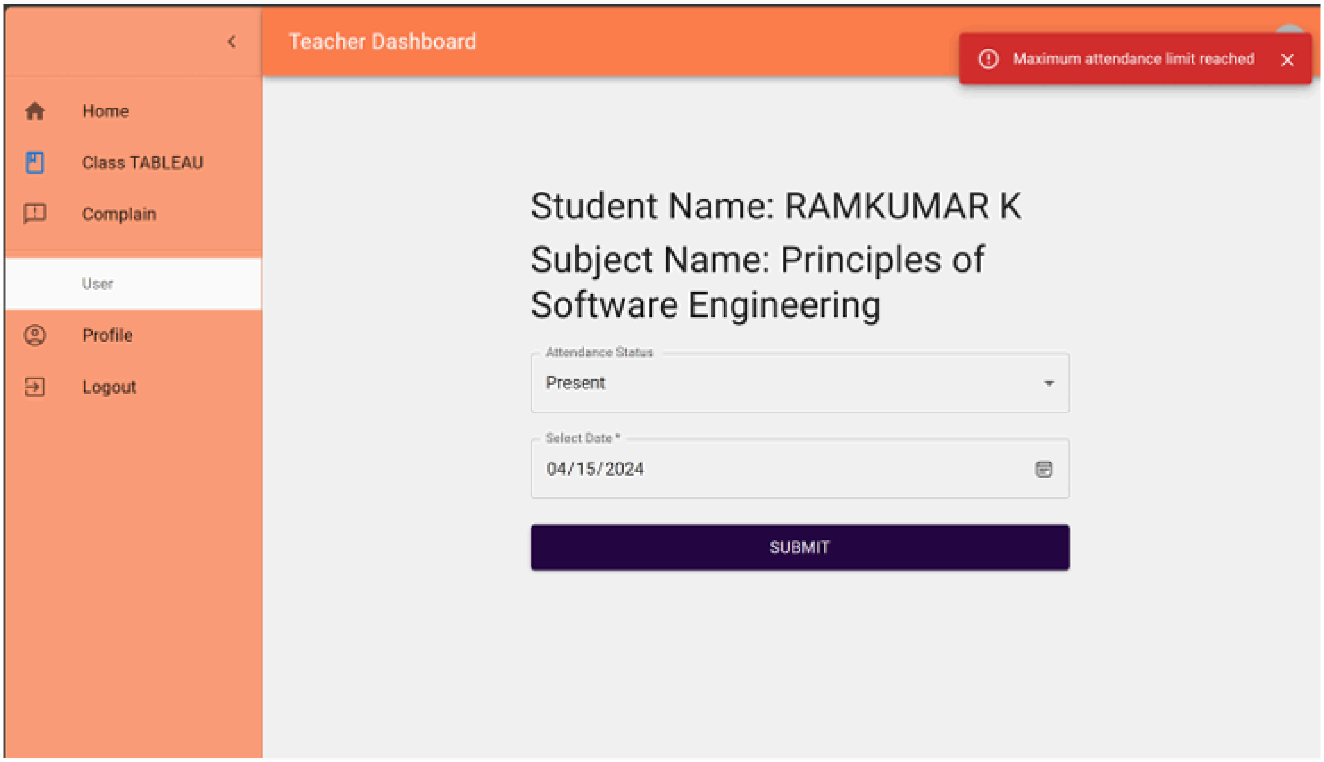
Test case: Check for marking present in the attendance.
Conclusion: Test case passed. Marking present in the attendance.



Test case: Check for marking absent in the attendance.
Conclusion: Test case passed. Marking absence in the attendance.

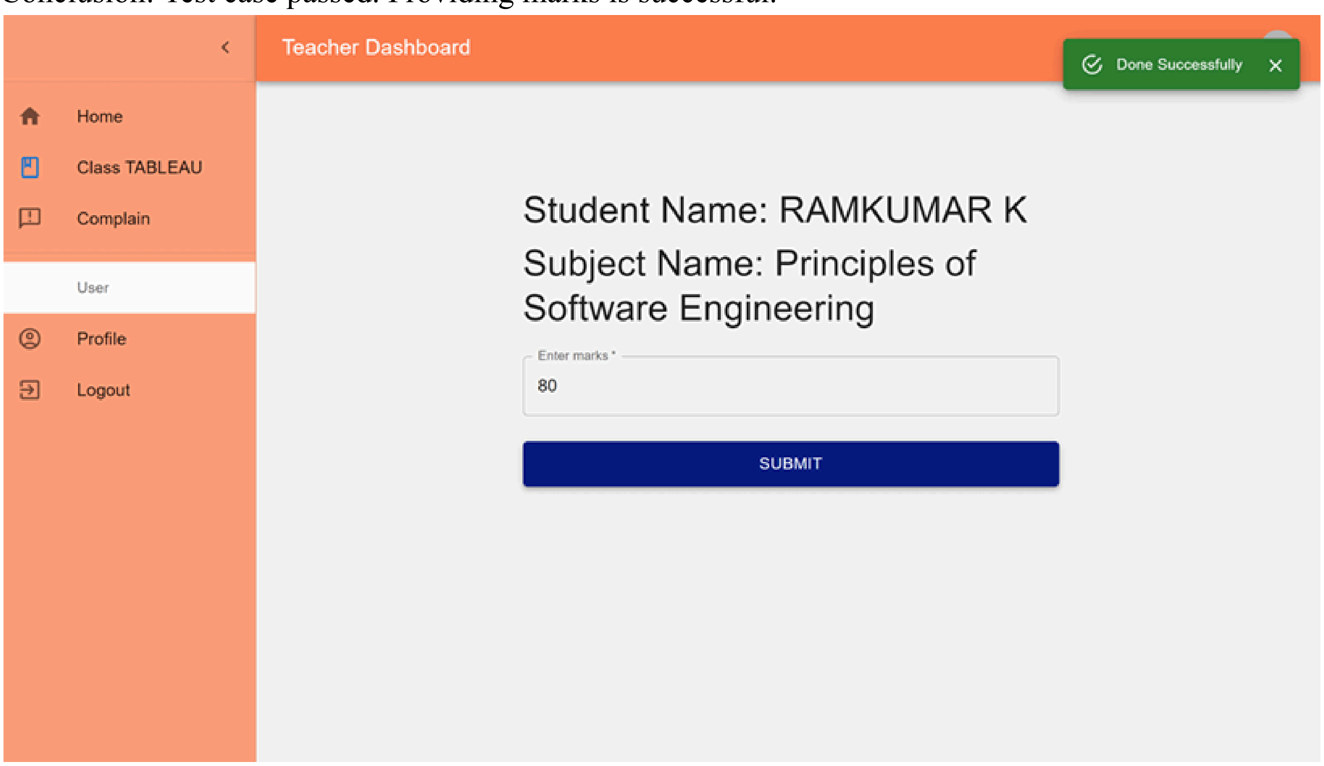


Test case: Check for maximum attendance marking.
Conclusion: Test case passed. Maximum attendance marking successful.



6.3.1.3 Marks

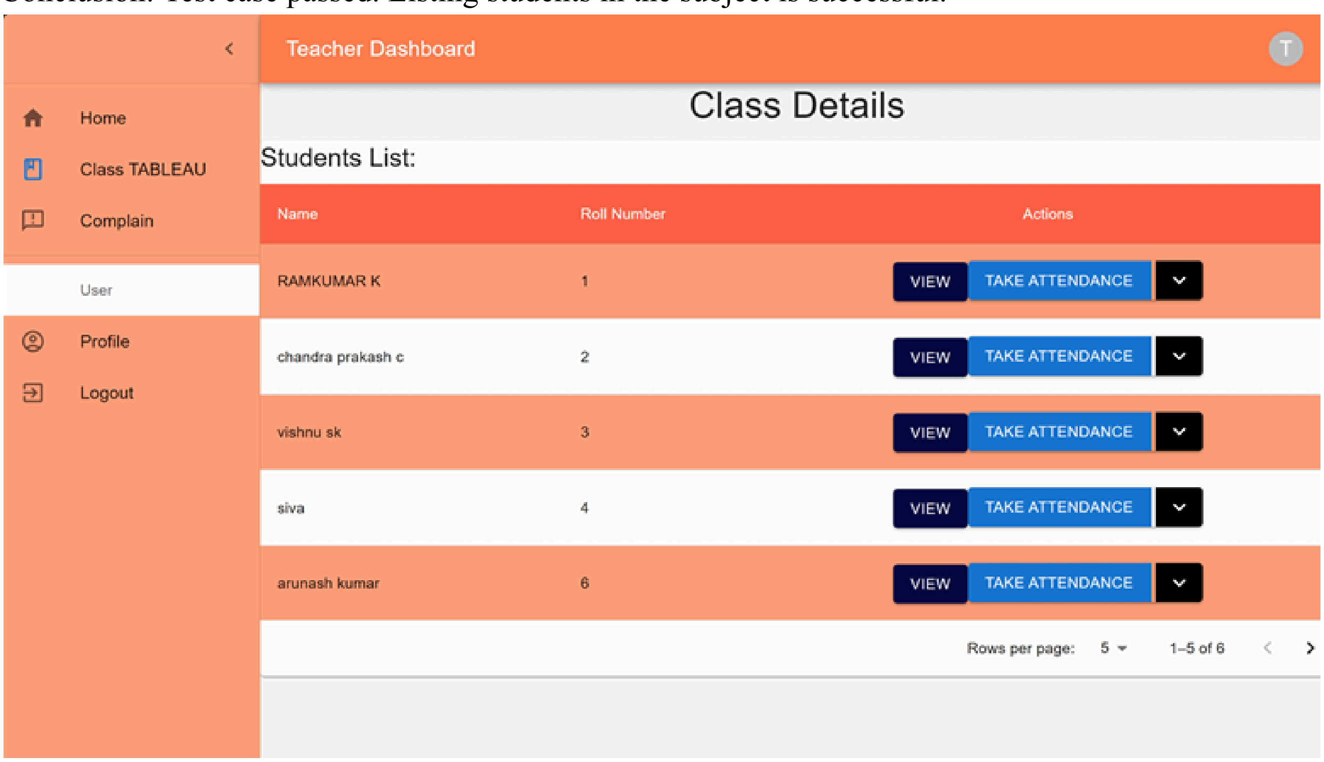
Test case: Check for providing marks.
Conclusion: Test case passed. Providing marks is successful.



Test case: Check for mark analysis.
Conclusion: Test case passed. Mark analysis successful.

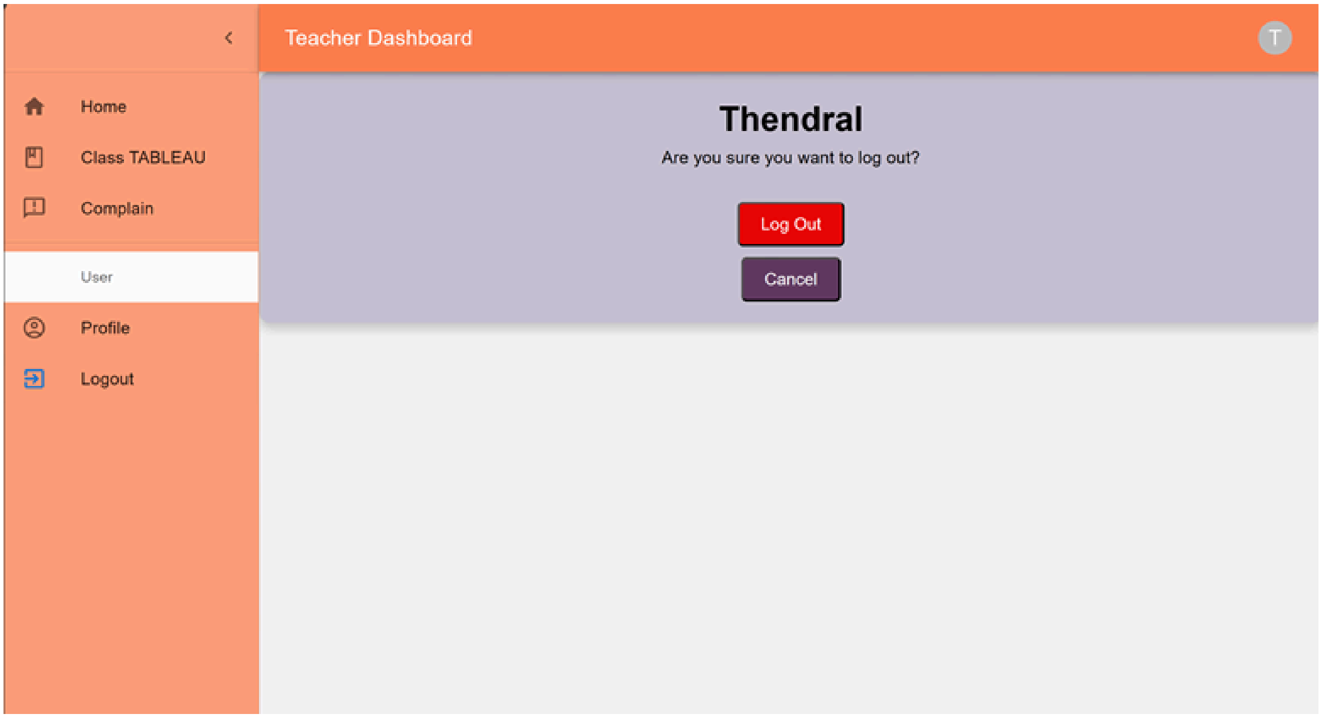
6.3.1.4 Subject

Test case: Check for listing students in the subject.
Conclusion: Test case passed. Listing students in the subject is successful.



6.3.1.5 Logout

Test case: Check for teacher logout.
Conclusion: Test case passed. Teacher logout successful.



7. System Design

7.1 Activity Diagram

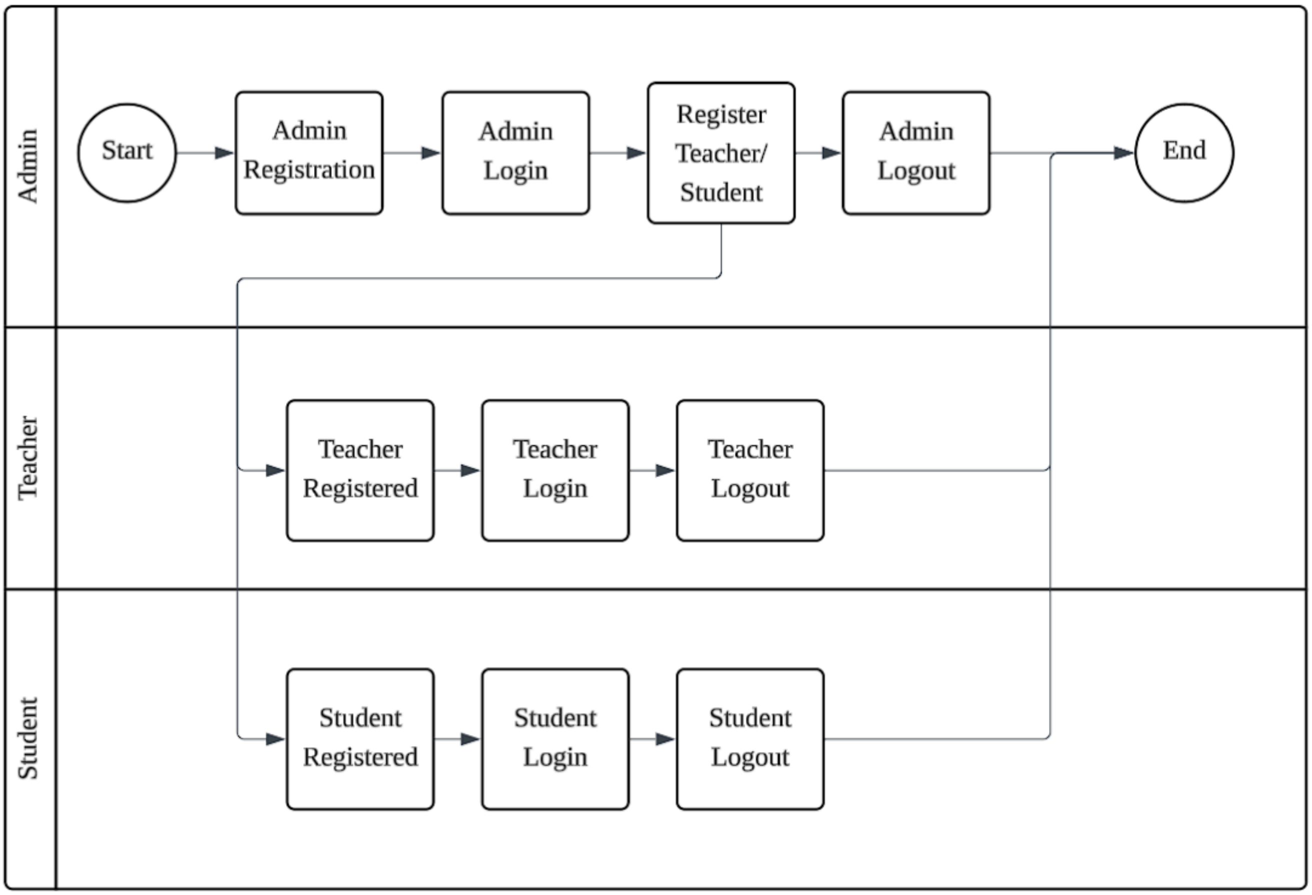


Figure 7.1.1: Signup, Login and Logout Activity Diagram

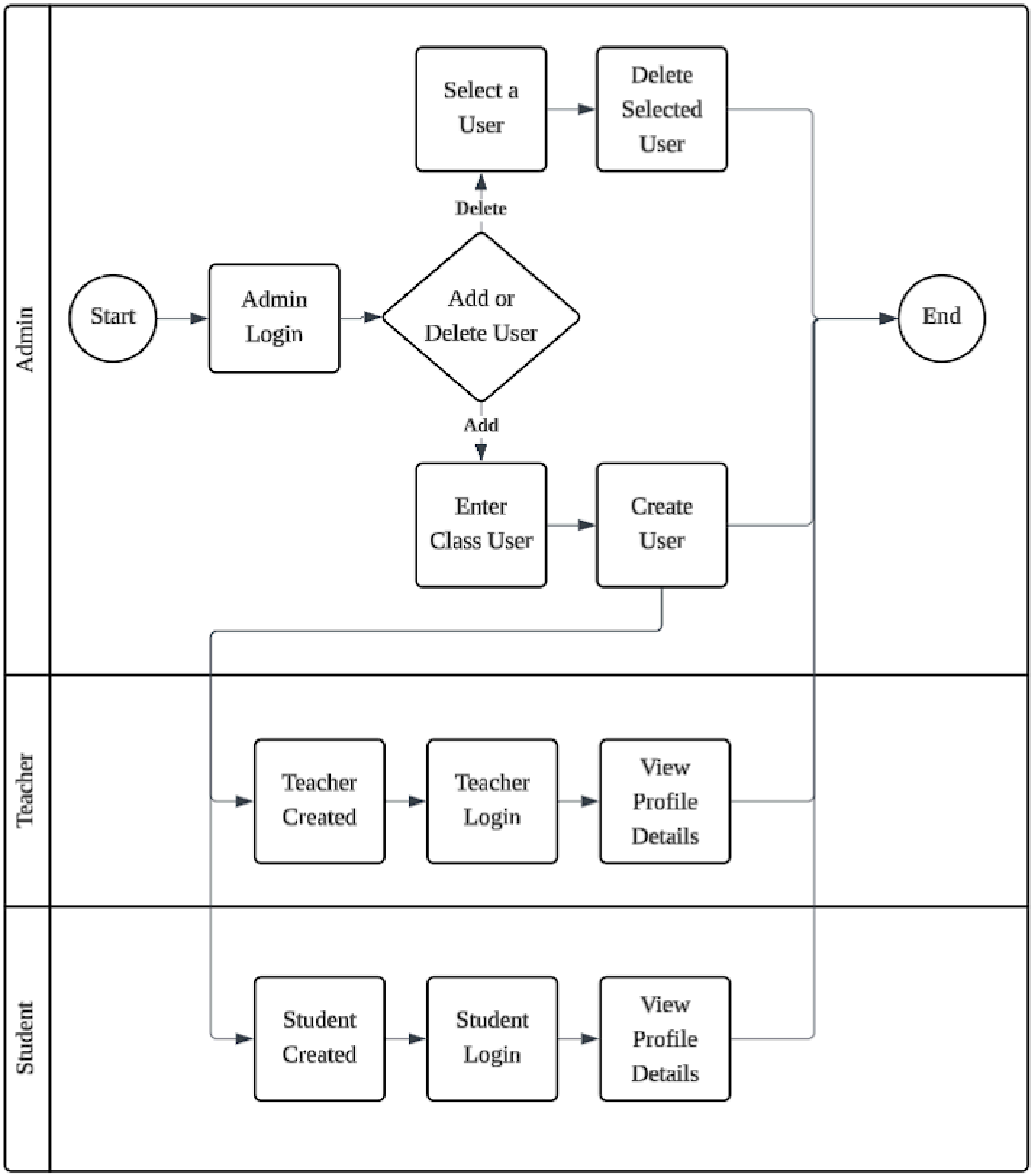


Figure 7.1.2: User Management Activity Diagram

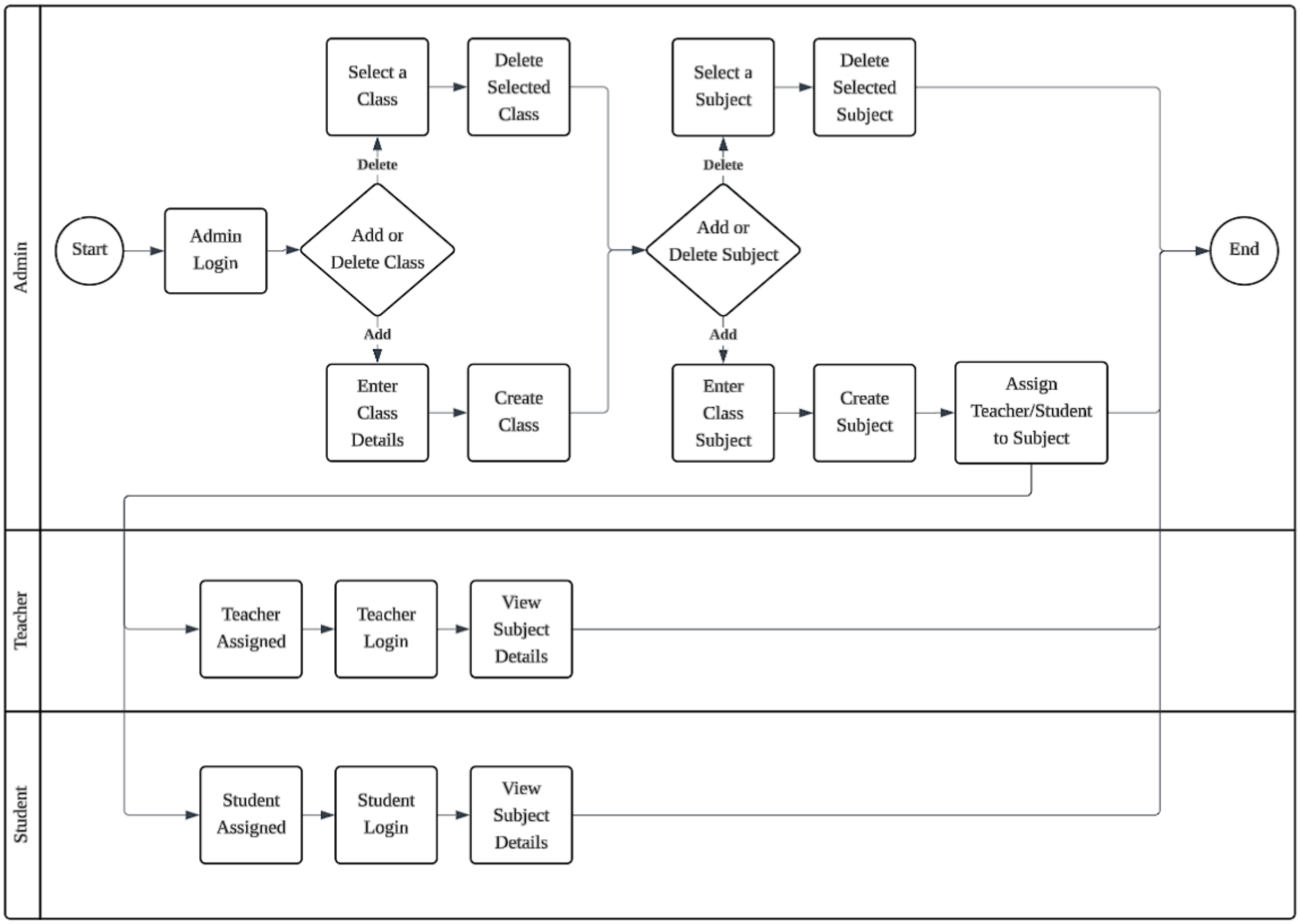


Figure 7.1.3: Classes and Subject Management Activity Diagram

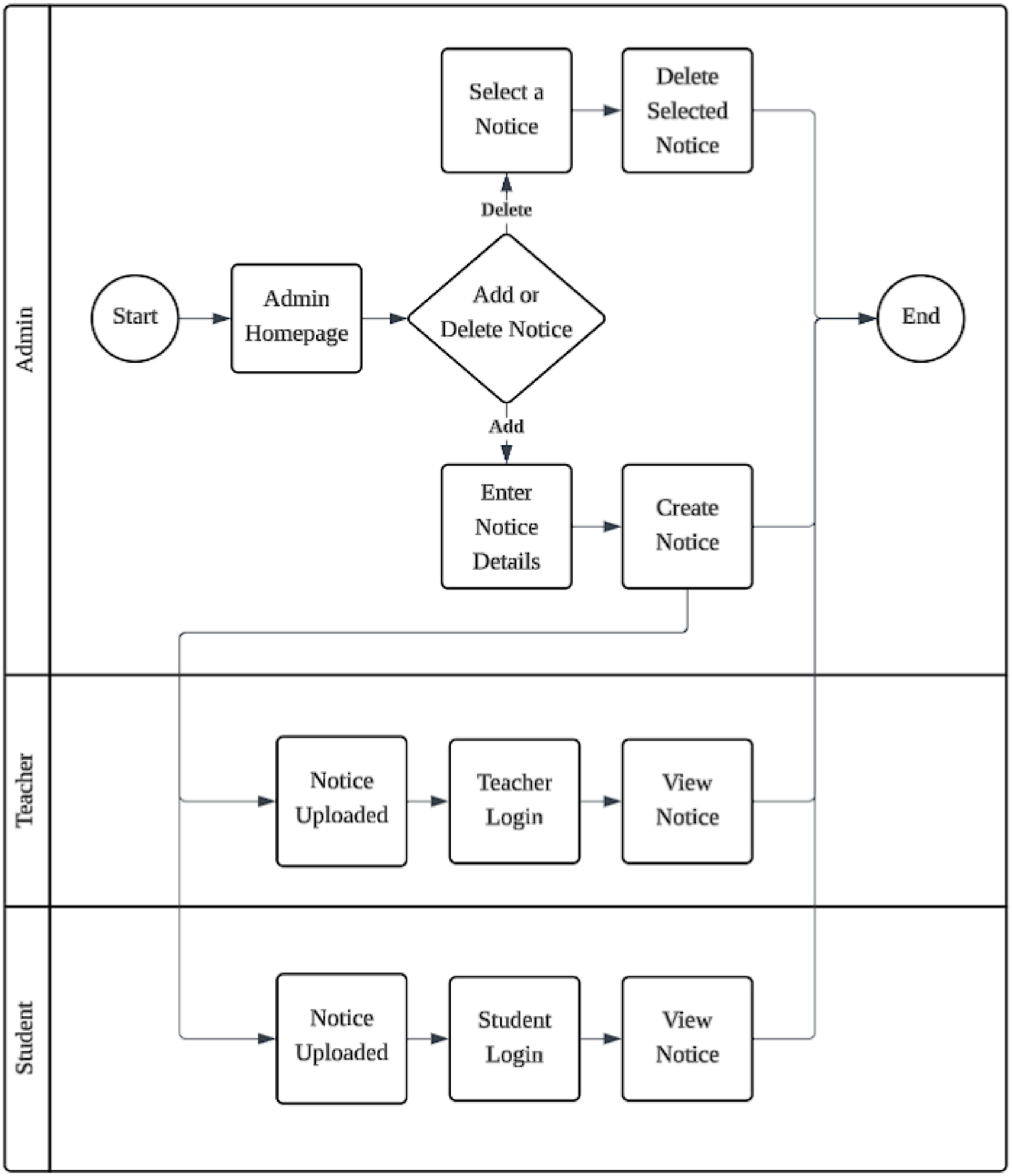


Figure 7.1.4: Notice Management Activity Diagram

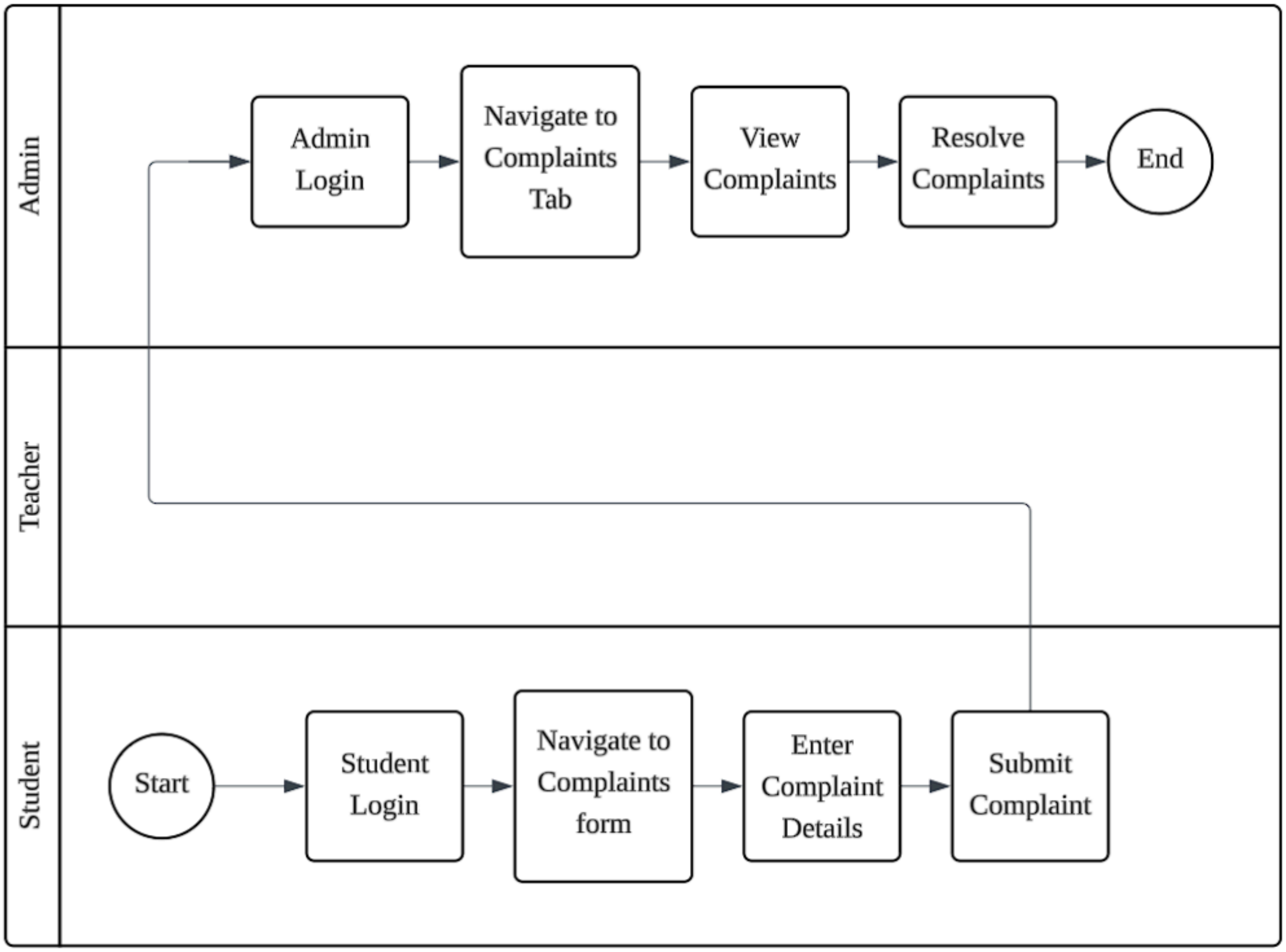


Figure 7.1.5: Complaints Management Activity Diagram

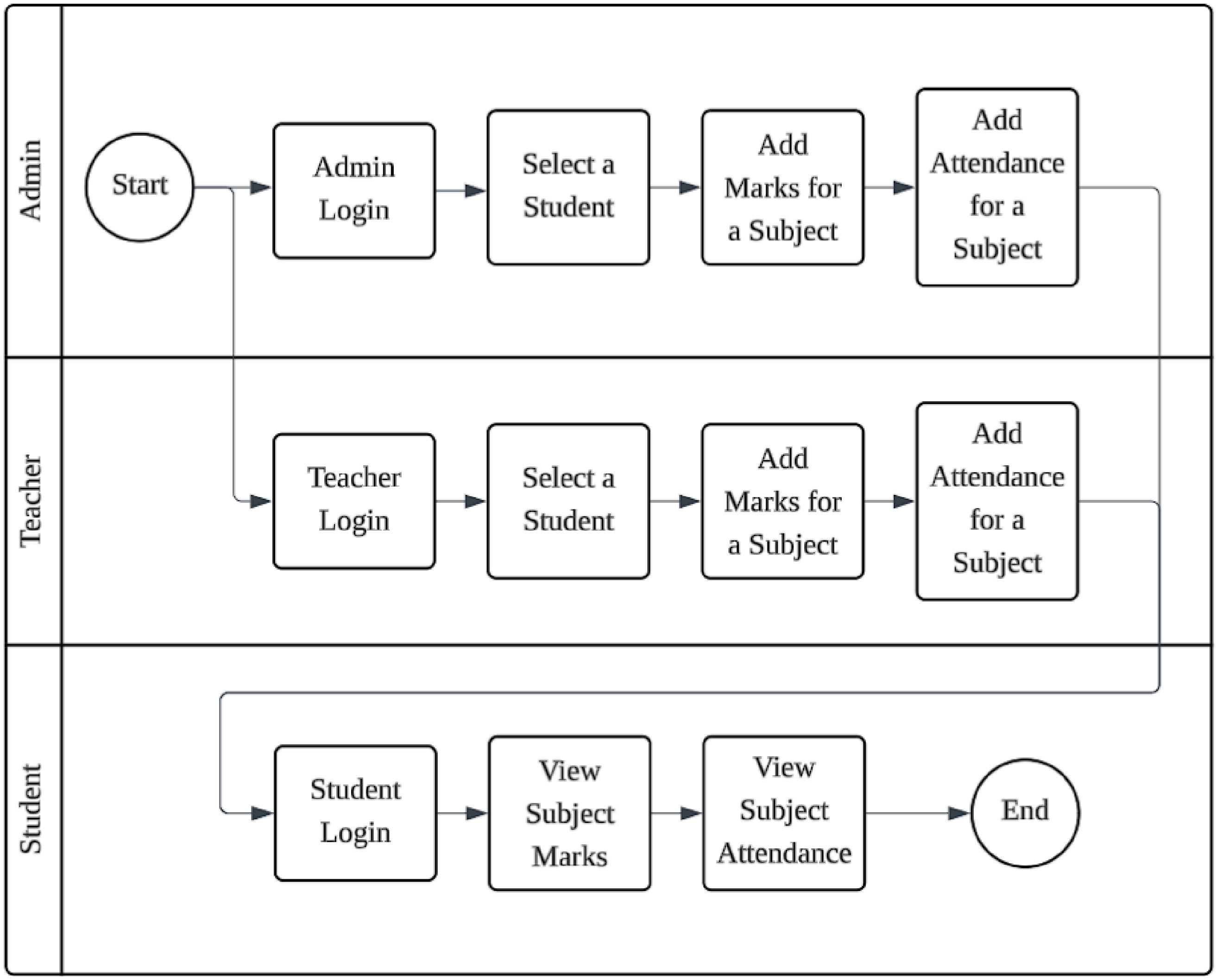


Figure 7.1.6: Marks and Attendance Management Activity Diagram

7.2 Class Diagram

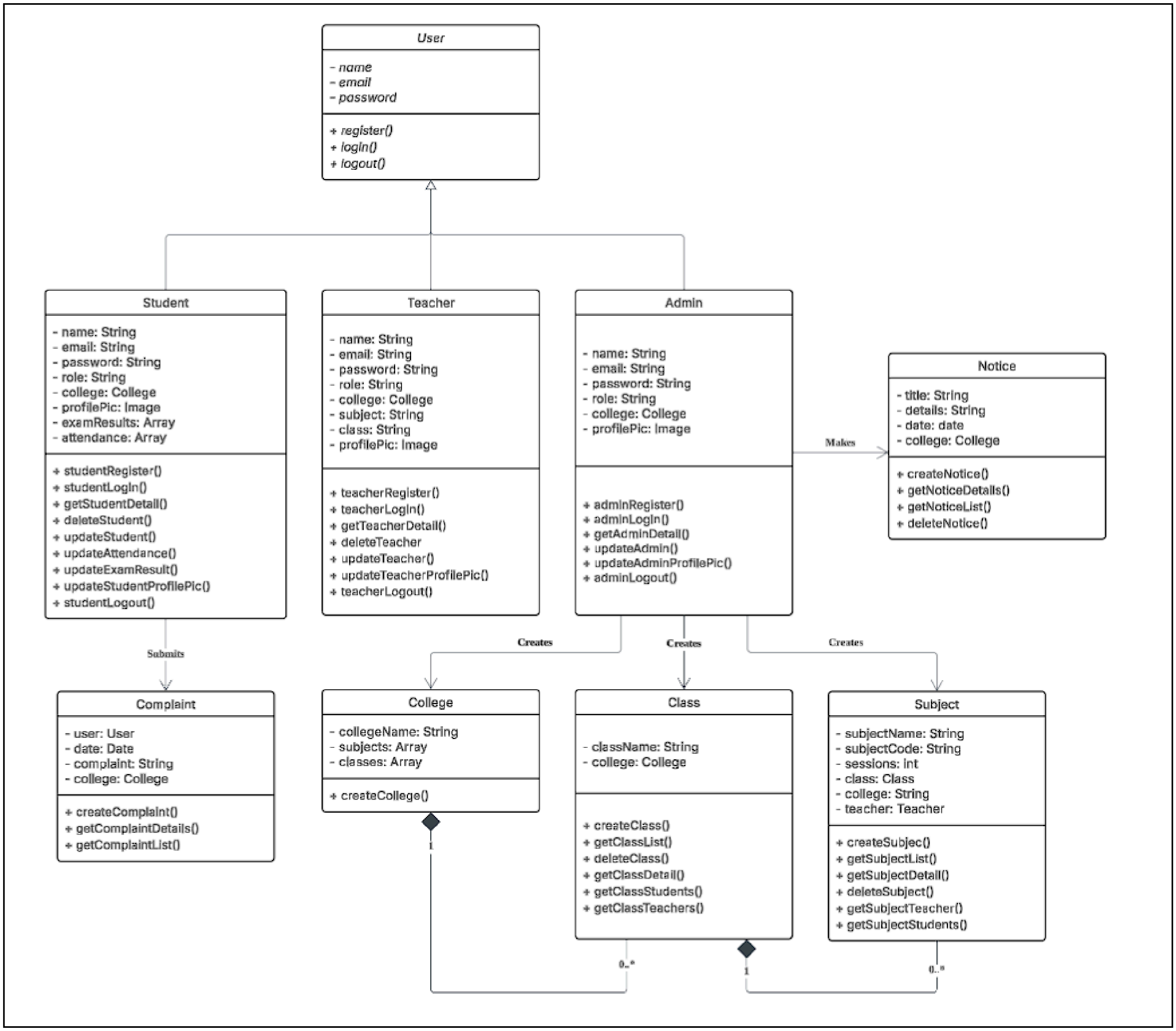


Figure 7.2.1: E-Learning Platform Class Diagram

7.3 Use Case Diagram

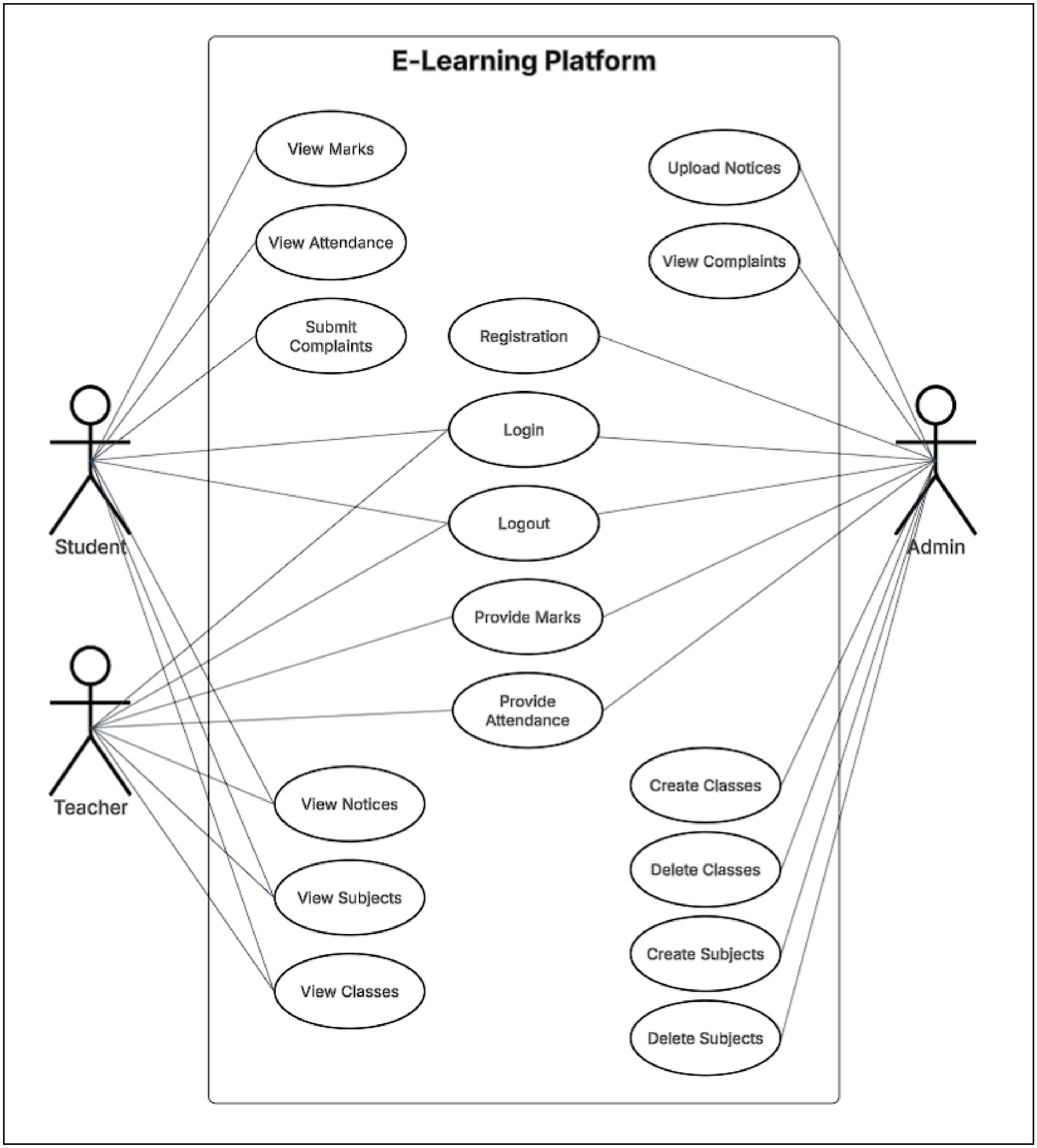


Figure 7.3.1: E-Learning Platform Class Diagram

7.4 Sequence Diagram

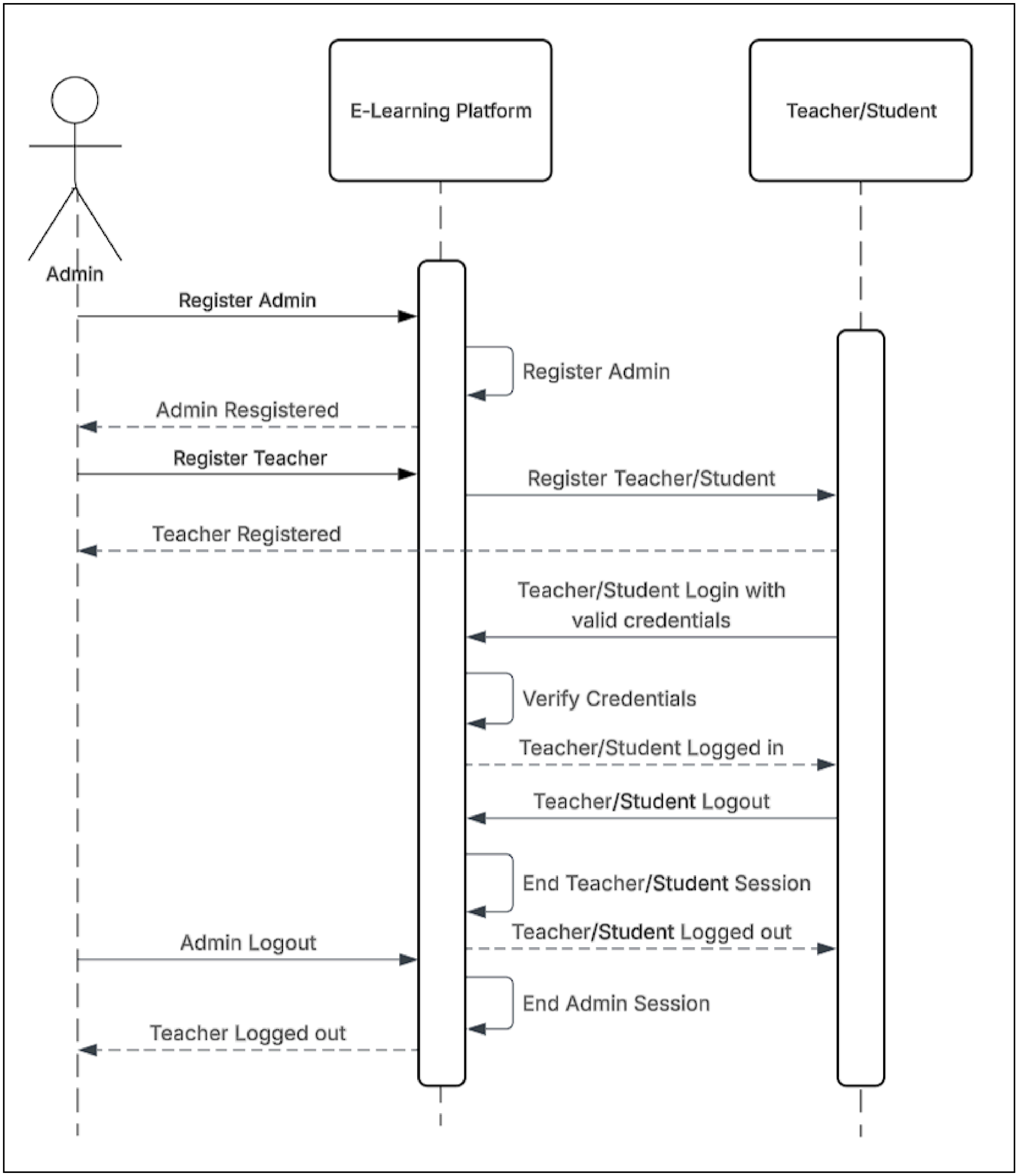


Figure 7.4.1: Signup, Login and Logout Sequence Diagram

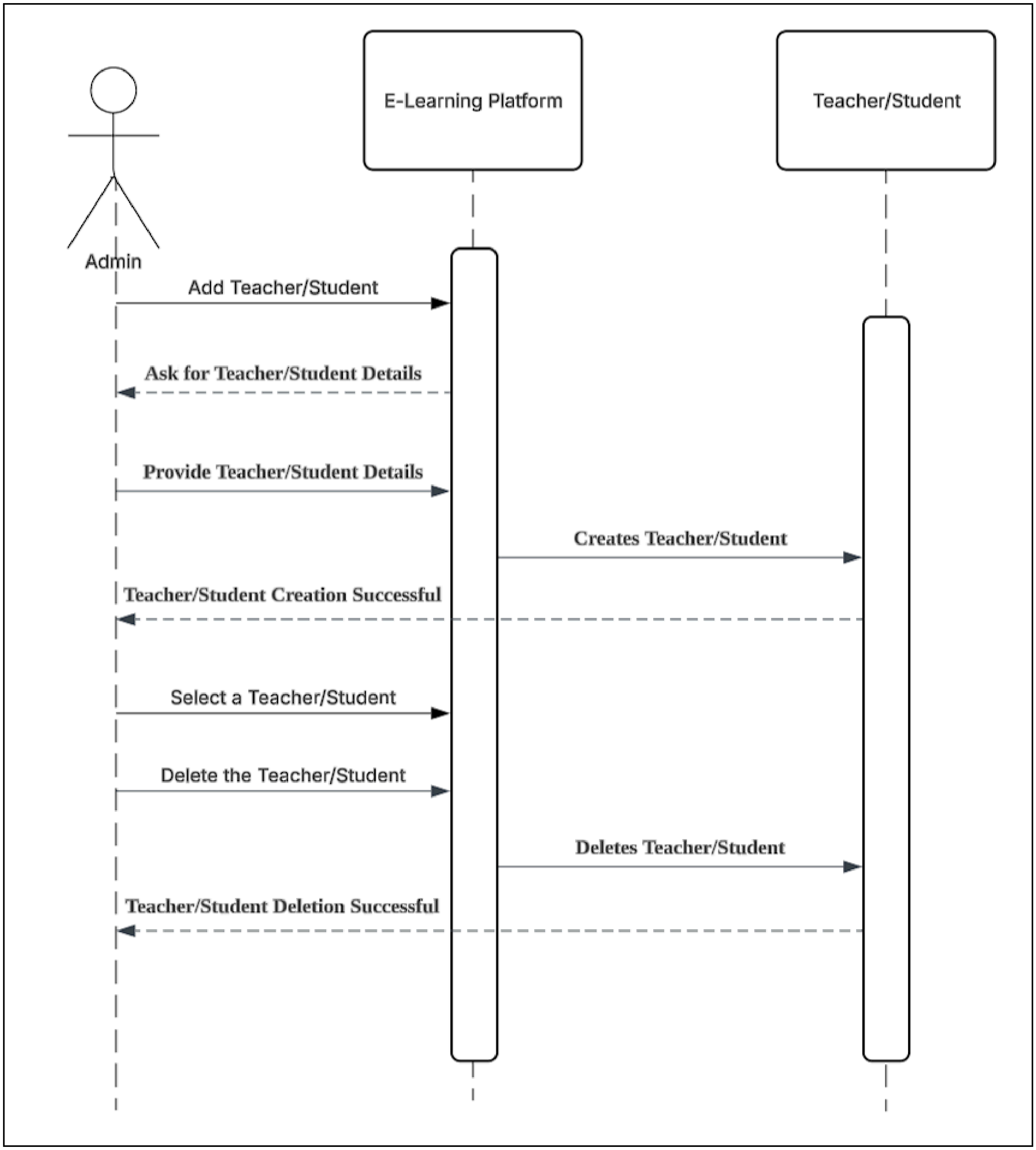


Figure 7.4.2: User Management Sequence Diagram

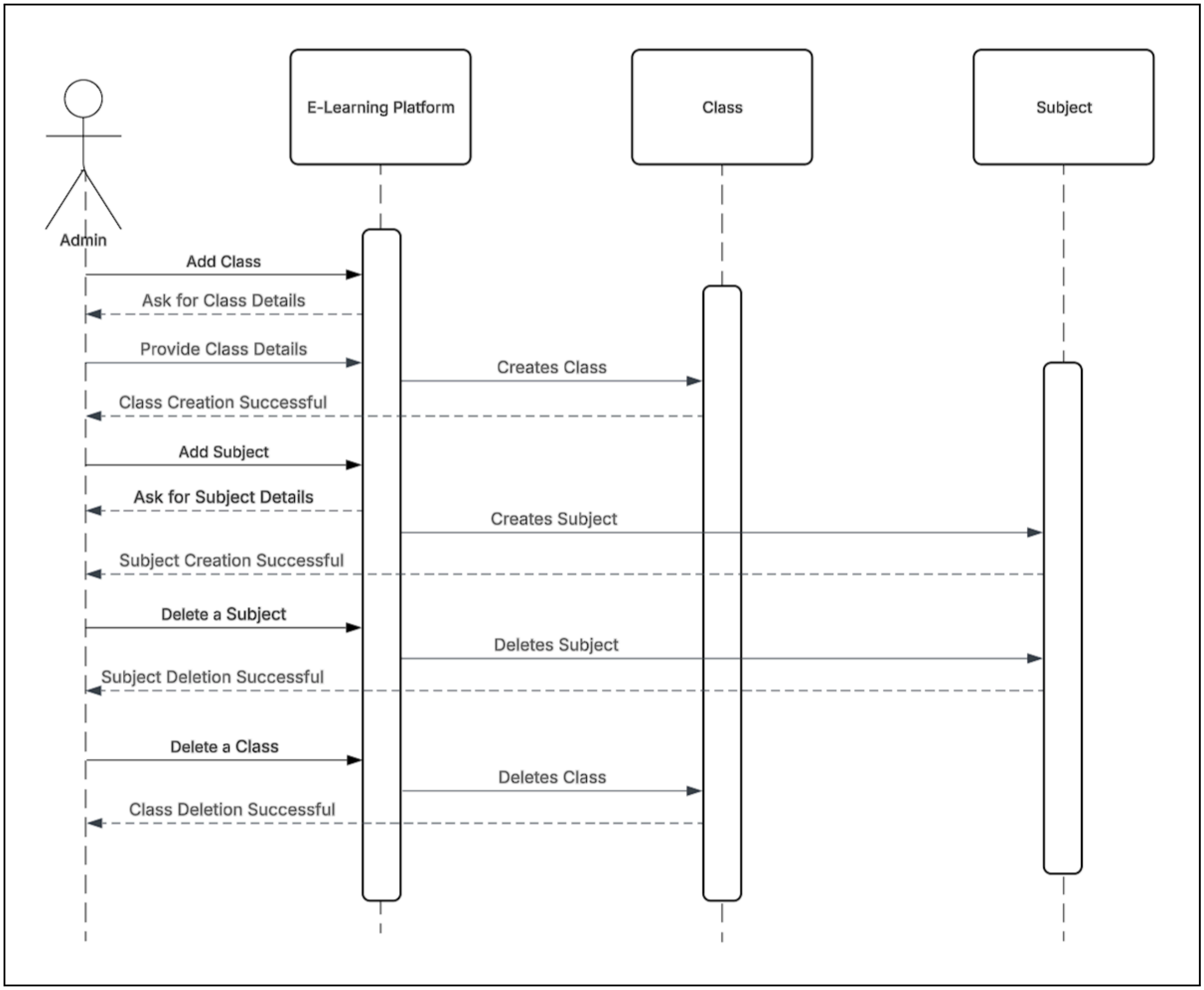


Figure 7.4.3: Classes and Subjects Management Sequence Diagram

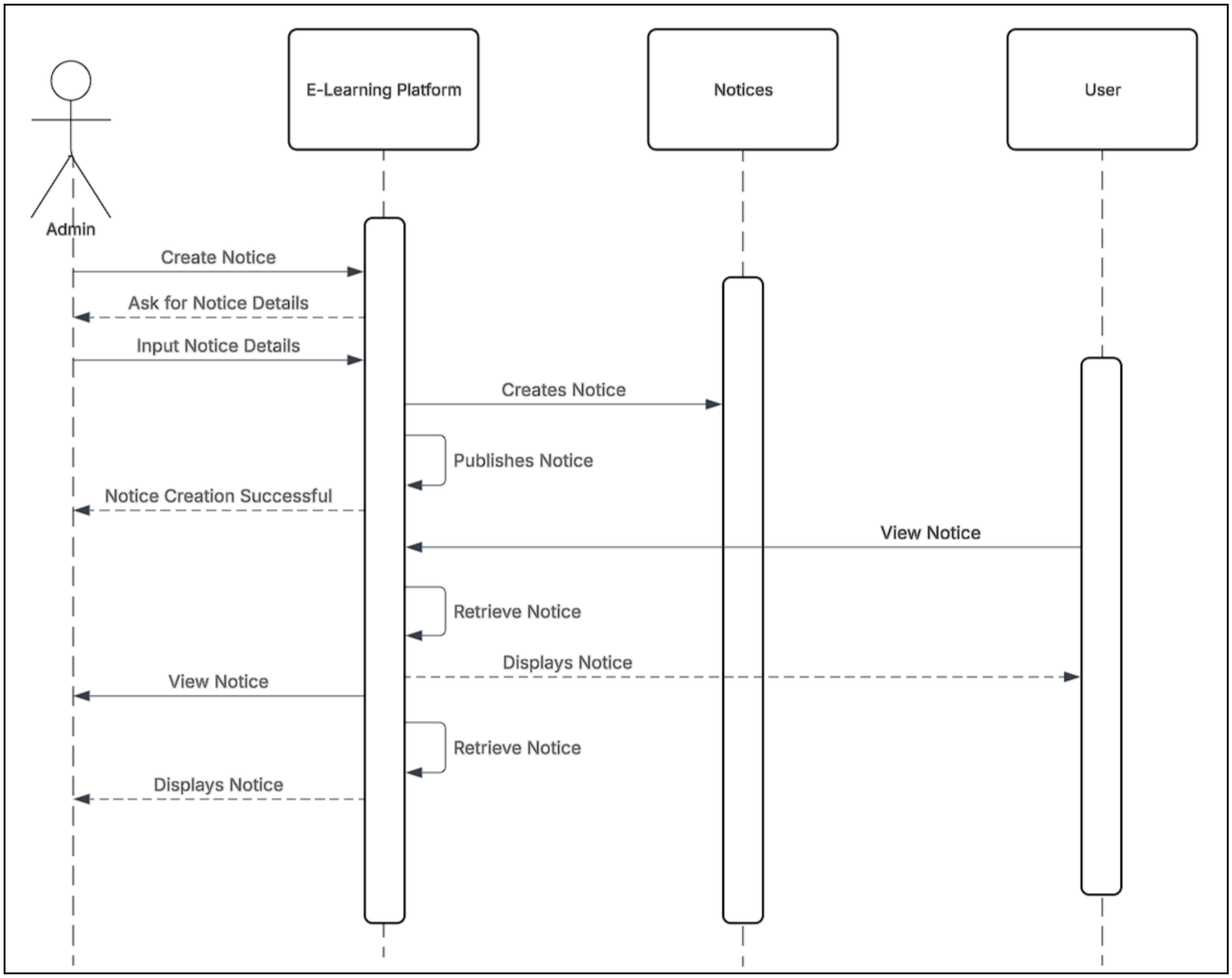


Figure 7.4.4: Notices Management Sequence Diagram

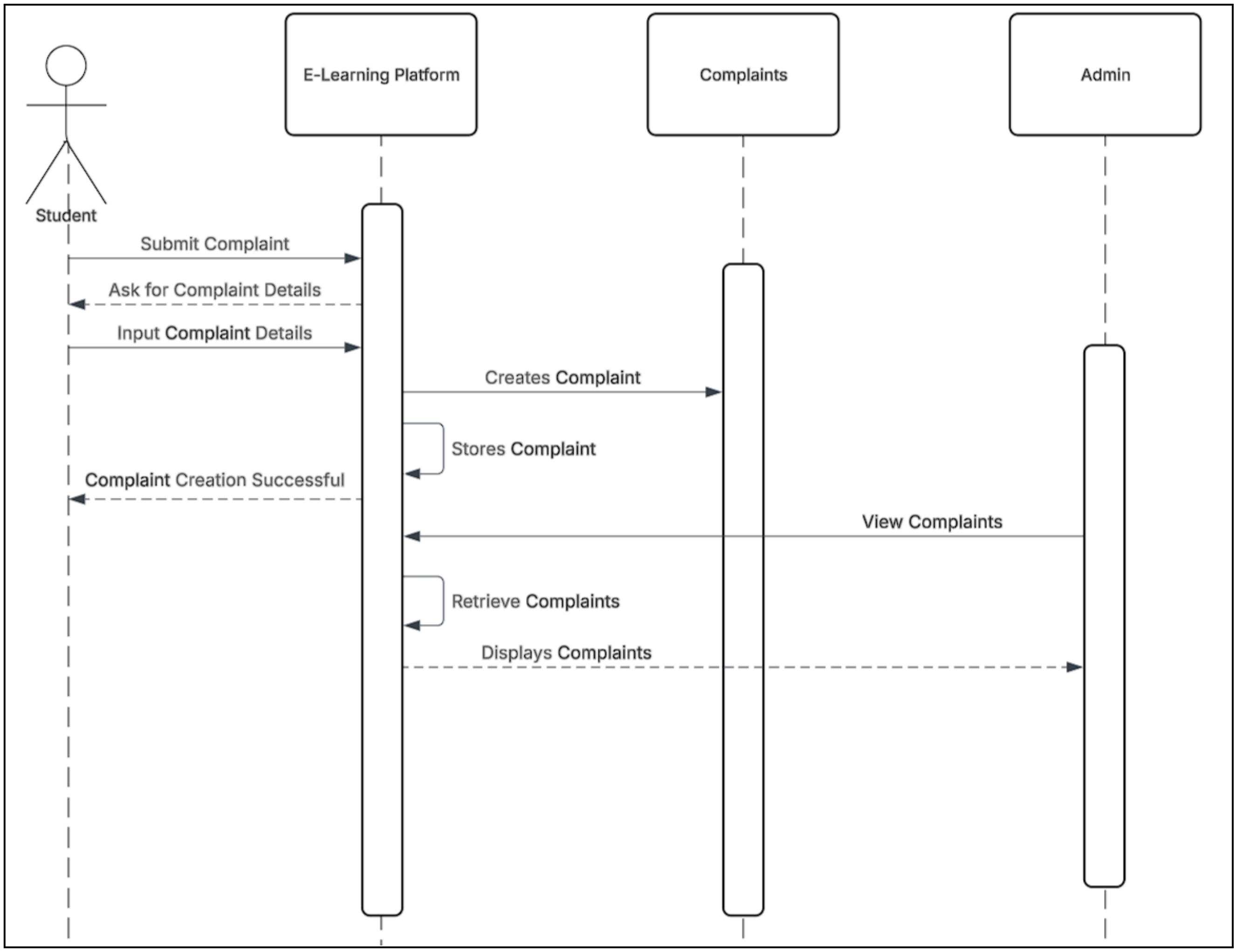


Figure 7.4.5: Complaints Management Sequence Diagram

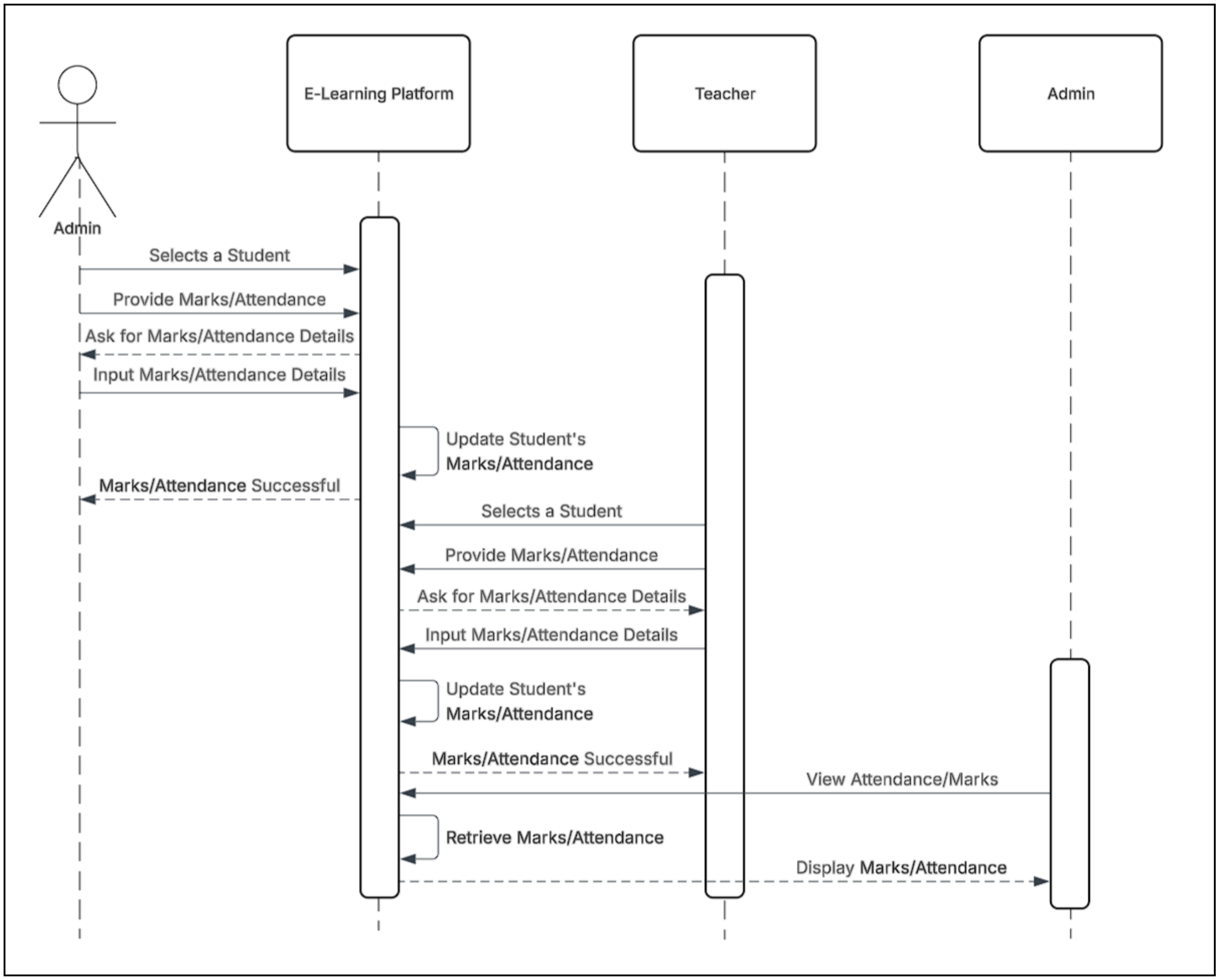


Figure 7.4.6: Marks and Attendance Management Sequence Diagram

7.5 State Diagram

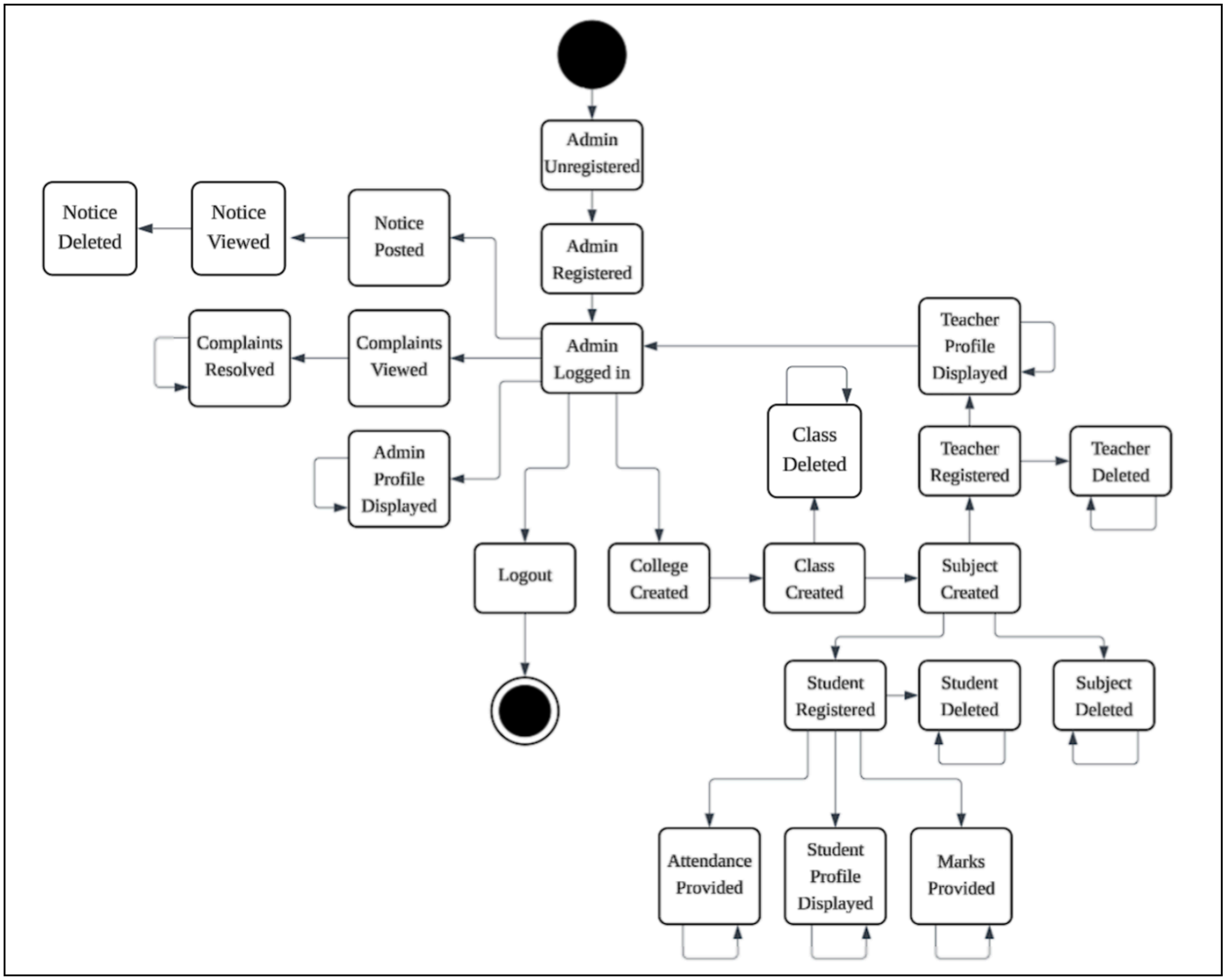


Figure 7.5.1: Admin State Diagram

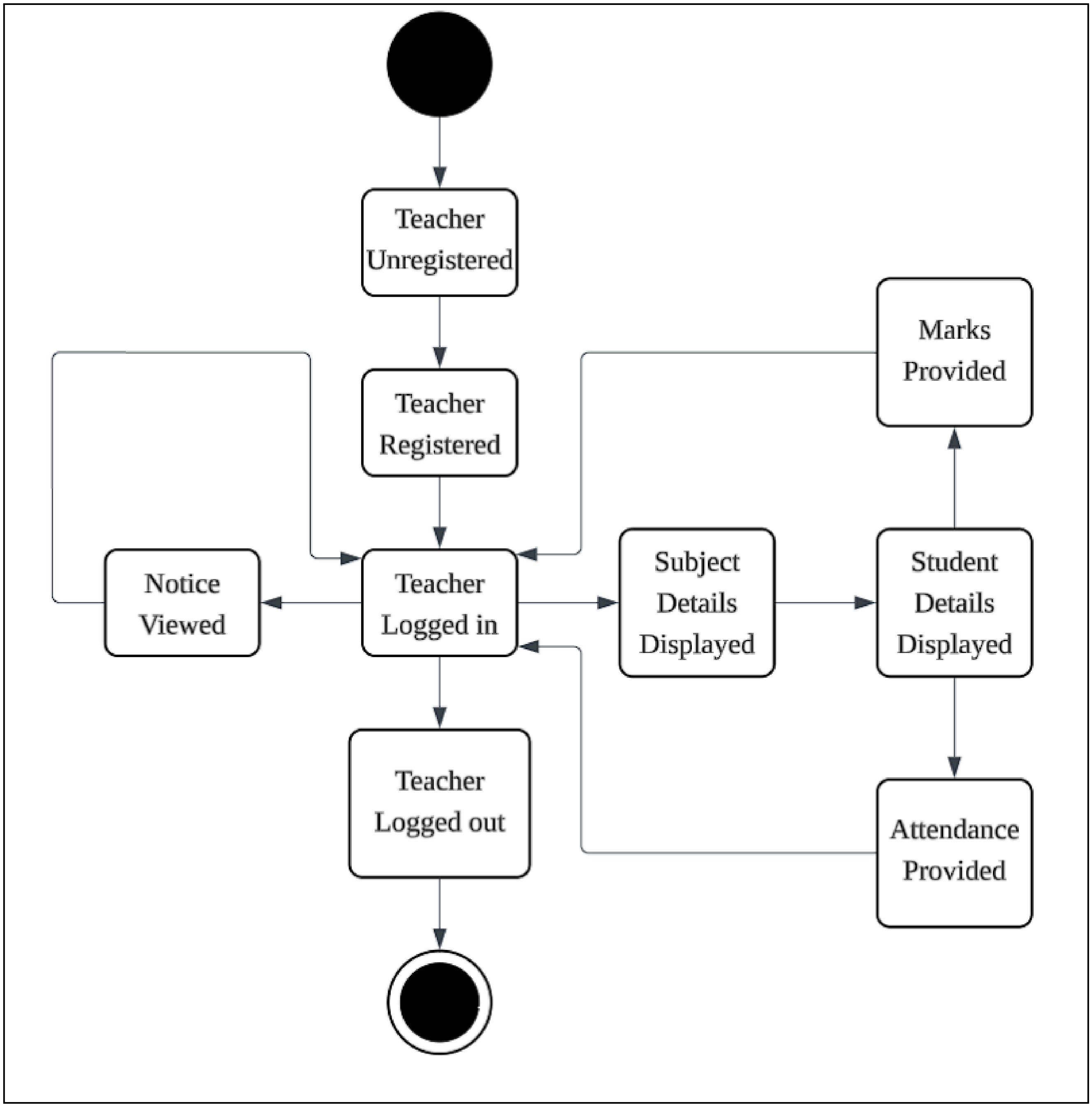


Figure 7.5.2: Teacher State Diagram

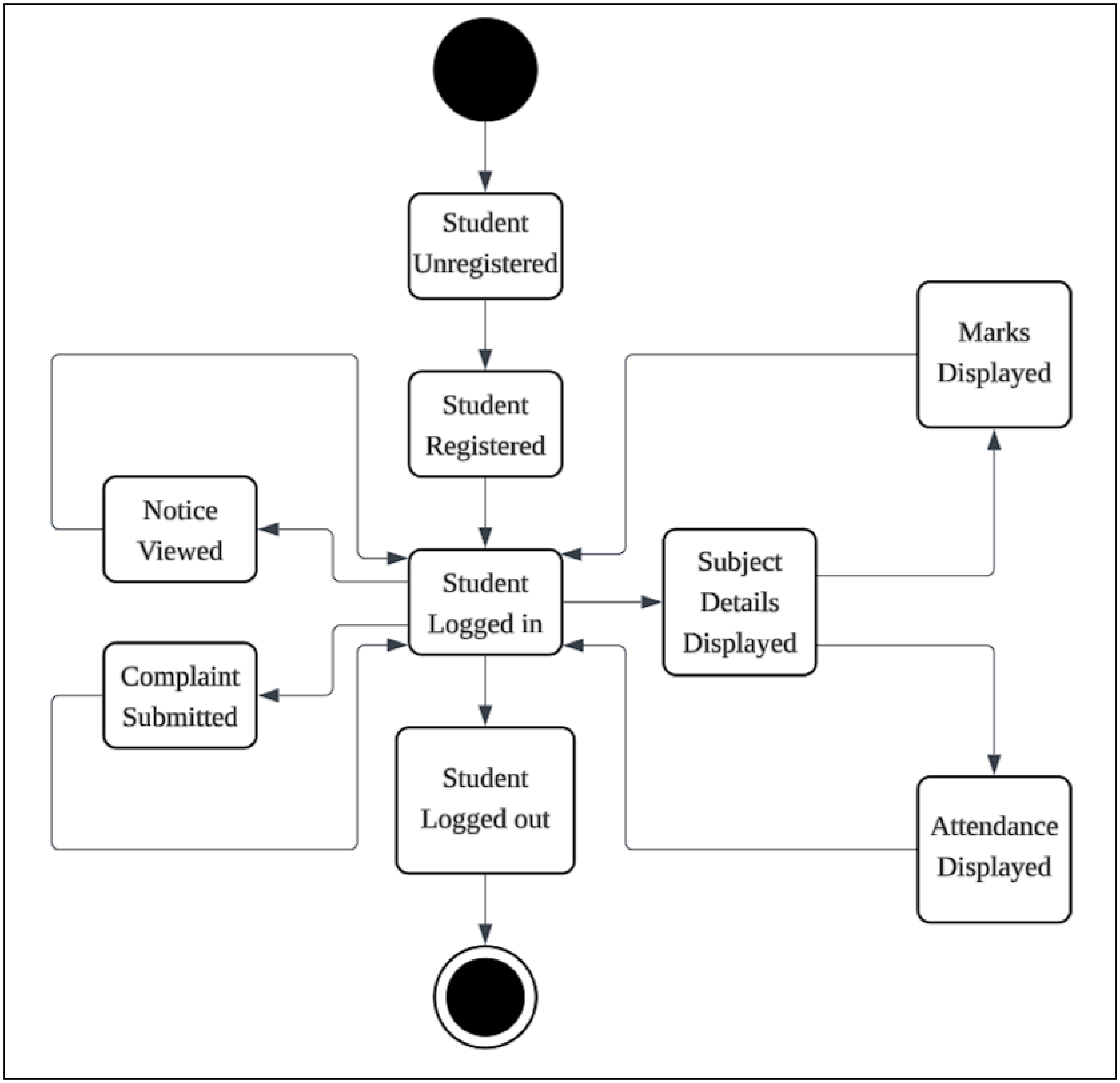
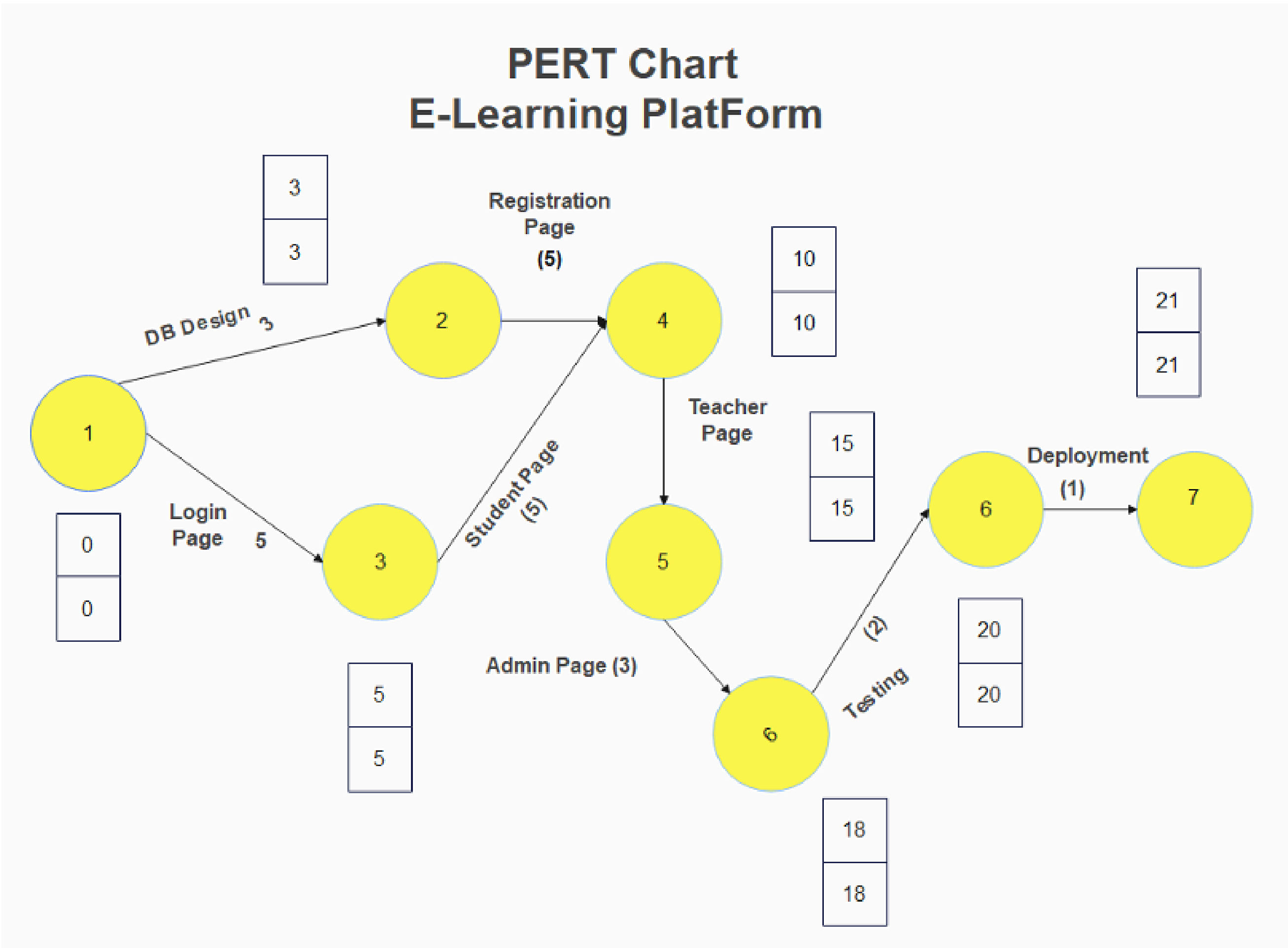


Figure 7.5.3: Student State Diagram

Appendix A: Source Code

The source code can be found through this GitHub link:
https://github.com/ZrfRz22/E-Learning-Platform-using-MERN-Stack-Software-Evolution.git

Appendix B: Project Management



GANTT CHART:

Task ID	Task Description	Start Date	Duration (Days)	End Date	Predecessor(s)
1.	Requirements Analysis	January 2, 2024	8	January 10, 2024	None
2.	Design	January 11, 2024	10	January 20, 2024	1
3.	Development	January 21, 2024	35	February 25, 2024	2
4.	Testing	February 26, 2024	4	February 29, 2024	3
5.	Deployment	March 1, 2024	31	March 31, 2024	4

January 2024 - January 2024 : [Requirements Analysis]

January 2024 - January 2024 : [Design]

January 2024 - February 2024 : [Development]

February 2024 - February 2024 : [Testing]

March 2024 - March 2024 : [Deployment]